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BOTTOM-LOADED PIVOTAL BONE ANCHOR ASSEMBLY

Abstract

A pivotal bone anchor assembly includes a receiver; an insert, an expandable open ring retainer, and a shank. During assembly, the insert and the open ring retainer are uploaded into a central bore and an inner cavity of a lower portion of the receiver, respectively, through a bottom opening. The shank head of the shank is then uploaded through the bottom opening and through the open ring retainer, causing the open ring retainer to expand to an expanded configuration as the shank head is uploaded therethrough and thereafter return to a non-expanded configuration and engage an interior seating surface of the inner cavity to capture the shank head within the lower portion of the receiver.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/514,962, filed Nov. 20, 2023, which is a divisional of U.S. application Ser. No. 18/045,113, now U.S. Pat. No. 11,819,249, which is a continuation of U.S. application Ser. No. 17/536,942, filed Nov. 29, 2021, now U.S. Pat. No. 11,464,549, which is a continuation of U.S. application Ser. No. 17/228,345, filed Apr. 12, 2021, now U.S. Pat. No. 11,185,352, which is a continuation of U.S. application Ser. No. 16/942,273, filed Jul. 29, 2020, now U.S. Pat. No. 10,973,555, which is a continuation of U.S. application Ser. No. 16/538,443, filed Aug. 12, 2019, now U.S. Pat. No. 10,765,456, which is a continuation of U.S. application Ser. No. 16/516,576, filed Jul. 19, 2019, now U.S. Pat. No. 10,765,455, which is continuation of U.S.

application Ser. No. 15/663,316, filed Jul. 28, 2017 now U.S. Pat. No. 10,441,319, which is a continuation of U.S. application Ser. No. 15/239,515, filed Aug. 17, 2016, now U.S. Pat. No. 9,956,006, which is a continuation of U.S. application Ser. No. 14/052,265, filed Oct. 11, 2013, now abandoned, which is a continuation of U.S. application Ser. No. 12/924,802, filed Oct. 5, 2010, now U.S. Pat. No. 8,556,938, each of which is incorporated by reference in its entirety herein and for all purposes. [0002] U.S. application Ser. No. 12/924,802 claims the benefit of priority from the following: U.S. Provisional Application No. 61/403,915, filed Sep. 23, 2010, U.S. Provisional Application No. 61/403,696, filed Sep. 20, 2010; U.S. Provisional Application No. 61/402,959, filed Sep. 8, 2010; U.S. Provisional Application No. 61/400,504, filed Jul. 29, 2010; U.S. Provisional Application No. 61/398,807, filed Jul. 1, 2010; U.S. Provisional Application No. 61/396,390, filed May 26, 2010; U.S. Provisional Application No. 61/395,752, filed May 17, 2010; U.S. Provisional Application No. 61/395,564, filed May 14, 2010, U.S. Provisional Application No. 61/343,737, filed May 3, 2010; U.S. Provisional Application No. 61/336,911, filed Jan. 28, 2010; and U.S. Provisional Application No. 61/278,240, filed Oct. 5, 2009; each of which is incorporated by reference in its entirety herein and for all purposes. [0003] U.S. application Ser. No. 14/052,265 is also a continuation-in-part of U.S. patent application Ser. No. 12/802,849, filed Jun. 15, 2010, now abandoned, which claims the benefit of priority from the following: U.S. Provisional Application No. 61/396,390, filed May 26, 2010; U.S. Provisional Application No. 61/395,752, filed May 17, 2010; U.S. Provisional Application No. 61/395,564, filed May 14, 2010; U.S. Provisional Application No. 61/336,911, filed Jan. 28, 2010; U.S. Provisional Application No. 61/270,754, filed Jul. 13, 2009; and U.S. Provisional Application No. 61/268,708, filed Jun. 15, 2009, each of which is incorporated by reference in its entirety herein and for all purposes.

BACKGROUND OF THE INVENTION

[0004] The present invention is directed to polyaxial bone screws for use in bone surgery, particularly spinal surgery.

[0005] Bone screws are utilized in many types of spinal surgery in order to secure various implants to vertebrae along the spinal column for the purpose of stabilizing and/or adjusting spinal alignment. Although both closed-ended and open-ended bone screws are known, open-ended screws are particularly well suited for connections to rods and connector arms, because such rods or arms do not need to be passed through a closed bore, but rather can be laid or urged into an open channel within a receiver or head of such a screw.

[0006] Typical open-ended bone screws include a threaded shank with a pair of parallel projecting branches or arms which form a yoke with a U-shaped slot or channel to receive a rod. Hooks and other types of connectors, as are used in spinal fixation techniques, may also include open ends for receiving rods or portions of other structure.

[0007] A common mechanism for providing vertebral support is to implant bone screws into certain bones which then in turn support a longitudinal structure such as a rod, or are supported by such a rod. Bone screws of this type may have a fixed head or receiver relative to a shank thereof. In the fixed bone screws, the rod receiver head cannot be moved relative to the shank and the rod must be favorably positioned in order for it to be placed within the receiver head. This is sometimes very difficult or impossible to do. Therefore, polyaxial bone screws are commonly preferred. Open-ended polyaxial bone screws typically allow for a loose or floppy rotation of the head or receiver about the shank until a desired rotational position of the head is achieved by fixing such position relative to the shank during a final stage of a medical procedure when a rod or other longitudinal connecting member is inserted into the head or receiver, followed by a locking screw or other closure.

SUMMARY OF THE INVENTION

[0008] A polyaxial bone anchor assembly according to the invention includes a receiver defining a chamber communicating with a channel, the channel sized and shaped for receiving a portion of a

longitudinal connecting member. The bone anchor further includes a shank having an upper portion and a retainer located in the chamber, the retainer being expandable in the chamber about the shank upper portion and receiving the upper portion therethrough to capture the upper portion in the chamber. The retainer is in a non-tapered locking engagement with the shank upper portion when the shank is in a locked orientation with respect to the receiver. The bone anchor assembly may include a variety of inserts, including compression inserts that may or may not have a lock and release feature as well as inserts having a super structure to provide a non-floppy friction fit between the insert and the shank upper portion when the shank is not otherwise locked in place with respect to the receiver. Furthermore, in some embodiments, the retainer may have super structure to provide a friction-fit insert.

[0009] A pre-assembled receiver, retainer and alternative insert may be “pushed-on”, “snapped-on” or “popped-on” to the shank head prior to or after implantation of the shank into a vertebra. Such a “snapping on” procedure includes the steps of uploading the shank head into the receiver lower opening, the shank head pressing against the retainer and expanding the resilient retainer portion out into an expansion portion of the receiver cavity followed by return of the retainer back to an original neutral shape thereof after the hemisphere of the shank head or upper portion passes through an open body portion of the retainer. The shank head may also enter into a friction fit super structure of either the retainer or an insert, panels or surfaces of the friction fit portion of the retainer or insert snapping or gripping onto the shank head as or after the retainer returns to a neutral or close to neutral orientation, providing a non-floppy connection between the retainer or insert and the shank head. The friction fit between the shank head and the retainer or insert is temporary. In several illustrated embodiments, when the shank is ultimately locked between the compression insert and the retainer non-tapered body, the friction fit portions of the retainer or insert typically are no longer in a friction fit engagement with the shank head. The final fixation typically occurs as a result of locking expansion type of contact between the shank head and the expandable retainer and expansion type of engagement between the retainer and the receiver cavity. In some embodiments, when the polyaxial mechanism is locked, an insert or a retainer portion is wedged against a surface of the receiver, allowing for adjustment or removal of the rod or other connecting member without loss of a desired angular relationship between the shank and the receiver.

[0010] Objects of the invention include providing apparatus and methods that are easy to use and especially adapted for the intended use thereof and wherein the tools are comparatively inexpensive to produce. Other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention.

[0011] The drawings constitute a part of this specification and include exemplary embodiments of the present invention and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is an enlarged and partial exploded perspective view of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a spring ring and a compression insert and also shown with a closure top and a longitudinal connecting member in the form of a rod.

[0013] FIG. 2 is an enlarged top plan view of the shank of FIG. 1.

[0014] FIG. 3 is a reduced cross-sectional view taken along the line 3-3 of FIG. 2.

[0015] FIG. 4 is an enlarged top plan view of the receiver of FIG. 1.

[0016] FIG. 5 is a bottom plan view of the receiver of FIG. 4.

[0017] FIG. 6 is a side elevational view of the receiver of FIG. 4.

[0018] FIG. 7 is a cross-sectional view taken along the line 7-7 of FIG. 4.

[0019] FIG. 8 is an enlarged perspective view of the retainer of FIG. 1.

[0020] FIG. 9 is a top plan view of the retainer of FIG. 8.

[0021] FIG. 10 is a bottom plan view of the retainer of FIG. 8.

[0022] FIG. 11 is a front elevational view of the retainer of FIG. 8.

[0023] FIG. 12 is a cross-sectional view taken along the line 12-12 of FIG. 9.

[0024] FIG. 13 is an enlarged perspective view of the compression insert of FIG. 1.

[0025] FIG. 14 is a front elevational view of the compression insert of FIG. 13.

[0026] FIG. 15 is a top plan view of the compression insert of FIG. 13.

[0027] FIG. 16 is a bottom plan view of the compression insert of FIG. 13.

[0028] FIG. 17 is a cross-sectional view taken along the line 17-17 of FIG. 14.

[0029] FIG. 18 is an enlarged and partial perspective view of the receiver and compression insert of FIG. 1 with portions broken away to show the detail thereof and shown in an early stage of assembly.

[0030] FIG. 19 is an enlarged partial front elevational view of the receiver, compression insert and retainer of FIG. 1 with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to that shown in FIG. 18.

[0031] FIG. 20 is a partial front elevational view, similar to FIG. 19, with portions broken away to show the detail thereof and showing the receiver, compression insert and retainer in a pre-assembled orientation with the compression insert and retainer captured within the receiver.

[0032] FIG. 21 is a partial front elevational view with portions broken away, similar to FIG. 20, illustrating capture of the compression insert within the receiver when the receiver is rotated or otherwise moved.

[0033] FIG. 22 is an enlarged and partial front elevational view of the shank of FIG. 1 with portions broken away to show the detail thereof, shown with a driving tool in a stage of implantation in a vertebra.

[0034] FIG. 23 is a partial front elevational view, similar to FIG. 22 and further showing an early stage of assembly of the shank with the pre-assembled receiver, compression insert and retainer of FIGS. 20 and 21.

[0035] FIG. 24 is an enlarged and partial front elevational view of the shank, receiver, compression insert and retainer of FIG. 1, with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to that shown in FIG. 23.

[0036] FIG. 25 is a partial front elevational view with portions broken away, similar to FIG. 24, showing a subsequent stage of assembly.

[0037] FIG. 26 is a partial front elevational view with portions broken away, similar to FIG. 25, showing the shank fully assembled with the receiver, compression insert and retainer and in a position ready to receive the longitudinal connecting member shown in FIG. 1 and further shown with a driving tool in phantom.

[0038] FIG. 27 is a partial side elevational view of the shank, receiver, compression insert and retainer of FIG. 26, with portions broken away to show the detail thereof and further shown with the shank disposed at an angle with respect to the receiver.

[0039] FIG. 28 is an enlarged perspective view of the entire assembly of FIG. 1 shown with the shank at an angle with respect to the receiver as shown in FIG. 27.

[0040] FIG. 29 is an enlarged and partial side elevational view of the assembly of FIG. 28 with portions broken away to show the detail thereof.

[0041] FIG. 30 is an enlarged and partial front elevational view of the entire assembly of FIG. 1 shown with the shank disposed axially aligned with the receiver as shown in FIG. 26 and further shown with a vertebra with portions broken away.

[0042] FIG. 31 is a partial front elevational view, similar to FIG. 30, with portions broken away to

show the detail thereof.

[0043] FIG. **32** is an enlarged and partial exploded front elevational view of another polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a spring ring and a compression insert.

[0044] FIG. **33** is an enlarged top plan view of the shank of FIG. **32**.

[0045] FIG. **34** is a reduced cross-sectional view taken along the line **34-34** of FIG. **33**.

[0046] FIG. **35** is an enlarged perspective view of the retainer of FIG. **32**.

[0047] FIG. **36** is another perspective view of the retainer of FIG. **32**.

[0048] FIG. **37** is a top plan view of the retainer of FIG. **35**.

[0049] FIG. **38** is a bottom plan view of the retainer of FIG. **35**.

[0050] FIG. **39** is a cross-sectional view taken along the line **39-39** of FIG. **37**.

[0051] FIG. **40** is an enlarged perspective view of the receiver of FIG. **32**.

[0052] FIG. **41** is a side elevational view of the receiver of FIG. **40**.

[0053] FIG. **42** is an enlarged cross-sectional view taken along the line **42-42** of FIG. **32**.

[0054] FIG. **43** is a cross-sectional view taken along the line **43-43** of FIG. **41**.

[0055] FIG. **44** is a reduced perspective view of the receiver of FIG. **40** with portions broken away to show the detail thereof.

[0056] FIG. **45** is an enlarged perspective view of the compression insert of FIG. **32**.

[0057] FIG. **46** is a top plan view of the compression insert of FIG. **45**.

[0058] FIG. **47** is a bottom plan view of the compression insert of FIG. **45**.

[0059] FIG. **48** is a front elevational view of the compression insert of FIG. **45**.

[0060] FIG. **49** is a cross-sectional view taken along the line **49-49** of FIG. **48**.

[0061] FIG. **50** is an enlarged and partial perspective view of the receiver and compression insert of FIG. **32** with portions of the receiver broken away to show the detail thereof and shown in an early stage of assembly.

[0062] FIG. **51** is an enlarged and partial front elevational view of the receiver, compression insert and retainer (shown in a compressed position) of FIG. **32** with portions of the receiver and compression insert broken away to show the detail thereof and shown in a stage of assembly subsequent to that shown in FIG. **50**.

[0063] FIG. **52** is a partial front elevational view, similar to FIG. **50**, with portions broken away to show the detail thereof and showing the receiver, compression insert and retainer in a pre-assembled orientation with the compression insert and retainer captured within the receiver.

[0064] FIG. **53** is an enlarged and partial front elevational view of the shank of FIG. **32** with portions broken away to show the detail thereof, shown with a driving tool in a stage of implantation in a vertebra.

[0065] FIG. **54** is a reduced and partial front elevational view of the implanted shank of FIG. **53** and further showing an early stage of assembly of the shank with the pre-assembled receiver, compression insert and retainer of FIG. **52**, also with portions broken away to show the detail thereof.

[0066] FIG. **55** is a partial front elevational view of the shank, receiver, compression insert and retainer of FIG. **54**, with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to that shown in FIG. **54**.

[0067] FIG. **56** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **55**, showing a subsequent stage of assembly.

[0068] FIG. **57** is a reduced and partial front elevational view with portions broken away, similar to FIG. **56**, showing the shank fully assembled with the receiver, compression insert and retainer and further including a longitudinal connecting member and a closure top.

[0069] FIG. **58** is a partial side elevational view of the shank, receiver, compression insert and retainer of FIG. **57**, with portions broken away to show the detail thereof and further shown with the shank disposed at an angle with respect to the receiver.

[0070] FIG. **59** is a partial side elevational view with portions broken away, similar to FIG. **58** showing the shank disposed at an alternative angle with respect to the receiver.

[0071] FIG. **60** is an exploded perspective view of a another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a spring ring and a compression insert.

[0072] FIG. **61** is an enlarged perspective view of the receiver of FIG. **60**.

[0073] FIG. **62** is a reduced side elevational view of the receiver of FIG. **61** with portions broken away to show the detail thereof.

[0074] FIG. **63** is a cross-sectional view taken along the line **63-63** of FIG. **62**.

[0075] FIG. **64** is an enlarged perspective view of the insert of FIG. **60**.

[0076] FIG. **65** is a front elevational view of the insert of FIG. **64** with portions broken away to show the detail thereof.

[0077] FIG. **66** is a top plan view of the insert of FIG. **64**.

[0078] FIG. **67** is a bottom plan view of the insert of FIG. **64**.

[0079] FIG. **68** is an enlarged front elevational view of the receiver of FIG. **60** shown in a stage of assembly with the insert of FIG. **60**, shown in enlarged side elevational view.

[0080] FIG. **69** is an enlarged front elevational view of the receiver of FIG. **60** with portions broken away to show the detail thereof shown in a stage of assembly with the insert subsequent to that shown in FIG. **68**, the insert in enlarged side elevational view with portions broken away to show the detail thereof.

[0081] FIG. **70** is an enlarged front elevational view of the receiver of FIG. **60** with portions broken away to show the detail thereof shown in a stage of assembly with the insert subsequent to that shown in FIG. **69**, the insert in enlarged side elevational view with portions broken away to show the detail thereof.

[0082] FIG. **71** is an enlarged front elevational view of the receiver of FIG. **60** with portions broken away to show the detail thereof shown in a stage of assembly with the insert subsequent to that shown in FIG. **70**, the insert in enlarged front elevational view.

[0083] FIG. **72** is an enlarged front elevational view of the receiver, insert and retainer of FIG. **60** with portions broken away to show the detail thereof shown in a pre-assembled orientation with the insert and retainer captured within the receiver.

[0084] FIG. **73** is a partial front elevational view of the receiver, insert and retainer with portions broken away to show the detail thereof, similar to FIG. **72**, and further showing a stage of assembly with the shank of FIG. **60**, shown in partial enlarged front elevational view.

[0085] FIG. **74** is a partial front elevational view with portions broken away, similar to FIG. **73** and showing a friction fit stage of assembly subsequent to that shown in FIG. **73**.

[0086] FIG. **75** is a partial front elevational view of the receiver, shank, retainer and insert with portions broken away, similar to FIG. **74**, further showing the rod and closure top of FIG. **60**, also in front elevational view with portions broken away, the assembly being in a locking stage of assembly subsequent to that shown in FIG. **74**.

[0087] FIG. **76** is an enlarged and partial front elevational view of the assembly shown in FIG. **74** with different portions broken away to show the detail thereof.

[0088] FIG. **77** is an enlarged and partial front elevational view of the assembly shown in FIG. **75** with different portions broken away to show the detail thereof.

[0089] FIG. **78** is a partial perspective view of the assembly of FIG. **75** with portions broken away to show the detail thereof.

[0090] FIG. **79** is a partial front elevational view with portions broken away, similar to FIG. **75** but showing the closure top and rod in a loosened position while the insert, shank, retainer and receiver remain in the locked position shown in FIG. **75**.

[0091] FIG. **80** is an enlarged and partial side elevational view of the assembly of FIG. **60** with portions broken away to show the detail thereof and the shank shown at an angle with respect to the

receiver.

[0092] FIG. **81** is an enlarged perspective view of an alternative compression insert for use with the assembly of FIG. **60**.

[0093] FIG. **82** is a front elevational view of the insert of FIG. **81** with portions broken away to show the detail thereof.

[0094] FIG. **83** is an enlarged and partial front elevational view of the assembly of FIG. **60** shown with the alternative insert of FIG. **81** and with portions broken away to show the detail thereof.

[0095] FIG. **84** is a reduced and partial side elevational view of two bone screw assemblies according to FIG. **60**, with portions broken away to show the detail thereof and shown with a multi-piece longitudinal connecting member, also shown with portions broken away, the connecting member having an inner cord and outer sleeves and spacers, also shown attached to a solid rod.

[0096] FIG. **85** is an enlarged front elevational view of an alternative closure top also shown in FIG. **84**.

[0097] FIG. **86** is a front elevational view of the closure top of FIG. **85** with portions broken away to show the detail thereof.

[0098] FIG. **87** is an enlarged front elevational view of another alternative closure top (not shown in FIG. **84**).

[0099] FIG. **88** is a front elevational view of the closure top of FIG. **87** with portions broken away to show the detail thereof.

[0100] FIG. **89** is an enlarged front elevational view of another closure top also shown in FIG. **84**.

[0101] FIG. **90** is a front elevational view of the closure top of FIG. **89** with portions broken away to show the detail thereof.

[0102] FIG. **91** is an exploded perspective view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a spring ring and a compression insert and shown with a closure top and a deformable rod.

[0103] FIG. **92** is an enlarged perspective view of the receiver of FIG. **91**.

[0104] FIG. **93** is a side elevational view of the receiver of FIG. **92**.

[0105] FIG. **94** is a cross-sectional view taken along the line **94-94** of FIG. **93**.

[0106] FIG. **95** is a cross-sectional view taken along the line **95-95** of FIG. **94**.

[0107] FIG. **96** is an enlarged perspective view of the insert of FIG. **91**.

[0108] FIG. **97** is a second perspective view of the insert of FIG. **96**.

[0109] FIG. **98** is a top plan view of the insert of FIG. **96**.

[0110] FIG. **99** is a bottom plan view of the insert of FIG. **96**.

[0111] FIG. **100** is a front elevational view of the insert of FIG. **96**.

[0112] FIG. **101** is a cross-sectional view taken along the line **101-101** of FIG. **98**.

[0113] FIG. **102** is an enlarged front elevational view of the receiver and an enlarged side elevational view of the insert of FIG. **91** shown in a stage of assembly.

[0114] FIG. **103** is a front elevational view, similar to FIG. **102** showing a later stage of assembly.

[0115] FIG. **104** is a perspective view showing the assembly step of FIG. **103**.

[0116] FIG. **105** is a front elevational view, similar to FIG. **103** showing a later stage of assembly.

[0117] FIG. **106** is an enlarged and partial perspective view of the receiver, insert and retainer of FIG. **91** with portions broken away to show the detail thereof.

[0118] FIG. **107** is an enlarged and partial perspective view of the receiver, insert and retainer and shown assembled with the shank of FIG. **91** and with portions broken away to show the detail thereof.

[0119] FIG. **108** is a reduced perspective view similar to FIG. **107** showing the shank at an angle with respect to the receiver.

[0120] FIG. **109** is an enlarged side elevational view similar to FIG. **108** with portions broken away to show the detail thereof.

[0121] FIG. **110** is an enlarged front elevational view of the assembly of FIG. **91** with portions

broken away showing a penultimate stage of assembly.

[0122] FIG. **111** is a front elevational view with portions broken away, similar to FIG. **110**, showing a final locked down stage of assembly.

[0123] FIG. **112** is an enlarged and partial view of the assembly as in FIG. **110** with portions broken away to show the detail thereof.

[0124] FIG. **113** is an enlarged and partial view of the assembly fully locked down as in FIG. **111** with portions broken away to show the detail thereof.

[0125] FIG. **114** is an enlarged and partial view, similar to FIG. **113** showing a loosened closure top with a fully locked down assembly.

[0126] FIG. **115** is an exploded perspective view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, upper and lower open retainer rings and a friction fit crown compression insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0127] FIG. **116** is an enlarged top plan view of the shank of FIG. **115**.

[0128] FIG. **117** is a reduced cross-sectional view taken along the line **117-117** of FIG. **116**.

[0129] FIG. **118** is an enlarged perspective view of the lower retainer of FIG. **115**.

[0130] FIG. **119** is another perspective view of the retainer of FIG. **118**.

[0131] FIG. **120** is a top plan view of the retainer of FIG. **118**

[0132] FIG. **121** is a bottom plan view of the retainer of FIG. **118**.

[0133] FIG. **122** is a cross-sectional view taken along the line **122-122** of FIG. **120**.

[0134] FIG. **123** is an enlarged perspective view of the friction fit crown insert of FIG. **115**.

[0135] FIG. **124** is a reduced front elevational view of the insert of FIG. **123**.

[0136] FIG. **125** is a reduced bottom plan view of the insert of FIG. **123**.

[0137] FIG. **126** is a reduced top plan view of the insert of FIG. **123**.

[0138] FIG. **127** is a cross-sectional view taken along the line **127-127** of FIG. **126**.

[0139] FIG. **128** is an enlarged perspective view of the receiver of FIG. **115**.

[0140] FIG. **129** is a second perspective view of the receiver of FIG. **128**.

[0141] FIG. **130** is a top plan view of the receiver of FIG. **128**.

[0142] FIG. **131** is a bottom plan view of the receiver of FIG. **128**.

[0143] FIG. **132** is an enlarged cross-sectional view taken along the line **132-132** of FIG. **130**.

[0144] FIG. **133** is an enlarged cross-sectional view taken along the line **133-133** of FIG. **130**.

[0145] FIG. **134** is an enlarged perspective view of the upper retainer of FIG. **115**.

[0146] FIG. **135** is an enlarged top plan view of the retainer of FIG. **134**.

[0147] FIG. **136** is a cross-sectional view taken along the line **136-136** of FIG. **135**.

[0148] FIG. **137** is an enlarged top plan view of the closure top of FIG. **115**.

[0149] FIG. **138** is a cross-sectional view taken along the line **138-138** of FIG. **137**.

[0150] FIG. **139** is an enlarged front elevational view of the receiver and upper retainer of FIG. **115** with portions of the receiver broken away to show the detail thereof, the upper retainer being shown in a compressed insertion stage of assembly.

[0151] FIG. **140** is a front elevational view with portions broken away, similar to FIG. **139**, showing the upper retainer in a neutral position, assembled within the receiver.

[0152] FIG. **141** is a front elevational view with portions broken away, similar to FIG. **140** and further showing the friction fit compression insert of FIG. **115** in an initial stage of assembly with the receiver.

[0153] FIG. **142** is a front elevational view with portions broken away, similar to FIG. **141**, showing the compression insert uploaded into the receiver and in engagement with the upper retainer, the upper retainer in an expanded position.

[0154] FIG. **143** is a front elevational view with portions broken away, similar to FIG. **142** and further showing the lower retainer of FIG. **115** in front elevation and in a compressed state, the lower retainer being shown in a stage of uploading into the receiver.

[0155] FIG. **144** is a front elevational view with portions broken away, similar to FIG. **143** showing the lower retainer within the receiver and in a neutral non-compressed state.

[0156] FIG. **145** is a front elevational view with portions broken away, similar to FIG. **144** and further showing a shank of FIG. **115** in partial front elevation.

[0157] FIG. **146** is a partial front elevational view with portions broken away, similar to FIG. **145** showing the shank in a stage of assembly with the lower retainer ring, the lower retainer ring being pushed up into engagement with the compression insert.

[0158] FIG. **147** is a partial front elevational view with portions broken away, similar to FIG. **146**, showing the lower retainer in an expanded state about an upper portion of the shank, the shank upper portion in a stage of assembly with the compression insert.

[0159] FIG. **148** is a partial front elevational view with portions broken away, similar to FIG. **147**, the shank upper portion in frictional engagement with the compression insert and the lower retainer in a substantially neutral state.

[0160] FIG. **149** is a partial front elevational view with portions broken away, similar to FIG. **148**, the shank upper portion and attached compression insert being in a downward, fully assembled position, the upper retainer being in a substantially neutral state.

[0161] FIG. **150** is a reduced partial front elevational view of the assembly of FIG. **149**, shown with the shank pivoted at an angle with respect to the receiver.

[0162] FIG. **151** is a front elevational view of the assembly of FIG. **150**, shown in a vertebra and in a locked position with the rod portion and closure top of FIG. **115**.

[0163] FIG. **152** is an enlarged and partial front elevational view of the assembly of FIG. **151** with portions broken away to show the detail thereof.

[0164] FIG. **153** is a partial front elevational view of an alternative embodiment of a bone screw assembly, substantially similar to the bone screw assembly shown in FIG. **115**, shown with portions broken away to show the detail thereof.

[0165] FIG. **154** is another partial front elevational view of the bone screw assembly of FIG. **153**, shown with the shank disposed at an angle with respect to the receiver.

[0166] FIG. **155** is a reduced front elevational view, similar to FIG. **154**, showing the bone screw assembly with a rod and closure top.

[0167] FIG. **156** is an enlarged front elevational view of the assembly of FIG. **155** with portions broken away to show the detail thereof.

[0168] FIG. **157** is an exploded perspective view of another polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of an open ring and a friction fit crown compression insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0169] FIG. **158** is an enlarged top plan view of the shank of FIG. **157**.

[0170] FIG. **159** is a reduced cross-sectional view taken along the line **159-159** of FIG. **158**.

[0171] FIG. **160** is an enlarged perspective view of the retainer of FIG. **157**.

[0172] FIG. **161** is another perspective view of the retainer of FIG. **160**.

[0173] FIG. **162** is a top plan view of the retainer of FIG. **160**.

[0174] FIG. **163** is a bottom plan view of the retainer of FIG. **160**.

[0175] FIG. **164** is a cross-sectional view taken along the line **164-164** of FIG. **162**.

[0176] FIG. **165** is an enlarged perspective view of the friction fit crown insert of FIG. **157**.

[0177] FIG. **166** is another perspective view of the insert of FIG. **165**.

[0178] FIG. **167** is a top plan view of the insert of FIG. **165**.

[0179] FIG. **168** is a bottom plan view of the insert of FIG. **165**.

[0180] FIG. **169** is a cross-sectional view taken along the line **169-169** of FIG. **167**.

[0181] FIG. **170** is a cross-sectional view taken along the line **170-170** of FIG. **167**.

[0182] FIG. **171** is an enlarged perspective view of the receiver of FIG. **157**.

[0183] FIG. **172** is a side elevational view of the receiver of FIG. **171** with portions broken away to

show the detail thereof.

[0184] FIG. **173** is a top plan view of the receiver of FIG. **171**.

[0185] FIG. **174** is a bottom plan view of the receiver of FIG. **171**.

[0186] FIG. **175** is an enlarged side elevational view of the insert of FIG. **157** and a front elevational view of the receiver of FIG. **157** with portions of the receiver broken away to show the detail thereof, the insert being shown downloaded into the receiver in an insertion stage of assembly.

[0187] FIG. **176** is a reduced front elevational view of the receiver, with portions broken away, similar to FIG. **175**, showing the insert of FIG. **175** also in reduced front elevational view, the insert having been lowered into the receiver and rotated there-within during an assembly stage subsequent to that shown in FIG. **175**.

[0188] FIG. **177** is a front elevational view with portions broken away, similar to FIG. **176** and further showing the retainer of FIG. **157** in front elevation and in a compressed state, the retainer being shown in a stage of uploading into the receiver.

[0189] FIG. **178** is a front elevational view with portions broken away, similar to FIG. **177** showing the retainer within the receiver and in a neutral non-compressed state and further showing a shank of FIG. **157** in partial front elevation being uploaded into the receiver.

[0190] FIG. **179** is a reduced front elevational view similar to FIG. **178** showing an alternative assembly stage in which the shank of FIG. **157** is first implanted in a vertebra, followed by assembly with the receiver, retainer and insert.

[0191] FIG. **180** is a partial front elevational view with portions broken away, similar to FIG. **178** showing the shank in a stage of assembly with the retainer, the retainer being pushed up into engagement with the crown insert.

[0192] FIG. **181** is a partial front elevational view with portions broken away, similar to FIG. **180**, showing the retainer in an expanded state about an upper portion of the shank, the shank upper portion in a stage of assembly with the insert.

[0193] FIG. **182** is a partial front elevational view with portions broken away, similar to FIG. **181**, the shank upper portion in frictional engagement with the insert and the retainer in a substantially neutral state.

[0194] FIG. **183** is a partial front elevational view with portions broken away, similar to FIG. **182**, the shank upper portion with attached insert being shown pulled down slightly from the position shown in FIG. **182**, the insert being placed into frictional engagement with the receiver.

[0195] FIG. **184** is a partial front elevational view with portions broken away, similar to FIG. **183**, the shank upper portion and attached insert being in a downward, fully assembled position, the insert being further wedged against inner surfaces of the receiver.

[0196] FIG. **185** is a partial front elevational view of the assembly of FIG. **184** with portions broken away to show the detail thereof and further shown in a locked position with a rod and closure top of FIG. **157**.

[0197] FIG. **186** is a partial front elevational view with portions broken away, similar to FIG. **185** showing the assembly remaining in the locked position of FIG. **185** when the rod and closure top are removed.

[0198] FIG. **187** is a reduced and partial front elevational view with portions broken away, similar to FIG. **185**, showing a locked assembly wherein the shank is disposed at an angle with respect to the receiver.

[0199] FIG. **188** is an exploded perspective view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a top-loadable open ring and a friction fit crown compression insert, the assembly further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0200] FIG. **189** is an enlarged top plan view of the shank of FIG. **188**.

[0201] FIG. **190** is a reduced cross-sectional view taken along the line **190-190** of FIG. **189**.

[0202] FIG. **191** is an enlarged perspective view of the retainer of FIG. **188**.
[0203] FIG. **192** is another perspective view of the retainer of FIG. **191**.
[0204] FIG. **193** is a top plan view of the retainer of FIG. **191**.
[0205] FIG. **194** is a bottom plan view of the retainer of FIG. **191**.
[0206] FIG. **195** is a cross-sectional view taken along the line **195-195** of FIG. **193**.
[0207] FIG. **196** is an enlarged perspective view of the friction fit crown insert of FIG. **188**.
[0208] FIG. **197** is another perspective view of the insert of FIG. **196**.
[0209] FIG. **198** is a top plan view of the insert of FIG. **196**.
[0210] FIG. **199** is a bottom plan view of the insert of FIG. **196**.
[0211] FIG. **200** is a cross-sectional view taken along the line **200-200** of FIG. **198**.
[0212] FIG. **201** is a cross-sectional view taken along the line **201-201** of FIG. **198**.
[0213] FIG. **202** is an enlarged perspective view of the receiver of FIG. **188**.
[0214] FIG. **203** is an enlarged side elevational view of the receiver of FIG. **202** with portions broken away to show the detail thereof.
[0215] FIG. **204** is a top plan view of the receiver of FIG. **202**.
[0216] FIG. **205** is a bottom plan view of the receiver of FIG. **202**.
[0217] FIG. **206** is an enlarged cross-sectional view taken along the line **206-206** of FIG. **203**.
[0218] FIG. **207** is an enlarged front elevational view of the receiver of FIG. **188** with portions broken away to show the detail thereof, shown with the retainer of FIG. **188** in perspective view being top loaded into the receiver during an early stage of assembly.
[0219] FIG. **208** is an enlarged side elevational view of the insert of FIG. **188** and a front elevational view of the receiver and retainer of FIG. **207** with portions broken away to show the detail thereof, the insert being shown top loaded into the receiver.
[0220] FIG. **209** is a front elevational view of the receiver and retainer, with portions broken away, similar to FIG. **208**, showing the insert of FIG. **208** in side elevation lowered into the receiver.
[0221] FIG. **210** is an enlarged front elevational view of the receiver, retainer and insert, with portions broken away, similar to FIG. **209**, the insert having been rotated into an assembled position within the receiver.
[0222] FIG. **211** is an enlarged and partial perspective view of the receiver, retainer and insert of FIG. **210**.
[0223] FIG. **212** is a partial perspective view, similar to FIG. **211** showing holding tabs of the receiver bent against the insert to prohibit further rotation thereof.
[0224] FIG. **213** is a cross-sectional view taken along the line **213-213** of FIG. **212**.
[0225] FIG. **214** is a reduced front elevational view with portions broken away, similar to FIG. **210** shown subsequent to the assembly step shown in FIGS. **212** and **213** and further showing the shank of FIG. **188** in partial front elevation being uploaded into the receiver.
[0226] FIG. **215** is a front elevational view with portions broken away, similar to FIG. **214** showing the shank in a stage of assembly with the retainer, the retainer being pushed up into engagement with the crown insert.
[0227] FIG. **216** is a partial front elevational view with portions broken away, similar to FIG. **215**, showing the retainer in an expanded state about an upper portion of the shank, the shank upper portion in a stage of assembly with the insert.
[0228] FIG. **217** is a partial front elevational view with portions broken away, similar to FIG. **216**, the shank upper portion in frictional engagement with the insert and the retainer in a substantially neutral state.
[0229] FIG. **218** is a partial front elevational view with portions broken away, similar to FIG. **217**, the shank upper portion with attached insert being shown pulled down slightly from the position shown in FIG. **217**.
[0230] FIG. **219** is a partial front elevational view with portions broken away, similar to FIG. **218**, the shank upper portion and attached insert being in a downward, fully assembled position, the

insert being allowed to expand under a ledge surface of the receiver.

[0231] FIG. **220** is a partial front elevational view of the assembly of FIG. **219** with portions broken away to show the detail thereof and further shown in a locked position with a rod and closure top of FIG. **188**.

[0232] FIG. **221** is a partial front elevational view with portions broken away, similar to FIG. **220**, showing a locked assembly wherein the shank is disposed at an angle with respect to the receiver.

[0233] FIG. **222** is an exploded front elevational view with portions broken away of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a retainer in the form of a top-loadable open ring and a lock and release friction fit crown compression insert, the assembly further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0234] FIG. **223** is an enlarged front elevational view, with portions broken away of the receiver and retainer of FIG. **222** showing stages of assembly of the retainer in phantom and with the insert of FIG. **222** shown in side elevational view prior to insertion and rotation into place within the receiver.

[0235] FIG. **224** is a front elevational view of the retainer and receiver of FIG. **223** with portions broken away and a side elevational view of the insert of FIG. **223** in a stage of assembly just prior to rotation within the receiver.

[0236] FIG. **225** is a front elevational view with portions broken away, similar to FIG. **224**, showing the insert after rotation thereof within the receiver.

[0237] FIG. **226** is a front elevational view with portions broken away, similar to FIG. **225** further showing the shank of FIG. **222** in partial front elevation and in an assembly step with the receiver and retainer.

[0238] FIG. **227** is a partial front elevational view with portions broken away showing an assembly step subsequent to that shown in FIG. **226**.

[0239] FIG. **228** is a partial front elevational view with portions broken away showing an assembly step subsequent to that shown in FIG. **227** with the rod and closure of FIG. **222**, also in front elevation.

[0240] FIG. **229** is a partial front elevational view with portions broken away, similar to FIG. **228**, showing the shank being retained and locked in place by the insert when the rod and closure top are removed.

[0241] FIG. **230** is a partial front elevational view with portions broken away, similar to FIG. **228**, showing the rod and closure top being replaced by an alternative deformable rod and cooperating alternative closure top.

[0242] FIG. **231** is an enlarged perspective view of an alternative locking insert for use with the assembly of FIG. **222** in lieu of the insert that is shown in FIG. **222**.

[0243] FIG. **232** is a reduced bottom plan view of the insert shown in FIG. **231**.

[0244] FIG. **233** is an enlarged cross-sectional view taken along the line **233-233** of FIG. **232**.

[0245] FIG. **234** is an enlarged perspective view of another alternative non-locking insert for use with the assembly of FIG. **222** in lieu of the insert shown in FIG. **222**.

[0246] FIG. **235** is an exploded perspective view of a receiver, retainer ring and insert of another embodiment of a polyaxial bone screw assembly according to the invention that is substantially similar to the assembly shown in FIG. **222**.

[0247] FIG. **236** is a front elevational view of the receiver of FIG. **235** shown with portions broken away to show the detail thereof.

[0248] FIG. **237** is a cross-sectional view taken along the line **237-237** of FIG. **236**.

[0249] FIG. **238** is a front elevational view of the retainer and receiver of FIG. **235** with portions broken away and a side elevational view of the insert of FIG. **235** in a stage of assembly just prior to rotation within the receiver.

[0250] FIG. **239** is a front elevational view with portions broken away, similar to FIG. **238**,

showing the insert being rotated within the receiver.

[0251] FIG. **240** is a front elevational view with portions broken away, similar to FIG. **239**, shown subsequent to rotation of the insert within the receiver.

[0252] FIG. **241** is a partial front elevational view with portions broken away, similar to FIG. **240** and further showing assembly with a shank, a rod and a closure top of FIG. **222**.

[0253] FIG. **242** is a partial front elevational view with portions broken away, similar to FIG. **241**, showing the rod and closure top removed and further showing unlocking of the insert from the receiver with a two-piece tool having an inner insert engaging portion and an outer tubular holding portion.

[0254] FIG. **243** is a reduced and partial front elevational view of the two-piece tool of FIG. **242**, holding prongs of the inner insert engaging portion being shown in phantom.

[0255] FIG. **244** is a partial front elevational view of the inner insert engaging portion of the tool shown in FIG. **242** with portions broken away to show the detail thereof.

[0256] FIG. **245** is an exploded front elevational view of another polyaxial bone screw assembly according to the present invention including a shank, a receiver, an open friction fit retainer and a compression insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0257] FIG. **246** is an enlarged top plan view of the shank of FIG. **245**.

[0258] FIG. **247** is a reduced cross-sectional view taken along the line **247-247** of FIG. **246**.

[0259] FIG. **248** is an enlarged side elevational view of the receiver of FIG. **245**.

[0260] FIG. **249** is a reduced perspective view of the receiver of FIG. **248**.

[0261] FIG. **250** is a reduced top plan view of the receiver of FIG. **248**.

[0262] FIG. **251** is a reduced bottom plan view of the receiver of FIG. **248**.

[0263] FIG. **252** is a reduced cross-sectional view taken along the line **252-252** of FIG. **250**.

[0264] FIG. **253** is an enlarged cross-sectional view taken along the line **253-253** of FIG. **250**.

[0265] FIG. **254** is an enlarged perspective view of the retainer of FIG. **245**.

[0266] FIG. **255** is an enlarged side elevational view of the retainer of FIG. **254**.

[0267] FIG. **256** is an enlarged front elevational view of the retainer of FIG. **254**.

[0268] FIG. **257** is an enlarged top plan view of the retainer of FIG. **254**.

[0269] FIG. **258** is an enlarged bottom plan view of the retainer of FIG. **254**.

[0270] FIG. **259** is a cross-sectional view taken along the line **259-259** of FIG. **257**.

[0271] FIG. **260** is an enlarged perspective view of the insert of FIG. **245**.

[0272] FIG. **261** is an enlarged side elevational view of the insert of FIG. **260**.

[0273] FIG. **262** is an insert of FIG. **260**.

[0274] FIG. **263** is an insert of FIG. **260**.

[0275] FIG. **264** is an enlarged top plan view of the enlarged bottom plan view of the cross-sectional view taken along the line **264-264** of FIG. **262**.

[0276] FIG. **265** is an enlarged front elevational view of an alternative insert according to the invention for use in lieu of the insert shown in FIG. **245**, with portions broken away to show the detail thereof.

[0277] FIG. **266** is an enlarged front elevational view of the retainer and receiver of FIG. **245** with portions of the receiver broken away (as illustrated in FIG. **271**) to show the detail thereof, the retainer being shown downloaded into the receiver (in phantom) to a partially inserted stage of assembly.

[0278] FIG. **267** is a front elevational view of the retainer and receiver with portions broken away, similar to that shown in FIG. **266**, further showing the retainer seated within the receiver and also showing the insert of FIG. **245** in side elevation (in phantom) above the receiver and then being downloaded into the receiver to a partially inserted stage of assembly.

[0279] FIG. **268** is a front elevational view with portions broken away, similar to FIG. **267**, showing the insert rotated into a position in alignment with the receiver.

[0280] FIG. **269** is a front elevational view with portions broken away, similar to FIG. **268** showing arms of the retainer being pinched (with a tool not shown) towards one another and the retainer partially moved upwardly within the receiver.

[0281] FIG. **270** is a front elevational view similar to FIG. **269** showing the retainer arms placed in a desired upward position within the receiver and the pinching tool removed so that the retainer pushes outwardly against the receiver and is held against the receiver during shipping.

[0282] FIG. **271** is a reduced perspective view with portions broken away of the assembly as shown in FIG. **270**.

[0283] FIG. **272** is a perspective view with portions broken away, similar to FIG. **271**, showing a portion of the receiver crimped against the insert.

[0284] FIG. **273** is an enlarged front elevational view with portions broken away, similar to FIG. **270**, also including the crimping of FIG. **272** and further showing an enlarged and partial shank of FIG. **245** in a first stage of assembly with the retainer, a hemisphere of the shank head and a vertebra portion are both shown in phantom.

[0285] FIG. **274** is a partial front elevational view with portions broken away, similar to FIG. **273**, showing the retainer lower portion in an expanded state about a mid-portion of the shank head, the head hemisphere shown in phantom.

[0286] FIG. **275** is a reduced partial front elevational view with portions broken away, similar to FIG. **274**, the shank upper portion or head in frictional engagement with an upper portion of the retainer.

[0287] FIG. **276** is a partial side elevational view with portions broken away of the assembly in a stage as shown in FIG. **275**.

[0288] FIG. **277** is a partial front elevational view with portions broken away, similar to FIG. **275**, the shank upper portion with attached retainer being shown pulled down into a seated position within the lower receiver cavity.

[0289] FIG. **278** is an enlarged and partial front elevational view with portions broken away of the entire assembly of FIG. **245**, the assembly shown in a locked position with the insert wedged against surfaces of the receiver.

[0290] FIG. **279** is an enlarged and partial side elevational view with portions broken away of the entire assembly of FIG. **245**, shown locked into position with the shank disposed at an angle with respect to the receiver, the rod being shown in phantom.

[0291] FIG. **280** is a reduced and partial front elevational view with portions broken away, similar to FIG. **278**, showing the insert retaining the assembly in a locked position when the closure top and the rod are removed.

[0292] FIG. **281** is an enlarged and partial front elevational view with portion broken away, similar to FIG. **280**, further showing the assembly with a replacement deformable rod and alternative closure top.

[0293] FIG. **282** is an enlarged perspective view of an alternative non-locking insert according to the invention for use with the assembly of FIG. **245**.

[0294] FIG. **283** is an enlarged and partial front elevational view of the assembly of FIG. **245** shown in a fully assembled locked position with the non-locking insert of FIG. **282** in lieu of the locking insert shown in FIG. **245**, with portions broken away to show the detail thereof.

[0295] FIG. **284** is an exploded perspective view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver and a retainer ring, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0296] FIG. **285** is an enlarged top plan view of the shank of FIG. **284**.

[0297] FIG. **286** is a reduced cross-sectional view taken along the line **286-286** of FIG. **285**.

[0298] FIG. **287** is an enlarged perspective view of the lower retainer of FIG. **284**.

[0299] FIG. **288** is another perspective view of the retainer of FIG. **287**.

[0300] FIG. **289** is a cross-sectional view taken along the line **289-289** of FIG. **287**.
[0301] FIG. **290** is an enlarged front elevational view of the receiver of FIG. **284**.
[0302] FIG. **291** is a side elevational view of the receiver of FIG. **290**.
[0303] FIG. **292** is a top plan view of the receiver of FIG. **290**.
[0304] FIG. **293** is a bottom plan view of the receiver of FIG. **290**.
[0305] FIG. **294** is a cross-sectional view taken along the line **294-294** of FIG. **292**.
[0306] FIG. **295** is a cross-sectional view taken along the line **295-295** of FIG. **292**.
[0307] FIG. **296** is an enlarged front elevational view of the receiver and retainer of FIG. **284** with portions of the receiver broken away to show the detail thereof, the retainer being shown in a compressed insertion stage of assembly.
[0308] FIG. **297** is a front elevational view with portions broken away, similar to FIG. **296**, showing the retainer in a neutral position, assembled with the receiver.
[0309] FIG. **298** is a front elevational view with portions broken away, similar to FIG. **297** and further showing the shank of FIG. **284** in partial front elevation and implanted in a vertebra.
[0310] FIG. **299** is a partial front elevational view with portions broken away, similar to FIG. **298** showing the shank in a stage of assembly with the lower retainer ring, the lower retainer ring being pushed up into engagement with the receiver.
[0311] FIG. **300** is a partial front elevational view with portions broken away, similar to FIG. **299**, showing the lower retainer in an expanded state about an upper portion of the shank.
[0312] FIG. **301** is a partial front elevational view with portions broken away, similar to FIG. **300**, the shank upper portion in engagement with a portion of the receiver and the retainer in a substantially neutral state.
[0313] FIG. **302** is a partial front elevational view with portions broken away, similar to FIG. **301**, the shank upper portion being in a downward, fully assembled position, the retainer being in a substantially neutral or slightly contracted state.
[0314] FIG. **303** is a partial front elevational view of the assembly of FIG. **302**, with portions broken away and shown in a locked position with the rod portion and closure top of FIG. **284**.
[0315] FIG. **304** is a partial perspective view of the assembly of FIG. **303** with the rod shown in phantom.
[0316] FIG. **305** is a perspective view of an alternative shank according to the invention that may be used with the assembly of FIG. **284** in lieu of the shank shown in FIG. **284**.
[0317] FIG. **306** is a perspective view of another alternative shank according to the invention that may be used with the assembly of FIG. **284** in lieu of the shank shown in FIG. **284**.
[0318] FIG. **307** is an exploded front elevational view of another polyaxial bone screw assembly according to the present invention including a shank, a receiver and a friction fit retainer, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.
[0319] FIG. **308** is an enlarged top plan view of the shank of FIG. **307**.
[0320] FIG. **309** is a reduced cross-sectional view taken along the line **309-309** of FIG. **308**.
[0321] FIG. **310** is an enlarged perspective view of the retainer of FIG. **307**.
[0322] FIG. **311** is a side elevational view of the retainer of FIG. **310**.
[0323] FIG. **312** is a front elevational view of the retainer of FIG. **310**.
[0324] FIG. **313** is a top plan view of the retainer of FIG. **310**.
[0325] FIG. **314** is a bottom plan view of the retainer of a cross-sectional view taken along the line **315-315** of FIG. **310**.
[0326] FIG. **315** is a bottom plan view of the retainer of a cross-sectional view taken along the line **315-315** of FIG. **313**.
[0327] FIG. **316** is an enlarged perspective view of the receiver of FIG. **307**.
[0328] FIG. **317** is a top plan view of the receiver of a bottom plan view of the receiver of FIG. **316**.
[0329] FIG. **318** is a bottom plan view of the receiver of FIG. **316**.

[0330] FIG. **319** is a cross-sectional view taken along the line **319-319** of FIG. **317**.
[0331] FIG. **320** is a cross-sectional view taken along the line **320-320** of FIG. **317**.
[0332] FIG. **321** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver.
[0333] FIG. **322** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **321**.
[0334] FIG. **323** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **322**.
[0335] FIG. **324** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **323**.
[0336] FIG. **325** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **324**.
[0337] FIG. **326** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **325**.
[0338] FIG. **327** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **326**.
[0339] FIG. **328** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **327**.
[0340] FIG. **329** is a reduced cross-sectional view of the receiver of FIG. **320** and a reduced front elevational view of the retainer of FIG. **312** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **328**.
[0341] FIG. **330** is an enlarged cross-sectional view of the receiver and front elevational view of the retainer, similar to FIG. **329** and further showing a partial front elevational view of the shank of FIG. **307** shown in a stage of assembly with the receiver and retainer.
[0342] FIG. **331** is an enlarged and partial front elevational view, similar to FIG. **330**, with portions broken away to show the detail thereof and showing the shank in a stage of assembly with the receiver and retainer subsequent to what is shown in FIG. **330**.
[0343] FIG. **332** is an enlarged and partial front elevational view with portions broken away, similar to FIG. **331**, showing the shank in a stage of assembly with the receiver and retainer subsequent to what is shown in FIG. **331**.
[0344] FIG. **333** is a reduced and partial front elevational view with portions broken away, similar to FIG. **332**, showing the retainer in a stage of assembly with the receiver subsequent to what is shown in FIG. **332**.
[0345] FIG. **334** is a partial front elevational view with portions broken away, similar to FIG. **333**, showing the retainer in a stage of assembly with the receiver subsequent to what is shown in FIG. **333**.
[0346] FIG. **335** is an enlarged and partial front elevational view, similar to FIG. **334**, with further portions broken away to shown the detail thereof.
[0347] FIG. **336** is an enlarged and partial front elevational view of a fully assembled shank, retainer, receiver, rod and closure top of FIG. **307** with portions broken away to show the detail thereof.
[0348] FIG. **337** is a partial side elevational view of the shank of FIG. **307** shown implanted in a vertebra and in an early stage of assembly with a retainer and receiver of FIG. **307**, also shown in

side elevation.

[0349] FIG. **338** is a partial side elevational view of the shank, retainer and receiver of FIG. **337**, shown fully assembled and further shown assembled with the rod and closure top of FIG. **307**, also in side elevation.

[0350] FIG. **339** is an enlarged and partial side elevational view of the shank, retainer, receiver, rod and closure top of FIG. **338** with portions broken away to show the detail thereof.

[0351] FIG. **340** is an exploded front elevational view of another embodiment of a polyaxial bone screw assembly according to the present invention including a shank, a receiver, a friction fit retainer and a lock and release insert, further shown with a portion of a longitudinal connecting member in the form of a rod and a closure top.

[0352] FIG. **341** is an enlarged top plan view of the shank of FIG. **340**.

[0353] FIG. **342** is a reduced cross-sectional view taken along the line **342-342** of FIG. **341**.

[0354] FIG. **343** is an enlarged perspective view of the retainer of FIG. **340**.

[0355] FIG. **344** is a side elevational view of the retainer of FIG. **343**.

[0356] FIG. **345** is a front elevational view of the retainer of FIG. **343**.

[0357] FIG. **346** is a top plan view of the retainer of FIG. **343**.

[0358] FIG. **347** is a bottom plan view of the retainer of FIG. **343**.

[0359] FIG. **348** is a cross-sectional view taken along the line **348-348** of FIG. **346**.

[0360] FIG. **349** is an enlarged perspective view of the receiver of FIG. **340**.

[0361] FIG. **350** is a top plan view of the receiver of FIG. **349**.

[0362] FIG. **351** is a bottom plan view of the receiver of FIG. **349**.

[0363] FIG. **352** is a cross-sectional view taken along the line **352-352** of FIG. **350**.

[0364] FIG. **353** is a cross-sectional view taken along the line **353-353** of FIG. **350**.

[0365] FIG. **354** is an enlarged side elevational view of the insert of FIG. **340**.

[0366] FIG. **355** is a front elevational view of the insert of FIG. **354**.

[0367] FIG. **356** is a top plan view of the insert of FIG. **354**.

[0368] FIG. **357** is a bottom plan view of the insert of FIG. **354**.

[0369] FIG. **358** is an enlarged perspective view of the insert of FIG. **354**.

[0370] FIG. **359** is another perspective view of the insert of FIG. **354**.

[0371] FIG. **360** is a cross-sectional view taken along the line **360-360** of FIG. **356**.

[0372] FIG. **361** is a cross-sectional view taken along the line **361-361** of FIG. **356**.

[0373] FIG. **362** is a reduced front-elevational view of the receiver of FIG. **349** and a reduced front elevational view of the retainer of FIG. **345** shown in a stage of assembly with the receiver.

[0374] FIG. **363** is a reduced cross-sectional view of the receiver as in FIG. **353** and a reduced front elevational view of the retainer of FIG. **345** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **362**.

[0375] FIG. **364** is a reduced cross-sectional view of the receiver of FIG. **353** and a reduced front elevational view of the retainer of FIG. **345** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **363**.

[0376] FIG. **365** is a reduced cross-sectional view of the receiver of FIG. **353** and a reduced front elevational view of the retainer of FIG. **345** shown in a stage of assembly with the receiver subsequent to what is shown in FIG. **364** and further shown in a first stage of loading with the insert of FIG. **354**, also in reduced front elevational view.

[0377] FIG. **366** is an enlarged front elevational view of the receiver, retainer and insert of FIG. **365** with portions broken away to show the detail thereof and shown in an initial stage of assembly of the insert into the receiver.

[0378] FIG. **367** is a front elevational view of the receiver, retainer and insert of FIG. **366** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **366**.

[0379] FIG. **368** is an enlarged front elevational view of the receiver, retainer and insert of FIG.

367 with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **367**.

[0380] FIG. **369** is a front elevational view of the receiver, retainer and insert of FIG. **368** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **368**.

[0381] FIG. **370** is a reduced front elevational view of the receiver, retainer and insert of FIG. **369** with portions broken away to show the detail thereof and shown in a first stage of assembly with a shank of FIG. **340**, in reduced and partial front elevational view and shown implanted in a vertebra.

[0382] FIG. **371** is an enlarged and partial front elevational view of the receiver, retainer, insert and shank of FIG. **370** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **370**.

[0383] FIG. **372** is a partial front elevational view of the receiver, retainer, insert and shank of FIG. **371** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **371**.

[0384] FIG. **373** is a reduced and partial front elevational view of the receiver, retainer, insert and shank of FIG. **372** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIG. **372**.

[0385] FIG. **374** is an enlarged and partial perspective view of the assembly of FIG. **373**.

[0386] FIG. **375** is an enlarged and partial front elevational view of the receiver, retainer, insert and shank of FIG. **373** with portions broken away to show the detail thereof and shown in a stage of assembly subsequent to what is shown in FIGS. **373** and **374**.

[0387] FIG. **376** is an enlarged and partial perspective view of the assembly of FIG. **375**.

[0388] FIG. **377** is an enlarged and partial perspective view, similar to FIG. **376** and further showing the rod (in phantom) and closure of FIG. **340** in a stage of assembly and also in enlarged perspective view.

[0389] FIG. **378** is an enlarged and partial front elevational view of the assembly of FIG. **377**, with a portion of the receiver broken away to show the detail thereof.

[0390] FIG. **379** is a reduced and partial front elevational view, similar to FIG. **378**, with a portion of the receiver broken away to show the detail thereof, and shown in a final stage of assembly subsequent to that shown in FIGS. **377** and **378**.

[0391] FIG. **380** is a reduced and partial side elevational view of the final assembly of FIG. **379**.

[0392] FIG. **381** is an enlarged and partial cross-sectional view taken along the line **381-381** of FIG. **379**.

[0393] FIG. **382** is a partial cross-sectional view, similar to FIG. **381**, showing an alternative position of the shank with respect to the receiver.

[0394] FIG. **383** is a reduced and partial perspective and partially exploded view of the assembly of FIG. **382**, showing the closure and rod being removed from the receiver and the insert retaining the shank in a locked position.

[0395] FIG. **384** is a partial perspective and partially exploded view similar to FIG. **383**, showing the closure and hard rod of FIG. **383** being replaced by an alternative closure and a deformable rod.

[0396] FIG. **385** is an enlarged and partial side elevational view of the assembly of FIG. **384** with portions broken away to show the detail thereof.

[0397] FIG. **386** is an exploded and partial perspective view of an alternative bone screw assembly according to the invention including a shank, a receiver, a retainer and a non-locking insert, and further shown with a rod and a closure top.

[0398] FIG. **387** is an enlarged front elevational view of the insert of FIG. **386**.

[0399] FIG. **388** is a side elevational view of the insert of FIG. **387**.

[0400] FIG. **389** is a top plan view of the insert of FIG. **387**.

[0401] FIG. **390** is a bottom plan view of the insert of FIG. **387**.

[0402] FIG. **391** is a perspective view of the insert of FIG. **387**.

[0403] FIG. **392** is an enlarged and partial front elevational view of the receiver, the shank, the retainer and the insert of FIG. **386** shown in a stage of assembly.

[0404] FIG. **393** is an enlarged and partial perspective view of the assembly of FIG. **386** shown fully assembled and with the rod in phantom.

[0405] FIG. **394** is an enlarged and partial front elevational view of the assembly of FIG. **393** with portions broken away to show the detail thereof.

[0406] FIG. **395** is a reduced perspective view, similar to FIG. **393**, showing an alternative angular position of the shank with respect to the receiver.

[0407] FIG. **396** is an enlarged and partial side elevational view of the assembly as shown in FIG. **395** with portions broken away to show the detail thereof.

[0408] FIG. **397** is a reduced and partial front elevational view of the assembly of FIG. **394** with further portions broken away to show the detail thereof.

[0409] FIG. **398** is a perspective view of an alternative insert for use with the assembly of FIG. **91**.

[0410] FIG. **399** is a front elevational view of the insert of FIG. **398**.

[0411] FIG. **400** is a side elevational view of the insert of FIG. **398**.

[0412] FIG. **401** is a front elevational view, similar to FIG. **399** with portions broken away to show the detail thereof.

[0413] FIG. **402** is a side elevational view, similar to FIG. **400** with portions broken away to show the detail thereof.

DETAILED DESCRIPTION OF THE INVENTION

[0414] As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in virtually any appropriately detailed structure. It is also noted that any reference to the words top, bottom, up and down, and the like, in this application refers to the alignment shown in the various drawings, as well as the normal connotations applied to such devices, and is not intended to restrict positioning of the bone attachment structures in actual use.

[0415] With reference to FIGS. **1-31** the reference number **1** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **1** includes a shank **4**, that further includes a body **6** integral with an upwardly extending upper portion or capture structure **8**; a receiver **10**; a retainer structure **12** and a compression or pressure insert **14**. The receiver **10**, retainer **12** and compression insert **14** are initially assembled and may be further assembled with the shank **4** either prior or subsequent to implantation of the shank body **6** into a vertebra **13**, as will be described in greater detail below. FIG. **1** further shows a closure structure **18** of the invention for capturing a longitudinal member, for example, a rod **21** which in turn engages the compression insert **14** that presses against the shank upper portion **8** into fixed frictional contact with the retainer **12**, so as to capture, and fix the longitudinal connecting member **21** within the receiver **10** and thus fix the member **21** relative to the vertebra **13**. The illustrated rod **21** is hard, stiff, nonelastic and cylindrical, having an outer cylindrical surface **22**. It is foreseen that in other embodiments, the rod **21** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **10** and the shank **4** cooperate in such a manner that the receiver **10** and the shank **4** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **10** with the shank **4** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0416] The shank **4**, best illustrated in FIGS. **1-3** and **22**, is elongate, with the shank body **6** having a helically wound bone implantable thread **24** (single or dual lead thread form) extending from near a neck **26** located adjacent to the upper portion or capture structure **8**, to a tip **28** of the body **6** and

extending radially outwardly therefrom. During use, the body **6** utilizing the thread **24** for gripping and advancement is implanted into the vertebra **13** leading with the tip **28** and driven down into the vertebra with an installation or driving tool **29**, so as to be implanted in the vertebra to near the neck **26**, as more fully described in the paragraphs below. The shank **4** has an elongate axis of rotation generally identified by the reference letter A.

[0417] The neck **26** extends axially upward from the shank body **6**. The neck **26** may be of the same or is typically of a slightly reduced radius as compared to an adjacent upper end or top **32** of the body **6** where the thread **24** terminates. Further extending axially and outwardly from the neck **26** is the shank upper portion **8** that provides a connective or capture apparatus disposed at a distance from the upper end **32** and thus at a distance from the vertebra **13** when the body **6** is implanted in such vertebra.

[0418] The shank upper portion **8** is configured for a pivotable connection between the shank **4** and the retainer **12** and receiver **10** prior to fixing of the shank **4** in a desired position with respect to the receiver **10**. The shank upper portion **8** has an outer, convex and substantially spherical surface **34** that extends outwardly and upwardly from the neck **26** and terminates at a substantially planar top surface **38**. The spherical surface **34** has an outer radius configured for sliding cooperation and ultimate frictional mating with a concave surface of the compression insert **14** having a substantially similar radius, and also a flat or, in some embodiments, curved surface of the retainer **12**, discussed more fully in the paragraphs below. The top surface **38** is substantially perpendicular to the axis A. The spherical surface **34** shown in the present embodiment is substantially smooth with the exception of a stepped or graduated upper surface portion **40** located adjacent to the top surface **38** and sized and shaped for cooperation and ultimate frictional engagement with the compression insert **14**. In the illustrated embodiment the surface portion **40** includes at least three graduated cylindrical surfaces disposed substantially parallel to the axis A and adjacent perpendicular step surfaces that are disposed generally perpendicular to the axis A. It is foreseen that the surface portion **40** may include greater or fewer number of stepped surfaces. It is foreseen that the surface portion **40** and also the rest of the surface **34** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the retainer **12** and/or the compression insert **14**.

[0419] A counter sunk substantially planar base or seating surface **45** partially defines an internal drive feature or imprint **46**. The illustrated internal drive feature **46** is an aperture formed in the top surface **38** and has a hex shape designed to receive the hex tool **29** of an Allen wrench type, into the aperture for rotating and driving the bone screw shank **4**. It is foreseen that such an internal tool engagement structure may take a variety of tool-engaging forms and may include one or more apertures of various shapes, such as a pair of spaced apart apertures or a multi-lobular or star-shaped aperture, such as those sold under the trademark TORX, or the like. The seat or base **45** of the drive feature **46** is disposed perpendicular to the axis A with the drive feature **46** otherwise being coaxial with the axis A. In operation, the driving tool **29** is received in the internal drive feature **46**, being seated at the base **45** and engaging the six faces of the drive feature **46** for both driving and rotating the shank body **6** into the vertebra **13**, either before the shank **4** is attached to the receiver **10** as shown in FIG. 22 or after the shank **4** is attached to the receiver **10** as shown in FIG. 26, with the shank body **6** being driven into the vertebra **13** with the driving tool extending into the receiver **10** as shown in phantom.

[0420] The shank **4** shown in the drawings is cannulated, having a small central bore **50** extending an entire length of the shank **4** along the axis A. The bore **50** is defined by an inner cylindrical wall of the shank **4** and has a circular opening at the shank tip **28** and an upper opening communicating with the external drive **46** at the surface **45**. The bore **50** is coaxial with the threaded body **6** and the upper portion **8**. The bore **50** provides a passage through the shank **4** interior for a length of wire (not shown) inserted into the vertebra **13** prior to the insertion of the shank body **6**, the wire providing a guide for insertion of the shank body **6** into the vertebra **13**.

[0421] To provide a biologically active interface with the bone, the threaded shank body **6** may be coated, perforated, made porous or otherwise treated. The treatment may include, but is not limited to a plasma spray coating or other type of coating of a metal or, for example, a calcium phosphate; or a roughening, perforation or indentation in the shank surface, such as by sputtering, sand blasting or acid etching, that allows for bony ingrowth or ongrowth. Certain metal coatings act as a scaffold for bone ingrowth. Bio-ceramic calcium phosphate coatings include, but are not limited to: alpha-tri-calcium phosphate and beta-tri-calcium phosphate ($\text{Ca}_{3}(\text{PO}_{4})_{2}$), tetra-calcium phosphate ($\text{Ca}_{4}(\text{P}_{2}\text{O}_{7})_{2}$), amorphous calcium phosphate and hydroxyapatite ($\text{Ca}_{10}(\text{PO}_{4})_{6}(\text{OH})_{2}$). Coating with hydroxyapatite, for example, is desirable as hydroxyapatite is chemically similar to bone with respect to mineral content and has been identified as being bioactive and thus not only supportive of bone ingrowth, but actively taking part in bone bonding.

[0422] With particular reference to FIGS. **1** and **4-7**, the receiver **10** has a generally U-shaped appearance with a partially discontinuous substantially cylindrical inner and outer profile. The receiver **10** has an axis of rotation B that is shown in FIG. **1** as being aligned with and the same as the axis of rotation A of the shank **4**, such orientation being desirable, but not required during assembly of the receiver **10** with the shank **4**. After the receiver **10** is pivotally attached to the shank **4**, either before or after the shank **4** is implanted in a vertebra **13**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIGS. **27-29**.

[0423] The receiver **10** includes a substantially cylindrical base **60** integral with a pair of opposed upstanding arms **62** forming a cradle and defining a channel **64** between the arms **62** with an upper opening, generally **66**, and a U-shaped lower seat **68**, the channel **64** having a width for operably snugly receiving the rod **21** between the arms **62**. Each of the arms **62** has an interior surface, generally **70**, that includes various inner cylindrical profiles, an upper of which is a partial helically wound guide and advancement structure **72** located adjacent top surfaces **73** of each of the arms **62**. In the illustrated embodiment, the guide and advancement structure **72** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **18**, as described more fully below. However, it is foreseen that the guide and advancement structure **72** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structure for operably guiding under rotation and advancing the closure structure **18** downward between the arms **62**, as well as eventual torquing when the closure structure **18** abuts against the rod **21**.

[0424] An opposed pair of tool receiving and engaging apertures **74** are formed on outer surfaces **76** of the arms **62**. Furthermore, two pair of tool receiving and engaging apertures **77** are formed in front and rear surfaces **78** of the arms **62**. Some or all of the apertures **74** and **77** may be used for holding the receiver **10** during assembly with the shank **4** and the retainer **12**, during the implantation of the shank body **6** into a vertebra when the shank is pre-assembled with the receiver **10**, and during assembly of the bone anchor assembly **1** with the rod **21** and the closure structure **18**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **62**.

[0425] Returning to the interior surface **70** of the receiver arms **62**, located below the guide and advancement structure **72** is a cylindrical surface **82** partially defining a run-out feature for the guide and advancement structure **72**. The cylindrical surface **82** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **72**. Moving downwardly, in a direction toward the base **60**, adjacent the cylindrical surface **82** is a run-out seat or surface **84** that extends inwardly toward the axis B and runs perpendicular to the axis B. Adjacent to and located below the surface **84** is another cylindrical surface **86** having a diameter smaller than the diameter of the surface **82**. A discontinuous annular surface **88** that provides an abutment surface or stop for capturing the compression insert **14** in the receiver **10** is located below and adjacent to the cylindrical surface **86**. The abutment surface **88** is disposed substantially perpendicular to the axis

B. Another cylindrical surface **90** is located below and adjacent to the surface **88**. The cylindrical surface **90** is oriented substantially parallel to the axis B and is sized and shaped to slidably receive the compression insert **14** as will be described in greater detail below. The surface **90** surrounds the U-shaped channel seat **68** and extends downwardly into the base **60**. Thus, a portion of the surface **90** extends upwardly into the arms **62**. The cylindrical surface **90** has a diameter greater than the diameter of the cylindrical surface **86**. A continuous annular surface **92** is located below and adjacent to the cylindrical surface **90**. The surface **92** is disposed in the base **60** and forms a stop for the resilient retainer **12**, prohibiting the retainer **12** (when in an uncompressed configuration) from moving upwardly into a space or cavity **91** defined by the cylindrical surface **90** that holds the compression insert **14**. Another cylindrical surface **94** is located below and adjacent to the surface **92**. The cylindrical surface **94** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **12** as will be described in greater detail below. The surfaces **92** and **94** partially define a circumferential recess or expansion chamber **95** that is sized and shaped to receive the retainer **12** as it expands around the shank upper portion **8** as the shank **8** moves upwardly toward the channel **64** during assembly, as well as form a restriction to prevent the expanded retainer **12** from moving upwardly with the shank portion **8**, the surface **92** preventing the retainer **12** from passing from the chamber **95** into the cavity **91** whether the retainer **12** is in an expanded position as shown in FIG. 24, or in a neutral or original operative position as shown in FIG. 25. A cylindrical surface **96** located below the cylindrical surface **94** of the expansion chamber is sized and shaped to closely receive the retainer **12** when the retainer is in a neutral interim position as shown in FIG. 26, for example, or expanded operative position as shown in FIG. 29, for example. Thus, the cylindrical surface **96** has a diameter smaller than the diameter of the cylindrical surface **94** that defines the expansion chamber **95**. The surface **96** also has a diameter larger than an outside diameter of the retainer **12** so that the retainer may expand outwardly into contact with the surface **96** when the bone screw shank upper portion **8** presses downwardly during locking of the shank **4** against the retainer **12**. The surface **96** is joined or connected to the surface **94** by one or more beveled, curved or conical surfaces **97**. The surfaces **97** allow for sliding gradual movement and/or contraction of the retainer **12** into the space defined by the surface **96** and ultimate seating of the retainer **12** on a lower annular seating surface **99** located below and adjacent to the cylindrical surface **96**. The surfaces **96** and **99** provide a seating chamber for the retainer **12** wherein the retainer expands out to the surface **96** when in a locked position as shown, for example, in FIG. 29. Located below and adjacent to the annular seating surface **99** is another cylindrical surface **100** that communicates with a beveled or flared bottom opening surface **102**, the surface **102** communicating with an exterior base surface **104** of the base **60**, defining a lower opening, generally **106**, of the receiver **10**. The illustrated surface **100** has a diameter that is substantially the same as the diameter of the surface **90**, allowing for slidable uploading of the compression insert **14** while requiring compression or squeezing of the retainer **12** during uploading of the retainer **12** through the lower opening **106** (see FIGS. 18 and 19).

[0426] With particular reference to FIGS. 1 and 8-12, the retainer **12** that operates to capture the shank upper portion **8** and the compression insert **14** within the receiver **10** has a central axis C that is operationally the same as the axis B associated with the receiver **10** when the shank upper portion **8** and the retainer **12** are installed within the receiver **10**. The retainer **12** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **12** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The retainer **12** has a central channel or hollow through bore, generally **121**, that passes entirely through the structure **12** from a top surface **122** to a bottom surface **124** thereof. Surfaces that define the channel or bore **121** include a discontinuous inner cylindrical surface **125** adjacent the top surface **122** and a discontinuous frusto-conical or beveled surface **127** adjacent the surface **125**, both surfaces coaxial with the axis C when the retainer **12** is in a neutral non-compressed, non-expanded orientation. The retainer **12** further includes an outer cylindrical surface **130** located

adjacent the top surface **122** and an outer beveled or frusto-conical surface **132** adjacent the bottom surface **124**. The surface **130** is oriented parallel to the axis C. Two or more evenly spaced notches **133** are formed in the cylindrical surface **130** to more evenly distribute stress across the entire retainer during contraction and expansion thereof. In other embodiments of the invention, the notches **133** may be on the inside of the ring or they may be omitted. The resilient retainer **12** further includes first and second end surfaces, **134** and **135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **134** and **135** are disposed substantially perpendicular to the top surface **122** and the bottom surface **124**. A width X between the surfaces **134** and **135** is determined by a desired amount of compressibility of the open retainer **12** when loaded into the receiver **10**. The space X shown in FIG. **8** provides adequate space between the surfaces **134** and **135** for the retainer **12** to be pinched, with the surfaces **134** and **135** compressed toward one another (as shown by arrows P and Q in FIG. **19**) to a closely spaced or even touching configuration, if necessary, to an extent that the compressed retainer **12** is up or bottom loadable through the receiver opening **106** as shown in FIG. **19**. After passing through the opening **106** and along a portion of the lower inner surface, the retainer **12** expands or springs back to an original uncompressed, rounded or collar-like configuration of FIGS. **8-12**, see, e.g., FIG. **21**. The embodiment shown in FIGS. **8-12** illustrates the surfaces **134** and **135** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer **12** into the receiver **10**.

[0427] With reference to FIGS. **1** and **13-17**, the compression insert **14** is illustrated that is sized and shaped to be received by and up-loaded into the receiver **10** at the lower opening **106**. The compression insert **14** has an operational central axis that is the same as the central axis B of the receiver **10**. The compression insert **14** has a central channel or through bore substantially defined by an inner cylindrical surface **141** coaxial with an inner partially spherical surface **142**. The compression insert **14** through bore is sized and shaped to receive the driving tool **29** therethrough that engages the shank drive feature **46** when the shank body **6** is driven into bone with the receiver **10** attached. The surface **142** is sized and shaped to slidably receive and ultimately frictionally engage the substantially spherical or domed surface **34** of the shank upper portion **8**, in particular the stepped or ridged surface **40** such that the surface **142** initially slidably and pivotally mates with the spherical surface **34** to create a ball-and-socket type joint. The surface **142** may include a roughening or surface finish to aid in frictional contact between the surface **142** and the surfaces **34** and/or **40**, once a desired angle of articulation of the shank **4** with respect to the receiver **10** is reached.

[0428] The compression insert **14** also includes an outer cylindrical surface **144** terminating at a top surface **145**. The top surface **145** engages the rod **21** or other longitudinal connecting member during operation of the assembly **1** and locates the rod above the lower seat **68** of the receiver. The top surface **145** may further include an outer bevel **147** that is located adjacent the outer cylindrical surface **144**. A bottom surface **149** extends between the spherical surface **142** and the outer cylindrical surface **144**.

[0429] It is foreseen that in some embodiments of the invention the compression insert **14** may further include upstanding arms that cradle the rod **21** or other connecting member. Such arms may be located spaced from the closure top **18** in some embodiments and may be sized and shaped to contact the closure top **18** in other embodiments in order to provide locking of the polyaxial mechanism of the assembly with capture but without fixing of the rod **21** or other longitudinal connecting member with respect to the closure top **18**.

[0430] The compression or pressure insert **14** ultimately seats on the shank upper portion **8** and is disposed substantially within the upper cylindrical wall **90**. In operation, the insert **14** extends at least partially in the channel **64** of the receiver **10** such that the top surface **145** substantially contacts and engages the outer surface **22** of the rod **21** when such rod is placed in the receiver **10**.

and the closure structure or top **18** is tightened thereon.

[0431] With reference to FIGS. **1** and **28-31**, the illustrated elongate rod or longitudinal connecting member **21** can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **22** of uniform diameter. The rod **21** may be made from a variety of metals, metal alloys and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials.

[0432] Longitudinal connecting members for use with the assembly **1** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the insert **14** may be modified so as to closely hold, and if desired, fix or slidingly capture the longitudinal connecting member to the assembly **1**. Some embodiments of the assembly **1** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the compression insert **14** of the receiver having a U-shaped channel (or rectangular- or other-shaped channel) for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **1**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **1**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from deformable plastics to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

[0433] With reference to FIGS. **1** and **28-31**, the closure structure or closure top **18** shown with the assembly **1** is rotatably received between the spaced arms **62**. It is noted that the closure **18** can be any of a variety of different types of closure structures for use in conjunction with the present invention with suitable mating structure on the upstanding arms **62**. It is also foreseen that the closure top could be a twist-in or slide-in closure structure. The illustrated closure structure **18** is substantially cylindrical and includes an outer helically wound guide and advancement structure **162** in the form of a flange form that operably joins with the guide and advancement structure **72** disposed on the arms **62** of the receiver **10**. The flange form utilized in accordance with the present invention may take a variety of forms, including those described in Applicant's U.S. Pat. No. 6,726,689, which is incorporated herein by reference. It is also foreseen that according to the invention the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure for operably guiding under rotation and advancing the closure structure **18** downward between the arms **62** and having such a nature as to resist splaying of the arms **62** when the closure structure **18** is advanced into the U-shaped channel **64**. The illustrated closure structure **18** also includes a top surface **164** with an internal drive **166** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **166** is used for both rotatable engagement and, if

needed, disengagement of the closure **18** from the receiver arms **62**. It is also foreseen that the closure structure **18** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **168** of the closure is planar and further includes a point **169** and a rim **170** for engagement and penetration into the surface **22** of the rod **21** in certain embodiments of the invention. The closure top **18** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **18** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **62**. [0434] Preferably, the receiver **10**, the retainer **12** and the compression insert **14** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **12**. In some circumstances, the shank **4** is also assembled with the receiver **10**, retainer **12** and compression insert **14** at the factory. In other instances, it is desirable to first implant the shank **4**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **4**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way.

[0435] Pre-assembly of the receiver **10**, retainer **12** and compression insert **14** is shown in FIGS. **18-20**. First, the compression insert **14** is uploaded into the receiver **10** through the lower opening **106** with the insert top surface **145** facing the receiver bottom surface **104**. The insert **14** is slid upwardly toward the channel seat **68** until the insert is within the cylindrical wall **90**. If pressed further upwardly, the insert top surface **145** abuts against the receiver surface **88**. Then, the resilient open retainer **12** is prepared for insertion into the receiver **10** by squeezing or pressing the retainer end surfaces **134** and **135** toward one another as shown by the arrows P and Q in FIG. **19**. The compressed retainer **12** is inserted into the lower opening **106** with the top surface **122** facing the receiver bottom surface **104**. The retainer **12** is typically moved upwardly into the receiver **10** and past the cylindrical surface **96** and allowed to expand to a neutral uncompressed state within the cylindrical surface **94** as shown in FIG. **21**. Also as shown in FIG. **21**, at this time, both the compression insert **14** and the retainer **12** are captured within the receiver **10**. The insert **14** is captured by the retainer **12** at the bottom surface **149** thereof and the top surface **145** abuts against the receiver **88** if the receiver **10** is tipped upside down as shown in FIG. **21**. The retainer **12** cannot move beyond the receiver surface **92** at the top **122** thereof and cannot move beyond the receiver surface **99** at the bottom **124** thereof when in a neutral, non-compressed state. At this time the receiver **10**, compression insert **14** and retainer **12** combination is pre-assembled and ready for assembly with the shank **4** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **4** as will be described herein.

[0436] As illustrated in FIG. **22**, the bone screw shank **4** (or as shown in FIG. **26**, an entire assembly **1** made up of the assembled shank **4**, receiver **10**, retainer **12** and compression insert **14**) is screwed into a bone, such as the vertebra **13**, by rotation of the shank **4** using a suitable driving tool **29** that operably drives and rotates the shank body **6** by engagement thereof at the internal drive **46**. Specifically, the vertebra **13** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **4** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank or assembly is threaded onto the guide wire utilizing the cannulation bore **50** by first threading the wire into the opening at the bottom **28** and then out of the top opening at the drive feature **46**. The shank **4** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **21** (also having a central lumen in some embodiments) and the closure top **18** (also with a central bore) can

be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires.

[0437] Again with respect to FIGS. 22 and 23, when the shank 4 is driven into the vertebra 13 without the remainder of the assembly 1, the shank 4 may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer. With reference to FIGS. 23-26, the pre-assembled receiver, insert and retainer are placed above the shank upper portion 8 until the shank upper portion is received within the opening 106. As the shank is moved into the interior of the receiver base, the shank upper portion 8 presses the retainer 12 upwardly into the groove 95 (if the retainer is not already located within such groove). As the portion 8 continues to move upwardly toward the channel 64, the retainer top surface 122 abuts against the annular surface 92 stopping upward movement of the retainer 12 and forcing outward movement of the retainer 12 towards the cylindrical surface 94 defining the expansion groove 95 as the spherical surface 34 continues in an upward direction. The retainer 12 begins to contract about the spherical surface 34 as the center of the sphere passes beyond the center of the retainer expansion groove 95 (see FIG. 25). The retainer 12 is then moved down into a final operative position shown in FIG. 26 by either an upward pull on the receiver 10 or, in some cases, by driving the shank 4 further into the vertebra 13 as shown in phantom in FIG. 26. Also, in some embodiments, when the receiver 10 is pre-assembled with the shank 4, the entire assembly 1 may be implanted at this time by inserting the driving tool 20 into the receiver and the shank drive 46 as shown in FIG. 26 and rotating and driving the shank 4 into a desired location of the vertebra 13. At this time, the receiver 10 may be articulated to a desired position with respect to the shank 4 as shown, for example, in FIG. 27.

[0438] With reference to FIGS. 28-31, the rod 21 is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies 1. The closure structure 18 is then inserted into and advanced between the arms 62 of each of the receivers 10. The closure structure 18 is rotated, using a tool engaged with the inner drive 166 until a selected pressure is reached at which point the rod 21 engages the flat top surface 145 of the compression insert 14, biasing the insert spherical surface 142 against the shank spherical surface 34. As shown in FIG. 29, when the shank 4 is articulated at an angle with respect to the receiver 10 both smooth and stepped 40 surface portions of the spherical surface 34 are in frictional engagement with the spherical surface 142 of the compression insert. When the shank 4 is axially aligned with the receiver 10 as shown in FIGS. 30 and 31, the surface 142 primarily engages the stepped surface portion 40 of the shank upper portion 8.

[0439] As the closure structure 18 rotates and moves downwardly into the respective receiver 10, the point 169 and rim 170 engage and penetrate the rod surface 22, the closure structure 18 pressing downwardly against and biasing the rod 21 (in a direction illustrated by the arrow M in FIG. 29) into engagement with the insert 14 that urges the shank upper portion 8 toward the retainer 12 and into locking engagement therewith, the retainer 12 frictionally abutting the surface 99 and expanding outwardly against the cylindrical surface 96. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank 6 with respect to the receiver 10.

[0440] If removal of the rod 21 from any of the bone screw assemblies 1 is necessary, or if it is desired to release the rod 21 at a particular location, disassembly is accomplished by using the driving tool (not shown) that mates with the internal drive 166 on the closure structure 18 to rotate and remove such closure structure from the cooperating receiver 10. Disassembly is then accomplished in reverse order to the procedure described previously herein for assembly.

[0441] With reference to FIGS. 32-59 the reference number 1001 generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly 1001 includes a shank 1004, that further includes a body 1006 integral with an upwardly extending upper portion or capture structure 1008; a receiver 1010; a retainer structure 1012 and a compression or pressure insert 1014. The receiver 1010, retainer 1012 and compression insert 1014 are initially assembled

and may be further assembled with the shank **1004** either prior or subsequent to implantation of the shank body **1006** into a vertebra **1013**. FIGS. **57-59** further show a closure structure **1018** of the invention for capturing a longitudinal connecting member, for example, a rod **1021** which in turn engages the compression insert **1014** that presses against the shank upper portion **1008** into fixed frictional contact with the retainer **1012**, so as to capture, and fix the longitudinal connecting member **1021** within the receiver **1010** and thus fix the member **1021** relative to the vertebra **1013**. The illustrated rod **1021** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **1022**. It is foreseen that in other embodiments, the rod **1021** may be elastic, deformable and/or of a different cross-sectional geometry as previously described herein with respect to the rod **21** of the assembly **1**. The receiver **1010** and the shank **1004** cooperate in such a manner that the receiver **1010** and the shank **1004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **1010** with the shank **1004** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0442] The shank **1004**, best illustrated in FIGS. **32-34** is substantially similar to the shank **4** previously described herein with respect to the assembly **1**. Thus, the shank **1004** includes the shank body **1006**, upper portion or head **1008**, a shank thread **1024**, a neck **1026**, a tip **1028**, a top of thread **1032**, an upper portion spherical surface **1034** a top surface **1038**, an internal drive **1046** with a base surface **1045** and an cannulation bore **1050** the same or substantially similar to the respective body **6**, upper portion or head **8**, shank thread **24**, neck **26**, tip **28**, top of thread **32**, spherical surface **34**, top surface **38**, internal drive **46** with base surface **45** and cannulation bore **50** previously described herein with respect to the shank **4** of the assembly **1**. Unlike the shank **4**, the shank **1004** does not include ridges **40** on the spherical surface **1034**. Rather ridges or gripping surfaces are located on the insert **1014** as will be described in greater detail below. To provide a biologically active interface with the bone, the threaded shank body **1006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**.

[0443] With particular reference to FIGS. **32** and **40-44**, the receiver **1010** has a generally squared-off U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **1010** has an axis of rotation B that is shown in FIG. **32** as being aligned with and the same as an axis of rotation A of the shank **1004**, such orientation being desirable, but not required during assembly of the receiver **1010** with the shank **1004**. After the receiver **1010** is pivotally attached to the shank **1004**, either before or after the shank **1004** is implanted in a vertebra **1013**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIGS. **58** and **59**.

[0444] The receiver **1010** includes a substantially cylindrical base **1060** defining an inner cavity **1061**, the base **1060** being integral with a pair of opposed upstanding arms **1062** forming a cradle and defining a channel **1064** between the arms **1062** with an upper opening, generally **1066**, and a squared-off U-shaped lower seat **1068**, the channel **1064** having a width for operably snugly receiving the rod **1021** between the arms **1062**, the channel **1064** communicating with the base cavity **1061**. The squared-off geometry of the channel **1064** and lower seat **1068** allow for use with a variety of longitudinal connecting members, including, but not limited to those with circular, oblong, oval, square and rectangular cross-sections. As compared to a U-shaped channel that includes a lower seat having a surface with a radius the same or slightly larger than a cooperating cylindrical rod or other connecting member, the squared-off seat **1068** of the present invention provides improved stress management, moving stress risers outwardly toward the two arms **1062** rather than being focused primarily at a center base line of the radiused lower seat. Furthermore, outer front and rear opposed substantially planar base surfaces **1069** that partially define the squared-off lower seat **1068** advantageously reduce the run on the rod (i.e., provide a more narrow

receiver that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and provide a planar surface for flush or close contact with other connecting member components in certain embodiments, such as for bumpers or spacers that surround a hard or deformable rod or provide support for cord-type connecting members.

[0445] Each of the arms **1062** has an interior surface, generally **1070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **1072** located adjacent top surfaces **1073** of each of the arms **1062**. In the illustrated embodiment, the guide and advancement structure **1072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **1018**. However, it is foreseen that the guide and advancement structure **1072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structure for operably guiding under rotation and advancing the closure structure **1018** downward between the arms **1062**, as well as eventual torquing when the closure structure **1018** abuts against the rod **1021** or other longitudinal connecting member.

[0446] An opposed pair of tool receiving and engaging apertures **1074** are formed on outer surfaces **1076** of the arms **1062**. Furthermore, two pair of tool receiving and engaging apertures **1077** are formed in front and rear surfaces **1078** of the arms **1062**. Some or all of the apertures **1074** and **1077** may be used for holding the receiver **1010** during assembly with the insert **1014**, the retainer **1012** and the shank **1004**, during the implantation of the shank body **1006** into a vertebra when the shank is pre-assembled with the receiver **1010**, and during assembly of the bone anchor assembly **1001** with the rod **1021** and the closure structure **1018**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **1062**.

[0447] Returning to the interior surface **1070** of the receiver arms **1062**, located below the guide and advancement structure **1072** is a discontinuous cylindrical surface **1082** partially defining a run-out feature for the guide and advancement structure **1072**. The cylindrical surface **1082** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **1072**. Moving downwardly, in a direction toward the base **1060**, adjacent the cylindrical surface **1082** is a run-out seat or surface **1084** that extends inwardly toward the axis B and runs perpendicular to the axis B. Adjacent to and located below the surface **1084** is another cylindrical surface **1086** having a diameter smaller than the diameter of the surface **1082**. A discontinuous annular surface **1088** that provides an upper abutment surface or stop for capturing the compression insert **1014** in the receiver **1010** is located below and adjacent to the cylindrical surface **1086**. The abutment surface **1088** is disposed substantially perpendicular to the axis B. As shown in FIG. 51 and discussed in greater detail below, the assembly **1001** is typically provided to a user with the insert **1014** being held within the receiver by a pair of spring tabs, generally **1090**, that resiliently hold the insert **1014** and keep the insert stationary with respect to the receiver **1010** and slightly spaced from the upper stop **1088** until the insert **1014** is pressed down by the user into a friction fit working position wherein the insert **1014** is in frictional contact with the shank upper portion **1008**, the shank still movable with respect to the insert **1014**, but not in a loose or floppy manner. Each spring tab **1090** generally extends from a location spaced from the surface **1088** and along one of the arms **1062** downwardly to the base **1060**; each spring tab **1090** being integral with the base **1060**. The opposed spring tabs **1090** include various surfaces for contacting the insert **1014** at different stages of assembly and will be discussed in greater detail in the paragraphs below. A continuous annular surface **1092** is located below and adjacent to the spring tabs **1090**. The surface **1092** is disposed in the base **1060**, partially forming the base cavity **1061** and forms a stop for the resilient retainer **1012**, prohibiting the retainer **1012** (when in an uncompressed configuration) from moving upwardly into a space or cavity **1091** defined by the spring tab **1090** inner surfaces that hold the compression insert **1014**. Another cylindrical surface is located below and adjacent to the

surface **1092**. The cylindrical surface **1094** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **1012**. The surfaces **1092** and **1094** define a circumferential recess, groove or chamber **1095** that is sized and shaped to receive the retainer **1012** as it expands around the shank upper portion **1008** as the shank **1008** moves upwardly toward the channel **1064** during assembly, as well as form a restriction to prevent the expanded retainer **1012** from moving upwardly with the shank portion **1008**, the surface **1092** preventing the retainer **1012** from passing from the groove **1095** into the cavity **1091** whether the retainer **1012** is in a partially or fully expanded position, or in a neutral or original or operative position (see, e.g., FIG. 54). A cylindrical surface **1096** located below the cylindrical surface **1094** is sized and shaped to closely receive the retainer **1012** when the retainer is in a neutral or slightly expanded position as shown in FIG. 57, for example. Thus, the cylindrical surface **1096** has a diameter smaller than the diameter of the cylindrical surface **1094** that defines the expansion chamber **1095**. The surface **1096** is joined or connected to the surface **1094** by one or more beveled, curved or conical surfaces **1097**. The surfaces **1097** allow for sliding gradual movement and/or contraction of the retainer **1012** into the space defined by the surface **1096** and ultimate seating and slight expansion of the retainer **1012** on a lower annular surface **1099** located below and adjacent to the cylindrical surface **1096**. Located below and adjacent to the annular seating surface **1099** is another cylindrical surface **1100** that communicates with a beveled or flared bottom opening surface **1102**, the surface **1102** communicating with an exterior base surface **1104** of the base **1060**, defining a lower opening, generally **1106**, into the base cavity **1061** of the receiver **1010**. The illustrated surface **1100** has a diameter that is substantially the same as an inner diameter of the spring tabs **1090**, when in a neutral, un-sprung position as will be described in greater detail below, allowing for slidable uploading of the compression insert **1014** while requiring compression or squeezing of the retainer **1012** during uploading of the retainer **1012** through the lower opening **1106** (see FIGS. 50 and 51, for example).

[0448] Returning to the spring tabs **1090**, each spring tab includes a top surface **1110** and a first radiused inner surface **1111** perpendicular to a lower lip or abutment surface **1112**. The abutment surface **1112** extends from the surface **1111** to another radiused surface **1114** having a radius larger than a radius of the surface **1111**. The surface **1114** is integral with a cylindrical surface **1115** that extends into the base **1060** defining an upper portion of the base cavity **1061**, the surface **1115** terminating at the annular abutment surface **1092**. Each spring tab **1090** is further defined by diverging side surfaces **1117** and an outer surface **1118**. The surfaces **1117** diverge at the inner surfaces **1111** and **1114** and converge toward the outer surface **1118**, the illustrated pairs of surfaces **1117** being at an acute angle with respect to one another. The top surface **1110** is spaced from the annular abutment surface **1088** and is substantially parallel thereto when the spring tabs **1090** are in a neutral, non-sprung state. Also, as will be described in greater detail below, when the tabs **1090** are in a neutral, non-sprung state, the surfaces **1111** form a discontinuous cylindrical surface having a diameter smaller than an outer diameter of the insert **1014**, while the inner surfaces **1114** form a discontinuous cylindrical surface having a diameter slightly larger than a largest outer diameter of the insert **1014**, the insert **1014** being snugly held thereby under the lip surface **1112** when in a fully assembled, friction fit position within the receiver **1010**. When the tabs **1090** are in an outwardly directed sprung state as shown on FIGS. 51 and 52, for example, the surfaces **1111** frictionally engage the insert **1014**, prohibiting both upward and downward movement of the insert **1014** within the receiver **1010**, advantageously keeping the insert **1014** clear of other tools and components prior to assembly with other components and during the insertion of the retainer **1012** into the receiver **1010** and the bone screw shank upper portion **1008** into the retainer **1012** within the receiver **1010**. As best shown in FIG. 54, the somewhat trapezoidal spring tabs **1090** are created by a machining process in which at least two cuts, at an acute angle to one another, are made in each receiver arm **1062**. The angular cuts advantageously create spring tabs **1090** having greater surface contact area with the insert **1014** than would occur with spring tabs having parallel side

surfaces formed by parallel cuts. A further advantage of angular cuts over parallel cuts is that angular cuts advantageously provide access to and removal of material from the inner receiver arms **1062** that then allow for the arms to receive the insert **1014** during the assembly step of springing the tabs **1090** outwardly and pushing the insert **1014** upwardly into frictional engagement with the surfaces **1111** that was mentioned above and will be described in greater detail below.

[0449] With particular reference to FIGS. **32**, **35-39** and **51-52**, the retainer **1012** that operates to capture the shank upper portion **1008** and the compression insert **1014** within the receiver **1010** has a central axis C that is operationally the same as the axis B associated with the receiver **1010** when the shank upper portion **1008** and the retainer **1012** are installed within the receiver **1010**. The retainer **1012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **1012** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The retainer **1012** has a central channel or hollow through bore, generally **1121**, that passes entirely through the structure **1012** from a top surface **1122** to a bottom surface **1124** thereof. Surfaces that define the channel or bore **1121** include a discontinuous inner cylindrical surface **1125** adjacent the top surface **1122** and a discontinuous frusto-conical or beveled surface **1127** adjacent the surface **1125**, both surfaces coaxial with the axis C when the retainer **1012** is in a neutral non-compressed, non-expanded orientation. The retainer **1012** further includes an outer cylindrical surface **1130** located adjacent the top surface **1122** and an outer beveled or frusto-conical surface **1132** adjacent the bottom surface **1124**. The surface **130** is oriented parallel to the axis C. A pair of spaced notches **1133** are formed in the cylindrical surface **1130**. The notches **1130** receive a holding and manipulation tool (not shown) used for contraction and insertion of the retainer **1012** into the receiver. In some embodiments further notches may be made to evenly distribute stress across the entire retainer during contraction and expansion thereof. In other embodiments of the invention, such notches may be on the inside of the ring. In some embodiments, the ring can have no notches. The resilient retainer **1012** further includes first and second end surfaces, **1134** and **1135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **1134** and **1135** are disposed substantially perpendicular to the top surface **1122** and the bottom surface **1124**. A width X between the surfaces **1134** and **1135** is determined by a desired amount of compressibility of the open retainer **1012** when loaded into the receiver **1010**. The space X shown in FIG. **35** provides adequate space between the surfaces **1134** and **1135** for the retainer **1012** to be pinched, with the surfaces **1134** and **1135** compressed toward one another (as shown in FIG. **51**) to a closely spaced or even touching configuration, if necessary, to an extent that the compressed retainer **1012** is up or bottom loadable through the receiver opening **1106** as shown in FIGS. **51** and **52**. After passing through the opening **1106** and along a portion of the lower inner surface, the retainer **1012** expands or springs back to an original uncompressed, rounded or collar-like configuration of FIGS. **35-39**, see, e.g., FIG. **52**. The embodiment shown in FIGS. **35-39** illustrates the surfaces **1134** and **1135** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer **1012** into the receiver **1010**.

[0450] With reference to FIGS. **32** and **45-51**, the compression insert **1014** is illustrated that is sized and shaped to be received by and up-loaded into the receiver **1010** at the lower opening **1106**. The compression insert **1014** has an operational central axis that is the same as the central axis B of the receiver **1010**. The compression insert **1014** has a central channel or through bore defined by an inner cylindrical surface **1141**, an inner partially spherical surface **1142** and a shank gripping surface portion, generally **1143**, extending between the surface **1141** and the surface **1142**. The gripping surface portion **1143** preferably includes two or more graduated cylindrical surfaces disposed substantially parallel to the axis B and adjacent perpendicular step surfaces that are disposed generally perpendicular to the axis B. It is foreseen that the stepped surface portion **1143** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping

surface portion **1143** and also the surface **1142** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **1008**.

[0451] The compression insert **1014** through bore is sized and shaped to receive the driving tool **1029** therethrough that engages the shank drive feature **1046** when the shank body **1006** is driven into bone with the receiver **1010** attached. The surfaces **1142** and **1143** are sized and shaped to initially slidingly receive and ultimately frictionally engage the substantially spherical or domed surface **1034** of the shank upper portion **1008**, in particular the stepped or ridged surface **1143** that will initially frictionally but slidingly and pivotally mate with the spherical surface **1034** to create a ball-and-socket type joint, but ultimately dig into and thus be securely fixed with the domed surface **1034**.

[0452] The compression insert **1014** also includes a first outer and upper cylindrical surface **1144** adjacent to a top surface **1145**. The top surface **1145** engages the rod **1021** or other longitudinal connecting member during operation of the assembly **1001** and locates the rod above the lower seat **1068** of the receiver. The insert **1014** also includes an outer lower cylindrical surface **1148** adjacent to a bottom surface **1149**. The cylindrical surfaces **1144** and **1148** have the same or substantially the same outer diameter, sized to be received by the receiver surface **1100** when loaded into the receiver **1010** and also be snugly received by spring tab **1090** surfaces **1114** when the spring tabs are in a neutral or relaxed state. Located between the surfaces **1144** and **1148** is a frusto-conical surface **1152** that extends from the surface **1144** inwardly toward the axis B and terminates at an annular ledge **1154**. The ledge **1154** extends from the frusto-conical surface **1152** to the surface **1148** and is substantially perpendicular to the surface **1148**. As will be described in greater detail below, during early stages of assembly, the insert **1014** outer surfaces **1144** and **1152** are resiliently gripped by the spring tab surfaces **1111** with the spring tab lower lip **1112** engaging the ledge **1154** to hold the insert **1014** in a desired stationary position with respect to the receiver **1010**. When the insert **1014** is lowered into a second or friction fit position in frictional engagement with the bone screw shank, the lower lip **1112** extends over the insert top surface **1145**.

[0453] It is foreseen that in some embodiments of the invention the compression insert **1014** may further include upstanding arms that cradle the rod **1021** or other connecting member. Such arms may be located spaced from the closure top **1018** in some embodiments and may be sized and shaped to contact the closure top **1018** in other embodiments in order to provide locking of the polyaxial mechanism of the assembly with capture but without fixing of the rod **1021** or other longitudinal connecting member with respect to the closure top **1018**.

[0454] The compression or pressure insert **1014** ultimately seats on the shank upper portion **1008** and is disposed substantially within the spring tab cylindrical wall **1114**. In operation, the insert **1014** extends at least partially in the channel **1064** of the receiver **1010** such that the top surface **1145** substantially contacts and engages the outer surface **1022** of the rod **1021** when such rod is placed in the receiver **1010** and the closure structure or top **1018** is tightened thereon.

[0455] With reference to FIGS. 57-59, the illustrated elongate rod or longitudinal connecting member **1021** is the same or substantially similar to the rod **21** previously described herein and thus can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **1022** of uniform diameter. The rod **1021** may be made from a variety of metals, metal alloys and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials. Longitudinal connecting members for use with the assembly **1001** may take a variety of shapes as previously described with respect to the assembly **1**, including outer sleeve and inner cord connecting member assemblies as shown and described, for example, in U.S. patent application Ser. No. 12/802,849 filed Jun. 15, 2010 that is incorporated by reference herein.

[0456] With reference to FIGS. 57-59, the closure structure or closure top **1018** shown with the

assembly **1001** is the same or substantially similar in form and function to the closure top **18** previously described herein with respect to the assembly **1** and can be any of a variety of different types of closure structures for use in conjunction with the present invention with suitable mating structure on the upstanding arms **1062**. The illustrated closure structure **1018** is substantially cylindrical and includes an outer helically wound guide and advancement structure **1162** in the form of a flange form that operably joins with the guide and advancement structure **1072** disposed on the arms **1062** of the receiver **1010**. The illustrated closure structure **1018** also includes a top surface **1164** with an internal drive **1166** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **1166** is used for both rotatable engagement and, if needed, disengagement of the closure **1018** from the receiver arms **1062**. It is also foreseen that the closure structure **1018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **1168** of the closure is planar and further includes a point **1169** and a rim **1170** for engagement and penetration into the surface **1022** of the rod **1021** in certain embodiments of the invention. The closure top **1018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **18** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **1062**.

[0457] Preferably, the receiver **1010**, the retainer **1012** and the compression insert **1014** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **1012** as well as spreading of the spring tabs **1090**. As described herein with respect to the assembly **1**, similarly, the shank **1004** may be assembled with the receiver **1010**, retainer **1012** and compression insert **1014** at the factory or it may be desirable to “pop” the shank **1004** into the receiver assembly at a later time, either before or after implantation of the shank **1004** in the vertebra **1013**.

[0458] Pre-assembly of the receiver **1010**, retainer **1012** and compression insert **1014** is shown in FIGS. 50-52. First, the compression insert **1014** is uploaded into the receiver **1010** through the lower opening **1106** with the insert top surface **1145** facing the receiver bottom surface **1104**. The insert **1014** is slid upwardly toward the channel seat **1068** until the insert is within the cylindrical walls **1114** of the spring tabs **1090**. If pressed further upwardly without expanding the spring tabs **1090**, the insert top surface **1145** would simply abut against the surface **1112**. A tool or tools (not shown) are used to pull or otherwise spread the spring tabs **1090** away from one another and allow the insert **1014** to be placed therebetween at the outer surfaces **1144** and/or **1152**. Then the resilient tabs **1090** are released and the surfaces **1111** of the tabs **1090** engage the surface **1144** or **1152**, preferably engaging the surface **1152** adjacent to the ledge **1154**. The surface **1112** advantageously abuts against the ledge **1154**, stopping the insert **1014** from any further upward movement towards the surface **1088** and providing adequate clearance for the later step of pushing the bone screw shank upper portion **1008** through the spring ring retainer **1112**. Although the surface **1088** would prohibit the insert **1014** from moving out the upper opening **1066**, engagement with the resilient spring tabs **1090** also prohibits downward movement of the insert **1014** and keeps the insert **1014** away from the lower opening **1106** during assembly with the retainer **1012** and subsequent assembly with the shank **1004**. Then, the resilient open retainer **1012** is prepared for insertion into the receiver **1010** by squeezing or pressing the retainer end surfaces **1134** and **1135** toward one another as shown in FIG. 51. The compressed retainer **1012** is inserted into the lower opening **1106** with the top surface **1122** facing the receiver bottom surface **1104**. The retainer **1012** is typically moved upwardly into the receiver **1010** and past the cylindrical surface **1096** and allowed to

expand to a neutral uncompressed state within the cylindrical surface **1096** as shown in FIG. 52. Also as shown in FIG. 52, at this time, both the compression insert **1014** and the retainer **1012** are captured within the receiver **1010**. The receiver **1010**, compression insert **1014** and the retainer **1012** combination is now pre-assembled and ready for assembly with the shank **1004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **1004**.

[0459] As illustrated in FIG. 53, the bone screw shank **1004** (or an entire assembly **1001** made up of the assembled shank **1004**, receiver **1010**, retainer **1012** and compression insert **1014**) is screwed into a bone, such as the vertebra **1013**, by rotation of the shank **1004** using a suitable driving tool **1029** that operably drives and rotates the shank body **1006** by engagement thereof at the internal drive **1046**. It is foreseen that the shank and other bone screw assembly parts, the rod **2021** (also having a central lumen in some embodiments) and the closure top **2018** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires.

[0460] Again with respect to FIGS. 53 and 54, when the shank **1004** is driven into the vertebra **1013** without the remainder of the assembly **1001**, the shank **1004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer. With reference to FIGS. 54-57, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **1008** until the shank upper portion is received within the opening **1106**. As the shank is moved into the interior of the receiver base, the shank upper portion **1008** presses the retainer **1012** upwardly into the chamber **1095** (if the retainer is not already located within such chamber). As the portion **1008** continues to move upwardly toward the channel **1064**, the retainer top surface **1122** abuts against the annular surface **1092** stopping upward movement of the retainer **1012** and forcing outward movement of the retainer **1012** towards the cylindrical surface **1094** defining the expansion chamber or groove **1095** as the spherical surface **1034** continues in an upward direction. The retainer **1012** begins to contract about the spherical surface **1034** as the center of the sphere passes beyond the center of the retainer expansion chamber **1095** (see FIG. 55). The retainer **1012** can then move down into a final operative location within the seating chamber or groove, shown in FIGS. 56 and 57 by either gravity and/or an upward pull on the receiver **1010** or, in some cases, by driving the shank **1004** further into the vertebra **1013**. Also, in some embodiments, when the receiver **1010** is pre-assembled with the shank **1004**, the entire assembly **1001** may be implanted at this time by inserting the driving tool **1020** into the receiver and the shank drive **1046** and rotating and driving the shank **1004** into a desired location of the vertebra **1013**.

[0461] With reference to FIG. 56, at this time, the compression insert **1014** is pressed downwardly with a tool (not shown), or with the rod, toward the shank upper portion **1008** and out of engagement with the spring tab surfaces **1111**. Once the insert surface **1144** clears the tab surfaces **1111**, the insert snaps into place and the spring tabs **1090** return to an original, relaxed orientation with the surfaces **1112** located over the insert top surface **1145** and frictionally engaging such surface. The spring tabs **1090** are sized such that when the surfaces **1112** frictionally engage the top surface **1145** of the insert, the insert **1014** surfaces **1142** and **1143** in turn press against the shank upper portion **1008** at the spherical surface **1034**. The friction fit between the compression insert **1014** and the shank upper portion **1008** is not totally locked or fixed, but at the same time not loose or floppy either, advantageously allowing the user to articulate the shank **1004** with respect to the receiver **1010**, but with some resistance, so that when the shank **1004** is placed in a desired orientation with respect to the receiver **1010**, the assembly **1001** remains substantially frictionally set in such desired orientation unless purposefully manipulated into another position. For example, at this time, the receiver **1010** may be articulated to a desired position with respect to the shank **1004**, for example, as shown in FIG. 58 or FIG. 59, but prior to locking of such position that is shown in those drawings.

[0462] With reference to FIGS. 57-59, the rod **1021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1001**. The closure

structure **1018** is then inserted into and advanced between the arms **1062** of each of the receivers **1010**. The closure structure **1018** is rotated, using a tool engaged with the inner drive **1166** until a selected pressure is reached at which point the rod **1021** engages the flat top surface **1145** of the compression insert **1014**, further pressing the insert spherical surface **1142** and stepped surfaces **1143** against the shank spherical surface **1034**, the edges of the stepped surfaces penetrating into the spherical surface **1034**.

[0463] As the closure structure **1018** rotates and moves downwardly into the respective receiver **1010**, the point **1169** and rim **1170** engage and penetrate the rod surface **1022**, the closure structure **1018** pressing downwardly against and biasing the rod **1021** into engagement with the insert **1014** that urges the shank upper portion **1008** toward the retainer **1012** and into locking engagement therewith, the retainer **1012** frictionally abutting the surface **1099** and expanding outwardly against the cylindrical surface **1096**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **1006** with respect to the receiver **1010**.

[0464] If removal of the rod **1021** from any of the bone screw assemblies **1001** is necessary, or if it is desired to release the rod **1021** at a particular location, disassembly is accomplished by using the driving tool (not shown) that mates with the internal drive **1166** on the closure structure **1018** to rotate and remove such closure structure from the cooperating receiver **1010**. Disassembly is then accomplished in reverse order to the procedure described previously herein for assembly.

[0465] With reference to FIGS. **60-83**, the reference numeral **1201** generally represents another embodiment of a polyaxial bone screw according to the invention. The assembly **1201** includes a shank **1204**, that further includes a body **1206** integral with an upwardly extending upper portion or capture structure **1208**; a receiver **1210**; a retainer structure **1212** and a compression or pressure insert **1214**. The receiver **1210**, retainer **1212** and compression insert **1214** are initially assembled and may be further assembled with the shank **1204** either prior or subsequent to implantation of the shank body **1206** into a vertebra, as will be described in greater detail below. FIG. **60** further shows a closure structure **1218** of the invention for capturing a longitudinal connecting member, for example, a rod **1221** which in turn engages the compression insert **1214** that presses against the shank upper portion **1208** into fixed frictional contact with the retainer **1212**, so as to capture, and fix the longitudinal connecting member **1221** within the receiver **1210** and thus fix the member **1221** relative to the vertebra. The illustrated rod **1221** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **1222**. It is foreseen that in other embodiments, the rod **1221** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **1210** and the shank **1204** cooperate in such a manner that the receiver **1210** and the shank **1204** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **1210** with the shank **1204** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0466] The shank **1204** is the same or substantially similar in form and function to the shank **1004** previously described herein and includes the body **1206**, upper portion or head **1208** having a spherical surface **1234** and an internal drive feature **1246** the same or similar to the respective body **1006**, upper portion **1008**, spherical surface **1034** and drive feature **1046** previously described herein with respect to the shank **1004** of the assembly **1001**.

[0467] The receiver **1210** is also substantially similar in form and function to the receiver **1010** previously described herein. However, there are differences between the two receivers as the receiver **1210** cooperates with the insert **1214** that varies in many respects from the insert **1014** previously described herein. Therefore, the receiver **1210** and the insert **1214** will be described in greater detail below.

[0468] With particular reference to FIGS. **60-63** and **72**, the receiver **1210** has a generally squared-off U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **1210** has an axis of rotation that is shown in FIG. **60** as being aligned with

and the same as an axis of rotation of the shank **1204**, such orientation being desirable, but not required during assembly of the receiver **1210** with the shank **1204**. The receiver **1210** includes a substantially cylindrical base **1260** defining an inner cavity **1261**, the base **1260** being integral with a pair of opposed upstanding arms **1262** forming a cradle and defining a channel **1264** between the arms **1262** with an upper opening, generally **1266**, and a squared-off U-shaped lower seat **1268**, the channel **1264** having a width for operably snugly receiving the rod **1221** between the arms **1262**, the channel **1264** communicating with the base inner cavity **1261**. The squared-off geometry of the channel **1264** and lower seat **1268** allow for use with a variety of longitudinal connecting members, including, but not limited to those with circular, square and rectangular cross-sections. As compared to a U-shaped channel that includes a lower seat having a surface with a radius the same or slightly larger than a cooperating cylindrical rod or other connecting member, the squared-off seat **1268** of the present invention provides improved stress management, moving stress risers outwardly toward the two arms **1262** rather than being focused primarily at a center base line of the radiused lower seat. Furthermore, outer front and rear opposed substantially planar base surfaces **1269** that partially define the squared-off lower seat **1268** advantageously reduce the run on the rod (i.e., provide a more narrow receiver that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and provide a planar surface for flush or close contact with other connecting member components in certain embodiments, such as for bumpers or spacers that surround a hard or deformable rod or provide support for cord-type connecting members. The planar surfaces can also better cooperate with compression and distraction tools.

[0469] Each of the arms **1262** has an interior surface, generally **1270**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **1272** located adjacent top surfaces **1273** of each of the arms **1262**. In the illustrated embodiment, the guide and advancement structure **1272** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **1218**. However, it is foreseen that the guide and advancement structure **1272** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structure for operably guiding under rotation and advancing the closure structure **1018** downward between the arms **1262**, as well as eventual torquing when the closure structure **1218** abuts against the rod **1221** or other longitudinal connecting member.

[0470] An opposed pair of tool receiving and engaging apertures **1274** are formed on outer surfaces **1276** of the arms **1262**. Furthermore, two pair of tool receiving and engaging apertures **1277** are formed in front and rear surfaces **1278** of the arms **1262**. Some or all of the apertures **1274** and **1277** may be used for holding the receiver **1210** during assembly with the insert **1214**, the retainer **1212** and the shank **1204**, during the implantation of the shank body **1206** into a vertebra when the shank is pre-assembled with the receiver **1210**, and during assembly of the bone anchor assembly **1201** with the rod **1221** and the closure structure **1218**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **1262**.

[0471] Returning to the interior surface **1270** of the receiver arms **1262**, located below the guide and advancement structure **1272**, adjacent a bottom surface **1280** thereof, is a discontinuous cylindrical surface **1282** partially defining a run-out feature for the guide and advancement structure **1272**. The cylindrical surface **1282** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **1272**. Moving downwardly, in a direction toward the base **1260**, adjacent the cylindrical surface **1282** is a run-out seat or surface **1284** that extends inwardly toward the central axis of the receiver and is perpendicular thereto. Adjacent to and located below the surface **1284** is another discontinuous cylindrical surface **1285** having a diameter smaller than the diameter of the surface **1282**. The surface **1285** terminates at a narrow

ledge **1286** that in turn partially defines another discontinuous cylindrical surface **1287** having a diameter slightly smaller than the diameter of the surface **1285**. With particular reference to FIGS. **76** and **77** an edge or rim **1288** that defines the junction of the ledge **286** and the cylindrical surface **287** is shown cooperating with the insert **214** as will be described in greater detail below, providing an advantageous shank lock and release feature of the assembly **201**.

[0472] As discussed in greater detail below, the assembly **1201** is typically provided to a user with the insert **1214** being held within the receiver by a pair of spring tabs, generally **1290**, that resiliently hold the insert **1214** and keep the insert stationary with respect to the receiver **1210** in an upward position between the arms **1262** until the insert **1214** is pressed by the user into a friction fit working position wherein the insert **1214** is in frictional contact with the shank upper portion **1208**, the shank still movable with respect to the insert **1214**, but not in a loose or floppy manner. In a later stage of assembly, the spring tabs **1290** advantageously hold the insert **1214** in a centered position (the insert arms being held in alignment with the receiver arms) during rotation and torquing of the closure top **1218** onto the rod **1221** or other connecting member. Each spring tab **1290** generally extends from a location spaced from the surface **1287** and along one of the arms **1262** downwardly to the base **1260**; each spring tab **1290** being integral with the base **1260**. The opposed spring tabs **1290** include various surfaces for contacting the insert **1214** at different stages of assembly and will be discussed in greater detail in the paragraphs below.

[0473] Returning to FIGS. **62** and **63**, the lower cavity **1261** within the base **1260** includes an inner cylindrical surface **1291**, portions of which extend up into and partially form the spring tabs **1290**. A continuous annular surface **1292** is located below and adjacent to the cylindrical surface **1291**. The surface **1292** is disposed in the base **1260**, partially defining the base cavity **1261** and providing a stop for the resilient retainer **1212**, prohibiting the retainer **1212** (when in an uncompressed configuration) from moving upwardly into a space or cavity defined by the cylindrical surface **1291** and the spring tab **1290** inner surfaces that hold the compression insert **1214**. Another cylindrical surface **1294** is located below and adjacent to the surface **1292**. The cylindrical surface **1294** is oriented substantially parallel to the receiver central axis and is sized and shaped to receive an expanded retainer **1212**. The surfaces **1292** and **1294** define a circumferential recess, groove or chamber **1295** that is sized and shaped to receive the retainer **1212** as it expands around the shank upper portion **1208** as the shank **1208** moves upwardly toward the channel **1264** during assembly, as well as form a restriction to prevent the expanded retainer **1212** from moving upwardly with the shank portion **1208**, the surface **1292** preventing the retainer **1212** from passing from the groove **1295** into the cavity defined by the surface **1291** whether the retainer **1212** is in a partially or fully expanded position, or in a neutral or original operative position. A cylindrical surface **1296** located below the cylindrical surface **1294** is sized and shaped to closely receive the retainer **1212** when the retainer is in a neutral or operative position, for example. Thus, the cylindrical surface **1296** has a diameter smaller than the diameter of the cylindrical surface **1294** that defines the expansion groove **1295**. The surface **1296** is joined or connected to the surface **1294** by one or more beveled, curved or conical surfaces **1297**. The surfaces **1297** allow for sliding gradual movement and/or contraction of the retainer **1212** into the space defined by the surface **1296** and ultimate seating of the retainer **1212** on a lower annular surface **1299** located below and adjacent to the cylindrical surface **1296**. Located below and adjacent to the annular seating surface **1299** is another cylindrical surface **1300** that communicates with a beveled or flared bottom opening surface **1302**, the surface **1302** communicating with an exterior base surface **1304** of the base **1260**, defining a lower opening, generally **1306**, of the receiver **1210**. The illustrated surface **1300** has a diameter that is substantially the same as an inner diameter of the surface **1291** that extends up into the spring tabs **1290**, when in a neutral, un-sprung position as will be described in greater detail below, allowing for slidable uploading of the compression insert **1214** (with minor squeezing of the insert arms toward one another) while requiring substantial compression or squeezing of the retainer **1212** during uploading of the retainer

1212 through the lower opening **1306**.

[0474] Returning to the spring tabs **1290**, each spring tab includes a top surface **1310** and a first radiused inner surface **1311** perpendicular to a lower lip or abutment surface **1312**. The abutment surface **1312** extends from the surface **1311** to the inner cylindrical surface **1291** that has a radius larger than a radius of the surface **1311**. Each spring tab **1290** is further defined by a pair of opposed parallel side surfaces **1314**, a pair of angled or diverging side surfaces **1315** and an outer surface **1317**. The parallel surfaces **1314** are located on either side of the inner surface **1311** and the top surface **1310**. The diverging side surfaces **1315** each run from the outer surface **1317** outwardly toward an adjacent surface **1314**, the illustrated pairs of surfaces **1317** being at an acute angle with respect to one another. The top surface **1310** is spaced from the cylindrical surface **1287**. When the tabs **1290** are in a neutral, non-sprung state, the surfaces **1311** define a diameter smaller than an outer diameter of the insert **1214**, while the inner surface **1291** forms a discontinuous cylindrical surface having a diameter slightly larger than a lower outer diameter of the insert **1214**, the insert **1214** being snugly held thereby and centered by the spring tabs **1290** that are positioned within a groove of the insert **1214** as will be discussed in greater detail below. When the tabs **1290** are in an outwardly sprung state as shown on FIG. 72, for example, the surfaces **1311** frictionally engage the insert **1214**, prohibiting both upward and downward movement of the insert **1214** within the receiver **1210**, advantageously keeping the insert **1214** clear of other tools and components prior to assembly with other components and during the insertion of the retainer **1212** into the receiver **1210** and the bone screw shank upper portion **1208** into the retainer **1212** within the receiver **1210**. When the spring tabs **1290** are later placed back into a neutral un-sprung state, the lip surface **1312** of the tabs **1290** press downwardly on the insert **1214**, holding the insert **1214** in a friction fit orientation with respect to the shank upper portion **1208**.

[0475] As best shown in FIG. 78, the somewhat trapezoidal spring tabs **1290** are created by a machining process in which at least two cuts, at an acute angle to one another, are made into each receiver arm **1262**. In order for the spring tabs **1290** to fit within grooves of the compression insert **1214**, two parallel cuts are also made to form the opposed side surfaces **1314** of the spring tab **1290**. Similar to that described with respect to the receiver **1010**, an advantage of making angular cuts into the receiver **1210** to create the spring tabs **1290** is that angular cuts advantageously provide access to and removal of material from the inner receiver arms **1262** that then allow for the arms **1262** to receive the insert **1214** during the assembly step of springing the tabs **1290** outwardly and pushing the insert **1214** upwardly into frictional engagement with the surfaces **1311**. This material clearing step is of special interest when the insert **1214** rather than the insert **1014** is being used according to the invention as the insert **1214** includes a pair of opposed arms that are taller than, and thus take up greater space within the receiver than the substantially cylindrical insert **1014**.

[0476] With reference to FIGS. 60, 72-75 and 79, the retainer **1212** is substantially similar in form and function to the retainer **1012** previously described herein. Therefore, the retainer **1212** includes a top surface **1322**, a bottom surface **1324**, an inner cylindrical surface **1325**, an inner frusto-conical surface **1327**, an outer cylindrical surface **1330** and opposed ends **1334** and **1335** that are the same or substantially similar to the respective top surface **1122**, bottom surface **1124**, inner cylindrical surface **1125**, inner frusto-conical surface **1127**, outer cylindrical surface **1130** and opposed ends **1134** and **1135** of the retainer **1012** previously described herein with respect to the assembly **1001**.

[0477] With reference to FIGS. 60 and 64-72, the compression insert **1214** is illustrated that is sized and shaped to be received by and up-loaded into the receiver at the lower opening **1306**. The compression insert **1214** has an operational central axis that is the same as the central axis of the receiver **1210**. The compression insert **1214** has a central channel or through bore defined by an inner cylindrical surface **1341**, an inner partially spherical surface **1342** and a shank gripping surface portion, generally **1343**, extending between the surface **1341** and the surface **1342**. The

gripping surface portion **1343** preferably includes two or more graduated cylindrical surfaces disposed substantially parallel to the insert central axis and adjacent perpendicular step surfaces that are disposed generally perpendicular to the insert central axis. It is foreseen that the stepped surface portion **1343** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping surface portion **1343** and also the surface **1342** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **1208**.

[0478] The compression insert **1214** through bore is sized and shaped to receive a driving tool (such as the driving tool **1029** shown with the assembly **1001**) therethrough that engages the shank drive feature **1246** when the shank body **1206** is driven into bone with the receiver **1210** attached. The surfaces **1342** and **1343** are sized and shaped to initially frictionally but slidably receive and ultimately frictionally engage and fix onto the substantially spherical or domed surface **1234** of the shank upper portion **1208**, in particular the stepped or ridged surface **1343** that will initially frictionally but slidably and pivotally mate with the spherical surface **1234** to create a ball-and-socket type joint, but ultimately dig or penetrate into and thus be securely fixed with the domed surface **1234**.

[0479] The compression insert **1214** also includes an outer and upper cylindrical surface **1344** that further extends upwardly on either side of the insert to form a pair of opposed arms **1345**. Each arm **1345** further includes a top surface **1346** and an outer frusto-conical surface portion **1348** terminating at a lower surface or lip **1350**. The frusto-conical surface portion **1348** flares outwardly and upwardly, having a largest radius thereof at the juncture of the surface **1348** with the top surface **1346**. As will be described in greater detail below, frictional engagement between the surface portion **1348** and the cylindrical surface **1287** associated with the driving and downward movement of the closure top **1218** on the rod **1221** results in a locking of the polyaxial screw mechanism of the assembly **1201** that remains locked even if the closure top **1218** and the rod **1221** are subsequently loosened, allowing for all type and manner of manipulation of the bone screw and/or the rod **1221** by the surgeon while the polyaxial mechanism of the assembly **1201** remains rigidly fixed in the desired orientation previously chosen and locked down by the surgeon. However, if it is desired to loosen the polyaxial mechanism, the surgeon may do so by squeezing the arms **1345** toward one another with a tool (not shown) and moving the insert **1214** away from the shank **1204**, thereby releasing the frusto-conical surface **1348** from the receiver cylindrical surface **1287** and thus loosening the polyaxial mechanism.

[0480] Extending between the insert arms **1345** is a U-shaped, saddle like surface **1352** that forms a seat for the rod **1221** or other longitudinal connecting member. Portions of the saddle surface **1352** communicate with the bore defined by the cylindrical surface **1341**. The surface **1352** is sized and shaped to closely receive the cylindrical rod **1221** at a location spaced from the lower seat **1268** of the receiver **1210**. A bottom surface **1354** communicates with the inner spherical surface **1342**, the insert **1214** being sized and shaped such that the surface **1354** is always spaced from the retainer **1212** as shown, for example, in FIGS. 75 and 80. The insert **1214** also includes an outer lower cylindrical surface **1356** adjacent to the bottom surface **1354**. The cylindrical surfaces **1344** and **1356** have the same or substantially the same outer diameter, sized to be received by the receiver surface **1300** when loaded into the receiver **1210** and also be snugly received by spring tab **1290** and receiver base inner surfaces **1291** when the spring tabs **1290** are in a neutral or relaxed state. Located between the surfaces **1344** and **1356** is a frusto-conical surface **1357** that extends from the surface **1344** inwardly toward the insert central axis and terminates at an annular ledge **1358**. The ledge **1358** extends from the frusto-conical surface **1357** to the surface **1356** and is substantially perpendicular to the surface **1356**. As will be described in greater detail below, during early stages of assembly, the insert **1214** outer surface **1357** is resiliently gripped by the spring tab surfaces **1311** with the spring tab lower lip **1312** engaging the ledge **1358** to hold the insert **1214** in a desired stationary position with respect to the receiver **1210**. When the insert **1214** is lowered into a second

or friction fit position in frictional engagement with the bone screw shank, the lower lip **1312** extends into one of a pair of opposed grooves **1359** as described below. The grooves **1359** are formed in the arm surfaces **1344** and extend upwardly into the upper frusto-conical surface **1348** and are located centrally with respect to each arm **1345**. Each illustrated groove **1359** is sized and shaped to cooperate with the spring tabs **1290** at the surfaces **1311**. The grooves **1359** are elongate, running parallel to a central axis of the insert **1214**. Each groove **1359** has a lower seat or shelf **1360** positioned to engage the spring tab surface **1312** when the insert **1214** is in friction fit working engagement with the shank upper portion **1208** as will be described in greater detail below.

[0481] In operation, the insert **1214** extends at least partially in the channel **1264** of the receiver **1210** such that the saddle **1352** surface substantially contacts and engages the outer surface **1222** of the rod **1221** when such rod is placed in the receiver **1210** and the closure structure or top **1218** is tightened thereon. As will also be described below, the cooperation between the insert grooves **1359** and the spring tabs **1290** prohibits additional rotation of the insert **1214** with respect to the receiver **1210** during rotation and torquing of the closure top **1218** against the rod **1221** within the receiver arms **1262**. The compression or pressure insert **1214** ultimately seats on the shank upper portion **1208** and is disposed partially within the spring tab cylindrical wall **1291** and partially between the receiver arms **1262**.

[0482] With reference to FIGS. **60** and **75-80**, the illustrated closure top **1218** and illustrated elongate rod or longitudinal connecting member **1221** are the same or substantially similar to the closure top **1018** and the rod **1021** previously described herein, and or alternatives also previously described herein. Thus, with respect to the closure top **218**, components of such closure top **1218** include a guide and advancement structure **1362**, a top surface **1364**, an internal drive **1366**, a bottom surface **1368**, a point **1369** and a rim **1370** that are the same or substantially similar to the respective guide and advancement structure **1162**, top surface **1164**, internal drive **1166**, bottom surface **1168**, point **1169** and rim **1170** of the closure top **1018** previously described herein with respect to the assembly **1001**.

[0483] Preferably, the receiver **1210**, the retainer **1212** and the compression insert **1214** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **1212** and pinching the insert **1214** as well as spreading of the spring tabs **1290**. In some circumstances, the shank **1204** is also assembled with the receiver **1210**, retainer **1212** and compression insert **1214** at the factory. In other instances, it is desirable to first implant the shank **1204**, followed by addition of the pre-assembled receiver **1210**, retainer **1212** and compression insert **1214** at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **1204**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. Furthermore, the assembly **1201** allows for manipulation of the rod **1221** subsequent to a complete lock down of the insert **1214** on the bone screw shank upper portion **1208** that completely locks the polyaxial mechanism of the assembly **1201**, as will be described in greater detail below. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size with the receiver **1210**, retainer **1212** and compression insert **1214**. Allowing the surgeon to choose the appropriately sized shank **1204** advantageously reduces inventory requirements, thus reducing overall cost.

[0484] Pre-assembly of the receiver **1210**, retainer **1212** and compression insert **1214** is shown in FIGS. **68-72**. First, the compression insert **1214** is prepared for uploading into the receiver **1210** through the lower opening **3106** using a holding tool (not shown) that squeezes or presses the insert arms **1345** toward one another at the upper frusto-conical surfaces **3148** so that the surfaces **1348** are received within the cylindrical surface **1300** during insertion of the insert **1214** into the receiver opening **1306**. As illustrated in FIG. **68**, the insert **1214** (in squeezed orientation) is inserted into the opening **1306** with the arms **1345** being aligned with the receiver channel **1264**. During insertion,

the spring tabs **2190** are also pulled apart as shown, for example in FIG. **70** until the spring tab surfaces **1311** are located adjacent the frusto-conical surface **1357**. At this time, the insert **1214**, no longer in a squeezed state, is rotated as illustrated by an arrow K in FIG. **70** about the central axis thereof until the insert arms **1345** are located within the receiver run-out defined by the cylindrical surface **1282** located directly below the guide and advancement structure **1272** as shown in FIG. **71**. As shown in FIG. **71**, the guide and advancement structure **1272** prohibits further upward movement of the insert **1214**. The spring tabs **1290** are allowed to resiliently move into contact with the insert surface **1357**, preferably engaging such surface **1357** adjacent to the ledge **1358**. The surface **1312** advantageously abuts against the ledge **1358**, stopping the insert **1214** from any further upward movement towards the guide and advancement structure **1272** and providing adequate clearance for the later step of pushing the bone screw shank upper portion **1208** through the spring ring retainer **1212**. Although the guide and advancement structure **1272** would prohibit the insert **1214** from moving out the upper opening **1266**, engagement with the resilient spring tabs **1290** also prohibits rotational and downward movement of the insert **1214** and keeps the insert **1214** away from the lower opening **1306** during assembly of the retainer **1212**.

[0485] With reference to FIGS. **72-75**, the resilient open retainer **1212** is prepared for insertion into the receiver **1210** by squeezing or pressing the retainer end surfaces **1334** and **1335** toward one another. The compressed retainer **1212** is inserted into the lower opening **1306** with the top surface **1322** facing the receiver bottom surface **1304**. The retainer **1212** is typically moved upwardly into the receiver **1210** and past the cylindrical surface **1296** and allowed to expand to a neutral uncompressed state within the cylindrical surface **1296** as shown in FIG. **72**. Also as shown in FIG. **72**, at this time, both the compression insert **1214** and the retainer **1212** are captured within the receiver **1210**. The receiver **1210**, compression insert **1214** and the retainer **1212** combination is now pre-assembled and ready for assembly with the shank **1204** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **1204**.

[0486] The bone screw shank **1204** (or an entire assembly **1201** made up of the assembled shank **1204**, receiver **1210**, retainer **1212** and compression insert **1214**) is screwed into a bone, such as the vertebra **1013** as previously described herein with respect to the shank **1004** and cooperating assembly **1001** and the shank **4** and assembly **1**. The pre-assembled receiver, insert and retainer are placed above the shank upper portion **1208** until the shank upper portion is received within the opening **1306**. As the shank is moved into the interior of the receiver base, the shank upper portion **1208** presses the retainer **1212** upwardly into the chamber **1295** (if the retainer is not already located within such chamber). As the portion **1208** continues to move upwardly toward the channel **1264**, the retainer top surface **1322** abuts against the annular surface **1292** stopping upward movement of the retainer **1212** and forcing outward movement of the retainer **1212** towards the cylindrical surface **1294** defining the expansion chamber **1295** as the spherical surface **1234** continues in an upward direction. The retainer **1212** begins to contract about the spherical surface **1234** as the center of the sphere passes beyond the center of the retainer expansion groove **1295** (see FIG. **73**). The retainer is then free to be moved down into an operative position by either gravity and/or an upward pull on the receiver **1210** or, in some cases, by driving the shank **1204** further into the vertebra **1013**. Also, in some embodiments, when the receiver **1210** is pre-assembled with the shank **1204**, the entire assembly **1201** may be implanted at this time by inserting the driving tool into the receiver and the shank drive **1246** and rotating and driving the shank **1204** into a desired location of the vertebra.

[0487] With reference to FIG. **74**, at this time, the compression insert **1214** is pressed downwardly manually, with a rod or with a tool (not shown) toward the shank upper portion **1208**, the insert surfaces **1357** and **1344** being moved out of engagement with the spring tab surfaces **1311** when the surfaces **1311** enter into the insert grooves **1359**. Once the spring tabs **1290** move into the grooves **1359**, the insert **1214** snaps into place and the spring tabs **1290** return to an original, relaxed or only slightly expanded orientation with the surfaces **1312** located over the groove seat

1360 and frictionally engaging such surface. The spring tabs **1290** are sized such that when the surfaces **1312** frictionally engage the surfaces **1360** of the insert, the insert **1214** surfaces **1342** and **1343** in turn press against the shank upper portion **1208** at the spherical surface **1234**. The friction fit between the compression insert **1214** and the shank upper portion **1208** is not fixed but at the same time not loose or floppy either, advantageously allowing the user to articulate the shank **1204** with respect to the receiver **1210**, but with some resistance, so that when the shank is placed in a desired orientation with respect to the receiver, the assembly **1201** remains substantially frictionally set in such desired orientation unless purposefully manipulated into another position. For example, at this time, the receiver **1210** may be articulated to a desired position with respect to the shank **1204** as shown, for example, as shown in FIG. **80**, but prior to locking of such position that is shown in those drawings.

[0488] With reference to FIGS. **75-78**, the rod **1221** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **1201**. The closure structure **1218** is then inserted into and advanced between the arms **1262** of the receiver **1210**. The closure structure **1218** is rotated, using a tool engaged with the inner drive **1366** until a selected pressure is reached at which point the rod **1221** engages the insert saddle **1352**, further pressing the insert spherical surface **1342** and stepped surfaces **1343** against the shank spherical surface **1234**, the edges of the stepped surfaces penetrating into the spherical surface **1234**. As the closure structure **1218** rotates and moves downwardly into the respective receiver **1210**, the point **1369** and rim **1370** engage and penetrate the rod surface **1222**, the closure structure **1218** pressing downwardly against and biasing the rod **1221** into full engagement with the insert **1214** that urges the shank upper portion **1208** toward the retainer **1212** and into locking engagement therewith, the retainer **1212** frictionally abutting the surface **1299** and expanding outwardly against the cylindrical surface **1296**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **1206** with respect to the receiver **1210**. As shown in FIG. **78**, during rotation and downward movement of the closure top **1218**, the insert **1214** arms **1345** are retained in alignment with the receiver arms **1262** and thus the saddle surface **1352** is retained in alignment within the receiver channel **1264** by the spring tabs **1290** located within the insert grooves **1359**. Also, as shown in FIGS. **76** and **77**, during rotation and downward movement of the closure top **1218**, the rod **1221** presses the insert **1214** in a direction towards the receiver base **1260**, pressing the frusto-conical insert surfaces **1348** into engagement with the cylindrical receiver surfaces **1287**, thereby wedging and compression locking the insert **1214** into and against the receiver **1210**. If the closure top **1218** is then loosened and rotated to an upward unlocked position, for example as shown in FIG. **79**, the rod **1221** is also loosened, but the insert **1214** remains in a downward position, wedged against the receiver walls **1287**. This advantageously allows the surgeon to slide or otherwise manipulate the bone anchor and/or the rod **1221** with respect to the assembly **1201** while the assembly **1201** is otherwise in a totally locked position with the shank **1204** in a desired fixed, unmovable angular orientation with respect to the receiver **1210**. Once any desirable movement or manipulation of the rod **1221** is completed, the closure top **1218** is simply rotated back into the position shown in FIG. **75**, locking the rod **1221** back into place. Furthermore, if the polyaxial mechanism needs to be unlocked, the insert surfaces **1348** are squeezed toward one another using a tool (not shown) that is inserted into the insert grooves **1359** at a location above the spring tabs **1290**. The squeezed insert **1214** is then pulled or moved slightly upwardly toward the opening **1266** disengaging the surfaces **1348** from the receiver walls **1287** and unlocking the polyaxial mechanism of the assembly **1201**.

[0489] With reference to FIGS. **81-83**, an alternative compression insert **1214'** is shown that is substantially similar to the insert **1214** with the exception that the insert **1214'** does not include the compression lock and squeeze release feature of the frusto-conical upper surfaces **1348**. Thus, the insert **1214'** may be utilized in embodiments wherein the lock and release feature is not desired. The insert **1214'** advantageously does not require any squeezing or other manipulation when

uploaded into the receiver **1210** as it includes a cylindrical outer surface **1344'** that is receivable within the receiver lower opening cylindrical surface **1300**.

[0490] With reference to FIG. **84** and to U.S. patent application Ser. No. 12/802,849 filed Jun. 15, 2010 (hereafter the '849 application) that is incorporated by reference herein, polyaxial bone screws **1**, **1001** and **1010** according to the invention (as well as the other polyaxial screws described later in this application) may be attached to a dynamic stabilization longitudinal connecting member assembly according to the present invention, generally **1401**. The connecting member assembly **1401** is elongate, having a substantially central axis. With particular reference to FIG. **84**, the connecting member assembly **1401** more fully described in the '849 application is illustrated that generally includes at least one inelastic sleeve, that may be flanged or not, such as, for example, the sleeves **1406** and **1406'** with spacers **1415** or spacer/liner **1416** combinations located between the bone screws and attached sleeves. The illustrated connector **1401** is further shown with a hard rod **1121'** and a rod cord connector **1424** as well as a cord **1422**. Two bone screws **1001** are shown, one of which is attached to the sleeve **1406'** and the other to the hard rod **1121'**. As more fully discussed in the '849 application, either a slide or slip closure top, such as the tops **18** and **1018** previously described herein or the break-off head closure tops **1430** and **1432** shown in FIGS. **87-90** (and shown in phantom in FIG. **84**) engage a respective sleeve (or a hard rod) but not the cord **1422**, allowing the cord to slip or slide within the polyaxial screw; or a grip closure top **1431** is used that extends through the sleeve and grips and fixes the cord **1422** against a surface of the sleeve and thus fixes the cord in relation to the polyaxial screw **1001**. The closure tops **1430**, **1431** and **1432** are shown in greater detail in FIGS. **85-90**.

[0491] With further reference to FIGS. **85-90**, various closure tops for use with the bone screw assemblies according to the invention and the connecting assembly **1401** are shown. The bone screw **1432** shown in FIGS. **89** and **90** is identical to the closure tops **18** and **1018** previously described herein with the exception that it includes a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Thus, the closure structure **1432** includes an outer helically wound guide and advancement structure **1502**, a top surface **1504** of the guide and advancement structure, an internal drive **1506**, a bottom surface **1508**, a point **1509** and a rim **1510**, the same or similar to the respective guide and advancement structure **1162**, top surface **1164**, internal drive **1166**, bottom surface **1168**, point **1169** and rim **1170** previously discussed herein with respect to the closure structure **1018**. Located above the guide and advancement structure top surface is a break-off head **1512**.

[0492] With reference to FIGS. **85** and **86**, also cooperating with the bone anchors **1** and **1001** is the closure top **1431** having an outer helically wound guide and advancement structure **1522**, a top surface **1524** of the guide and advancement structure, an internal drive **1526** and a break-off head **1532**, the same or similar to the respective guide and advancement structure **1502**, top surface **1504**, internal drive **1506** and break-off head **1512** previously discussed herein with respect to the closure top **1432**. In lieu of the point and rim of the closure top **1432**, the closure top **1431** has a lower cylindrical portion **1527** having a substantially planar bottom surface **1528**. The portion **1527** is sized and shaped to be received by a bore of the cooperating sleeve, for example, the sleeve **1406'**, the bottom surface **1528** pressing the cord **1422** into fixed engagement with the sleeve.

[0493] With reference to FIGS. **87** and **88**, also cooperating with the bone anchors **1** and **1001** is the closure top **1430** having a an outer helically wound guide and advancement structure **1542**, a top surface **1544** of the guide and advancement structure, an internal drive **1546** and a break-off head **1552**, the same or similar to the respective guide and advancement structure **1522**, top surface **1524**, internal drive **1526** and break-off head **1532** previously discussed herein with respect to the closure top **1431**. The closure top **1430** includes a planar bottom surface **1548** adjacent the guide and advancement structure **1542**. The planar bottom surface **1548** remains flush with a corresponding sleeve surface and does not enter into the bore of the sleeve, allowing sliding movement of the cord **1422** with respect to the bone screw receivers **1010** cooperating with the

closure top **1430**.

[0494] With reference to FIGS. **91-114**, the reference numeral **1601** generally represents another embodiment of a polyaxial bone screw according to the invention. The assembly **1601** includes a shank **1604**, that further includes a body **1606** integral with an upwardly extending upper portion or capture structure **1608**; a receiver **1610**; a retainer structure **1612** and a compression or pressure insert **1614**. The receiver **1610**, retainer **1612** and compression insert **1614** are initially assembled and may be further assembled with the shank **1604** either prior or subsequent to implantation of the shank body **1606** into a vertebra. The shank **1604** and the retainer **1612** are substantially the same in form and function as the respective shank **1204** and retainer **1212** previously discussed herein. With particular reference to FIGS. **106-109**, the receiver **1610** is also similar in form and function to the receiver **1210** and other receivers previously discussed herein in that the receiver **1610** provides an expansion chamber **1695** for the retainer to expand about the shank upper portion **1608** allowing the shank to “pop” or “snap” on to the assembly, and a receiver lower seat **1696** for the retainer **1612** to slightly expand into when the shank upper portion **1608** is locked against the retainer **1612**. The receiver **1610** differs from the receiver **1210** in that the receiver **1610** does not include spring tabs, but rather has a blocking feature **1623** as will be described below and crimping walls **1625**. Furthermore, the receiver **1610** includes surfaces **1640** and **1641** for engagement with the insert **1614**, first to hold the insert **1614** in an upper portion of the receiver during shipping and assembly with the shank **1604**, the surfaces facilitating a friction fit between the insert **1614** and the shank head **1608** during manipulation of the bone screw **1601** and then locking of the shank **1604** with respect to the receiver by the insert **1614** even if a rod and closure top is loosened or removed from the assembly **1601**.

[0495] FIGS. **91** and **110-114** further show a closure structure **1618** that is the same as the closure **1018** previously described herein with the exception that the point and rim have been replaced by a bottom outer planar annular rim **1768**, a central point or knob **1769** and a domed surface **1770** running from the point **1769** to the rim **1768**, the closure **1618** for capturing a longitudinal connecting member, for example, a deformable rod **1621** in the form of a PEEK rod which in turn engages the compression insert **1614** that presses against the shank upper portion **1608** into fixed frictional contact with the retainer **1612**, so as to capture, and fix the longitudinal connecting member **1621** within the receiver **1610** and thus fix the member **1621** relative to the vertebra. Furthermore, the insert **1614** includes top surfaces **1620** of arms thereof that engage the closure top **1618** at an annular bottom rim **1768**, providing for a locked polyaxial mechanism in the event that the deformable rod **1621** loosens within the receiver **1610**. The receiver **1610** and the shank **1604** cooperate in such a manner that the receiver **1610** and the shank **1604** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **1610** with the shank **1604** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0496] Features of the assembly **1601** include, but are not limited to the downloaded lock and release insert **1614** that includes a rotation block feature **1622** that abuts against the stop or wall **1623** of the receiver **1610** upon insertion, placing the insert **1614** into alignment with the receiver **1610**. With reference to FIGS. **102-105**, the block feature **1622** and the drop-down insertion of the insert into the receiver followed rotation thereof until the insert feature abuts the wall **1623** of the receiver is described in greater detail in the '849 patent application that is fully incorporated by reference herein. With reference to FIG. **106**, thereafter, thin crimp walls **1625** of the receiver **1610** are pressed inwardly into grooves **1626** of the insert **1614** to block reverse rotation of the insert **1614** out of the receiver and to also frictionally hold the receiver in a desired location, including an upward location shown in FIG. **106** during shipping and early assembly and a down, shank engaging location shown in FIG. **109**, for example. Furthermore, in some embodiments of the invention, the insert arms have some flexibility and the arm surfaces **1630** abutting against surfaces

1640 of the receiver may also aid in keeping the insert in the upward location shown in FIG. **106** until the insert is pushed downwardly toward the receiver base in a later stage of assembly.

[0497] With further particular reference to FIGS. **110-114**, the insert **1614** upper surface **1630** is frusto-conical or otherwise tapered, sized and shaped for wedging against the receiver cylindrical surface **1640** and the insert further includes a lower frusto-conical surface **1631** sized and shaped for wedging into the lower cylindrical surface **1641** of the receiver **1610**. Similar to the assembly **1200**, as the closure top is advanced downwardly, the frusto-conical surfaces of the insert **1614** wedge into the cylindrical surfaces of the receiver **1610**, locking the insert against the shank **1604** and thus locking the polyaxial mechanism, even if the closure top **1618** is later backed out as shown in FIG. **114**, allowing for manipulation of the rod **1621** with an advantageously fully locked polyaxial mechanism. If it is desired to loosen the polyaxial mechanism, a tool, not shown may be inserted into the receiver **1610** to push arms of the insert **1614** toward one another and upwardly, loosening the surfaces **1630** and **1631** from the respective receiver surfaces **1640** and **1641**.

[0498] Furthermore, prior to locking of the insert **1614** against the receiver **1610**, the insert may be pressed downwardly into engagement with the shank upper portion **1608** to provide a friction fit between the insert **1615** and the upper portion **1608**, either one or both of the upper and lower receiver engagement surfaces **1640** and **1641** engaging with the respective insert surfaces **1630** and **1631** to provide enough downward force or frictional fit between the insert inner stepped surfaces **1643** and/or spherical surface **1644** to provide a non-floppy friction fit with the shank spherical upper portion or head **1608** when the surgeon is manipulating the unlocked assembly **1601** during surgery. Upon locking the shank in place, the stepped surfaces **1643** engage and penetrate the shank spherical head **1608**. A squeeze release feature or aperture **1632** located on each insert arm may be accessed through the receiver **1610** apertures to press the insert arms toward one another to lift the insert away from the shank upper portion **1608** and thus unlock the polyaxial mechanism if desired.

[0499] With reference to FIGS. **398-402**, an alternative insert **1614'** for use with the assembly **1601** is substantially identical to the insert **1614** (having the same reference numerals marked with a "' to indicate features the same or similar to the features identified on the insert **1614**). The insert **1614'** further includes optional lower slots or slits **1650** for enhancing friction fit with the shank upper portion **1608** and for ease of removal from a locking fit with the receiver **1610**, if required.

[0500] With reference to FIGS. **115-152** the reference number **2001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **2001** includes a shank **2004**, that further includes a body **2006** integral with an upwardly extending upper portion or head-like capture structure **2008**; a receiver **2010**; a lower retainer structure illustrated as a resilient open ring **2012**, a friction fit crown collet compression or pressure insert **2014**, and an upper retainer structure illustrated as an open resilient snap ring **2016**. The receiver **2010**, retainer structures **2012** and **2016** and compression insert **2014** are initially assembled and may be further assembled with the shank **2004** either prior or subsequent to implantation of the shank body **2006** into a vertebra **2017**, as will be described in greater detail below.

[0501] FIGS. **115** and **151-152** further show a closure structure **2018** for capturing a longitudinal connecting member, for example, a rod **2021** which in turn engages the compression insert **2014** that presses against the shank upper portion **2008** into fixed frictional contact with the lower retainer **2012**, so as to capture, and fix the longitudinal connecting member **2021** within the receiver **2010** and thus fix the member **2021** relative to the vertebra **2017**. The illustrated rod **2021** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **2022**. It is foreseen that in other embodiments, the rod **2021** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **2010** and the shank **2004** cooperate in such a manner that the receiver **2010** and the shank **2004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **2010** with the shank **2004** until both are locked or fixed relative to each other near the end of an implantation

procedure.

[0502] The shank **2004**, best illustrated in FIGS. **115-117** is substantially similar to the shank **1004** previously described herein with respect to the assembly **1000**. Thus, the shank **2004** includes the shank body **2006**, upper portion or head **2008**, a shank thread **2024**, a neck **2026**, a tip **2028**, a top of thread **2032**, an upper portion spherical surface **2034** a top surface **2038**, an internal drive **2046** with a base surface **2045** and an cannulation bore **2050** the same or substantially similar to the respective body **1006**, upper portion or head **1008**, shank thread **1024**, neck **1026**, tip **1028**, top of thread **1032**, spherical surface **1034**, top surface **1038**, internal drive **1046** with base surface **1045** and cannulation bore **1050** previously described herein with respect to the shank **1004** of the assembly **1001**. To provide a biologically active interface with the bone, the threaded shank body **2006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**. The shank spherical surface **2034** has an outer radius configured for frictional, non-floppy, sliding cooperation with a discontinuous concave surface **2142** of the compression insert **2014** having a substantially similar or slightly smaller radius, as well as ultimate frictional engagement and penetration by a stepped, gripping portion **2143** of the insert **2014**, as will be discussed more fully in the paragraphs below. The spherical surface **2034** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and ultimate frictional engagement with the compression insert **2014** as well as ultimate frictional engagement with the lower retainer **2012**. The shank spherical surface is locked into place exclusively by the insert **2014** and the retainer **2012** and not by inner surfaces defining the receiver cavity.

[0503] With particular reference to FIGS. **115** and **128-133**, the receiver **2010** has a generally squared-off U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **2010** has an axis of rotation B that is shown in FIG. **115** as being aligned with and the same as an axis of rotation A of the shank **2004**, such orientation being desirable, but not required during assembly of the receiver **2010** with the shank **2004**.

[0504] The receiver **2010** includes a substantially cylindrical base **2060** defining a bore or inner cavity **2061**, the base **2060** being integral with a pair of opposed upstanding arms **2062** forming a cradle and defining a channel **2064** between the arms **2062** with an upper opening, generally **2066**, and a squared off lower channel portion including a substantially planar lower seat **2068**, the channel **2064** having a width for operably snugly receiving the rod **2021** or portion of another longitudinal connector between the arms **2062**, the channel **2064** communicating with the base cavity **2061**. The squared-off geometry of the channel **2064** and lower seat **2068** allow for use with a variety of longitudinal connecting members, including, but not limited to those with circular, square and rectangular cross-sections. As compared to a U-shaped channel that includes a lower seat having a surface with a radius the same or slightly larger than a cooperating cylindrical rod or other connecting member, the squared-off seat **2068** of the present invention provides improved stress management, moving stress risers outwardly toward the two arms **2062** rather than being focused primarily at a center base line of the radiused lower seat. Furthermore, outer front and rear opposed substantially planar base surfaces **2069** that partially define the squared-off lower seat **2068** advantageously reduce the run on the rod (i.e., provide a more narrow receiver that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and provide the planar surface **2069** for flush or close contact with other connecting member components in certain embodiments, such as for bumpers or spacers that surround a hard or deformable rod or provide support for cord-type connecting members.

[0505] Each of the arms **2062** has an interior surface, generally **2070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **2072** located adjacent top surfaces **2073** of each of the arms **2062**. In the illustrated embodiment, the guide and advancement structure **2072** is a partial helically wound interlocking

flangeform configured to mate under rotation with a similar structure on the closure structure **2018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, for example, when the receiver **2010** includes a thicker body having a U-shaped channel (as compared to the squared-off channel of the illustrated receiver), the guide and advancement structure **2072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **2018** downward between the arms **2062**, as well as eventual torquing when the closure structure **2018** abuts against the rod **2021** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0506] As an example of an alternative closure mechanism, with reference to FIGS. **153-156**, an alternative embodiment or assembly **2001'** cooperating with a reverse-angle thread form closure top **2018'** is shown that is substantially similar to the assembly **2001** with the exception that reverse angle threads **2072'** are used in lieu of the flange form **2072** with a receiver **2010'** that is substantially similar to the receiver **2010** with the exception of having a U-shaped channel **2064'**. The assembly **2001'** otherwise includes a shank **2004'**, a receiver **2010'**, a lower retainer structure illustrated as a resilient open ring **2012'**, a friction fit crown collet compression or pressure insert **2014'**, and an upper retainer structure illustrated as an open resilient snap ring **2016'** that are the same or substantially similar in form and function to the respective shank **2004**, receiver **2010**, lower retainer **2012'**, friction fit insert **2014** and upper retainer **2016** of the assembly **2001**. The assembly **2001'** is shown with a rod **2021'** that is the same or substantially similar to the rod **2021** shown with the assembly **2001**. A more detailed description of the assembly **2001'** utilizing the reverse angle thread closure top **2018'** is provided in Applicants' Provisional Application Ser. No. 61/343,737 filed May 3, 2010 that is incorporated herein by reference.

[0507] Returning to the assembly **2001** shown in FIGS. **115-152**, and in particular to FIGS. **128-133**, an opposed pair of tool receiving and engaging apertures **2074** are formed on outer surfaces **2076** of the arms **2062**. Furthermore, two pair of tool receiving and engaging apertures **2077** are formed in front and rear surfaces **2078** of the arms **2062**. Transition base surfaces **2079** span between the surfaces **2078** and the planar base surfaces **2069**, the surfaces **2069** and **2078** both running substantially parallel to the receiver axis B, the surfaces **2079** sloping downwardly toward the base **2060** at an angle with respect to the axis B. Some or all of the apertures **2074** and **2077** may be used for holding the receiver **2010** during assembly with the insert **2014**, the retainers **2012** and **2016** and the shank **2004**, during the implantation of the shank body **2006** into a vertebra when the shank is pre-assembled with the receiver **2010**, and during assembly of the bone anchor assembly **2001** with the rod **2021** and the closure structure **2018**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **2062**.

[0508] Returning to the interior surface **2070** of the receiver arms **2062**, located below the guide and advancement structure **2072** is a discontinuous cylindrical surface **2082** partially defining a run-out feature for the guide and advancement structure **2072**. The cylindrical surface **2082** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **2072**. Moving downwardly, in a direction toward the base **2060**, adjacent the cylindrical surface **2082** of each arm is a run-out seat or surface **2084** that extends inwardly toward the axis B and runs perpendicular to the axis B. Adjacent to and located below the surface **2084** is another cylindrical surface **2086** having a diameter smaller than the diameter of the surface **2082**. A discontinuous annular surface **2088** that provides an upper abutment surface or stop for capturing the compression insert **2014** in the receiver **2010** is located below and adjacent to the cylindrical surface **2086**. The abutment surface **2088** is disposed substantially perpendicular to the axis B. As shown in FIG. **144** and discussed in greater detail below, the assembly **2001** is typically provided to a user with the insert **2014** being held within the receiver by the upper snap-ring retainer **2016** that resiliently holds

the insert **2014** and keeps the insert stationary with respect to the receiver **2010** and abutting against or slightly spaced from the upper stop **2088** until the insert **2014** is friction fitted about the shank upper portion **2008**, also described in greater detail below. The insert **2014** and the shank **2004** are then moved downwardly toward the base **2060** into a working position shown in FIGS. **149** and **150** wherein the insert **2014** is in frictional contact with the shank upper portion **2008**, the shank still being movable, with some force, with respect to the insert **2014**, and thus advantageously placeable and then held in a selected angular position with respect to the insert **2014** and the receiver **2010**, due to the friction fit between the insert **2014** and the shank upper portion **2008**.

[0509] The inner surfaces **2070** of the arms **2062** include an additional discontinuous cylindrical surface **2090** adjacent the annular surface **2088** and extending downwardly toward the receiver base **2060**. The surface **2090** is disposed parallel to the receiver axis B. The surface **2090** has a diameter greater than the diameter of the surface **2086** but less than the diameter of the surface **2082**. In some embodiments of the invention, the surface **2090** terminates near a discontinuous annular surface **2092**. In the present invention, another cylindrical surface **2093** spans between the surface **2090** and the annular surface **2092**, the surface **2093** having a diameter slightly larger than the diameter of the surface **2090**. The surfaces **2090** and **2093** are sized and shaped to receive the compression insert **2014** as shown, for example, in FIGS. **144** and **145** when in a pre-assembled configuration, and also during assembly with the shank **2004** as shown in FIGS. **146-148**. The surface **2092** is perpendicular to the receiver axis B. A cylindrical surface **2094** adjacent and perpendicular to the surface **2092** is formed in the arm surfaces **2070** and also partially extends into the base **2060**. The surface **2094** has a diameter greater than the diameters of the surfaces **2090** and **2093** and also greater than the diameter of the surface **2082**. The surface **2094** terminates at a continuous annular seating surface **2095** formed in the receiver base **2060**. The surface **2095** is substantially parallel to the surface **2092**. The surfaces **2092**, **2094** and **2095** form a recess in each arm **2062** for holding the open retainer **2016**. As shown in FIGS. **139-149** and discussed in greater detail below, the open snap ring **2016** of the assembly **2001** is compressed and inserted into the channel **2064** from the top opening **2066** and then allowed to expand to a neutral state at a location beneath the surface **2092** and above the surface **2095**, the retainer having room to expand outwardly to or near the cylindrical surface **2094** when required. As the compression insert **2014** is placed in different stages of assembly with the shank **2004** (see, e.g., FIGS. **148** and **149**), the retainer **2016** expands into the discontinuous recess formed by the surfaces **2092**, **2094** and **2095** in the arm surfaces **2070** and then returns to a neutral state during operation of the assembly **2001**, the surface **2092** serving as an upper stop, capturing the retainer **2016** in the lower portion of the arms **2062** near the channel seat **2068** and the continuous surface **2095** serving as a stable lower seat for the open retainer **2016**.

[0510] A cylindrical surface **2096** formed in the base **2060** and partially defining the base cavity **2061** is adjacent to the annular surface **2095** and perpendicular thereto. A diameter of the surface **2096** is smaller than the diameter of the surface **2094**. A continuous annular upper rim or stop **2098** is located below and adjacent to the cylindrical surface **2096**. The surface **2098** is disposed in the base **2060**, partially forming the base cavity **2061** and forms an abutment stop for the resilient retainer **2012**, prohibiting the retainer **2012** (when in an uncompressed configuration) from moving upwardly into the space defined by the cylindrical surface **2096** and the channel **2064**. Another cylindrical surface **2099** is located below and adjacent to the surface **2098**. The cylindrical surface **2099** is oriented substantially parallel to the axis B and is sized and shaped to provide an expansion chamber for receiving an expanded retainer **2012**. The surfaces **2098** and **2099** define a circumferential recess that is sized and shaped to receive the retainer **2012** as it expands around the shank upper portion **2008** as the shank **2008** moves upwardly toward the channel **2064** during assembly, as well as form a restriction to prevent the expanded retainer **2012** from moving upwardly with the shank portion **2008**, the surface **2098** preventing the retainer **2012** from passing upwardly out of the cavity **2061** whether the retainer **2012** is in a partially or fully expanded

position or state, or in a neutral or original or operative position or state (see, e.g., FIGS. **146** and **147**). A cylindrical surface **2101** located below the cylindrical surface **2099** is sized and shaped to closely receive the retainer **2012** when the retainer is in a neutral or slightly expanded operative position as shown in FIG. **152**, for example. Thus, the cylindrical surface **2101** has a diameter smaller than the diameter of the cylindrical surface **2099** that defines the expansion area for the retainer **2012**. The surface **2101** is joined or connected to the surface **2099** by one or more beveled, curved or conical surfaces **2102**. The surfaces **2102** allow for sliding gradual movement and/or contraction of the retainer **2012** into the space defined by the surface **2101** and ultimate seating of the retainer **2012** on a lower annular surface **2104** located below and adjacent to the cylindrical surface **2101**.

[0511] Located below and adjacent to the annular seating surface **2104** is another substantially cylindrical surface **2106** that communicates with a beveled or flared bottom opening surface **2107**, the surface **2107** communicating with an exterior base surface **2108** of the base **2060**, defining a lower opening, generally **2110**, into the base cavity **2061** of the receiver **2010**. The illustrated surface **2100** has a diameter allowing for slidable uploading of the compression insert **2014** (with some compression of a portion of the insert **2014** as will be described below) while requiring compression or squeezing of the retainer **2012** during uploading of the retainer **2012** through the lower opening **2110** (see FIGS. **141** and **143**, for example).

[0512] With particular reference to FIGS. **115** and **118-122**, the lower open retainer ring **2012** that operates to capture the shank upper portion **2008** and attached compression insert **2014** within the receiver **2010** has a central axis that is operationally the same as the axis B associated with the receiver **2010** when the shank upper portion **2008** and the retainer **2012** are installed within the receiver **2010**. The retainer ring **2012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **2012** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The lower retainer **2012** has a central channel or hollow through bore, generally **2121**, that passes entirely through the ring **2012** from a top surface **2122** to a bottom surface **2124** thereof. Surfaces that define the channel or bore **2121** include a discontinuous inner cylindrical surface adjacent the top surface **2122** and a discontinuous frusto-conical or beveled surface **2127** adjacent the surface **2125**, both surfaces coaxial when the retainer **2012** is in a neutral non-compressed, non-expanded orientation. The retainer **2012** further includes an outer cylindrical surface **2130** located adjacent the top surface **2122** and an outer beveled or frusto-conical surface **2132** adjacent the bottom surface **2124**. The surface **2130** is oriented parallel to the central axis of the retainer **2012**. In some embodiments of the invention spaced notches (not shown) may be formed in the cylindrical surface **2130** to receive a holding and manipulation tool (not shown) used for contraction and insertion of the retainer **2012** into the receiver **2010**. In some embodiments further notches may be made to evenly distribute stress across the entire retainer **2012** during contraction and expansion thereof. In other embodiments of the invention, such notches may be on the inside of the retainer **2012** ring. The resilient retainer **2012** further includes first and second end surfaces, **2134** and **2135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **2134** and **2135** are disposed substantially perpendicular to the top surface **2122** and the bottom surface **2124**. A width X between the surfaces **2134** and **2135** is determined by a desired amount of compressibility of the open retainer **2012** when loaded into the receiver **2010**. The space X shown in FIG. **118** provides adequate space between the surfaces **2134** and **2135** for the retainer **2012** to be pinched, with the surfaces **2134** and **2135** compressed toward one another (as shown in FIG. **143**) to a closely spaced or even touching configuration, if necessary, to an extent that the compressed retainer **2012** is up or bottom loadable through the receiver opening **2110**. After passing through the opening **2110** and along a portion of the lower inner surface **2106**, the retainer **2012** expands or springs back to an original uncompressed, rounded or collar-like configuration of FIGS. **118-122**, see, e.g., FIG. **144**. The embodiment shown in FIGS. **118-122** illustrates the surfaces **2134** and

2135 as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer **2012** into the receiver **2010**.

[0513] With particular reference to FIGS. **115** and **123-127**, the friction fit crown compression insert **2014** is illustrated that is sized and shaped to be received by and up-loaded into the receiver **2010** at the lower opening **2110**. The compression insert **2014** has an operational central axis that is the same as the central axis B of the receiver **2010**. In operation, the insert advantageously frictionally engages the bone screw shank upper portion **2008**, allowing for unlocked but non-floppy placement of the angle of the shank **2004** with respect to the receiver **2010** during surgery prior to locking of the shank with respect to the receiver near the end of the procedure. The insert **2014** is thus preferably made from a resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be expanded about and then contracted, snapped or popped onto the shank upper portion **2008**. Furthermore, in operation, the insert **2014** is suspended within the receiver **2010**, being frictionally held in place by the shank upper portion at a lower end thereof and prohibited from moving upward by the upper resilient retainer **2016**. As will be explained in greater detail below, after initial assembly and during operation of the assembly **2001**, neither the retainer **2016** nor the inner surfaces of the receiver **2010** that define the cavity **2061** place any compressive force on the insert **2014** to hold the shank portion **2008** therein.

[0514] The crown collet compression insert **2014** has a central channel or through bore, generally **2138** running from an annular planar top surface **2139** to an annular planar and discontinuous bottom surface **2140** thereof, the bore **2138** defined by an inner cylindrical surface **2141**, an inner partially spherical surface **2142** and a shank gripping surface portion, generally **2143**, extending between the surface **2141** and the surface **2142**. The gripping surface portion **2143** preferably includes two or more graduated cylindrical surfaces disposed substantially parallel to the axis B and adjacent perpendicular step surfaces that are disposed generally perpendicular to the axis B when the insert **2014** is mounted within the receiver **2010**. It is foreseen that the stepped surface portion **2143** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping surface portion **2143** and also the surface **2142** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **2008**. A plurality of slits or slots **2145** are formed in the spherical surface **2142**, running through the bottom surface **2140** and terminating near or slightly extending into the graduated surface portion **2143**. The illustrated embodiment includes six slots **2145**. It is foreseen that other embodiments of the invention may include more or fewer slots **2145**. Each pair of slots **2145** forms a distinct resilient, partially spherical finger, tab or panel **2146** that extends from the shank gripping portion **2143** to the bottom surface **2140**. In other words, the inner spherical surface **2142** is separated into six surface portions **2146**, each being partially spherical and sized and shaped to resiliently expand about the spherical surface **2034** of the shank upper portion **2008** and then snap on and frictionally grip the surface **2034**. Preferably, the spherical surface **2142** is designed such that the gripping tabs or panels **2146** have a neutral or non-expanded radius that is slightly smaller than a radius of the shank surface **2034** so that when the tabs or panels **2146** are gripping the surface **2034**, the insert is in a slightly expanded state. When the shank **2004** is locked into position by a rod **2021** or other connecting member being pressed downwardly on the insert top surface **2139** by the closure top **2018**, the insert **2014** shank gripping portion **2143** that is initially slidable along the shank surface **2034** then digs or penetrates into the surface **2034** and thus securely fixes the shank upper portion **2008** to the insert at the portion **2143**.

[0515] The compression insert **2014** through bore **2138** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **2046** when the shank body **2006** is driven into bone with the receiver **2010** attached. The compression insert **2014** also includes a first outer and upper cylindrical surface **2148** adjacent to the top surface **2139**. The top surface

2139 engages the rod **2021** or other longitudinal connecting member during operation of the assembly **2001** and locates the rod above the lower seat **2068** of the receiver. The insert **2014** also includes an outer lower and discontinuous cylindrical surface **2150** adjacent to the bottom surface **2140**. A discontinuous annular ledge **2151** extends between and connects the upper and lower cylindrical surfaces **2148** and **2150**. The cylindrical surface **2148** is sized and shaped to be received within the receiver surface **2106** when loaded through the receiver bottom opening **2110** as shown, for example, in FIG. **141**. The surface **2150**, on the other hand, has a neutral diameter that is larger than the diameter of the receiver surface **2106**. Therefore, during assembly, the resilient insert fingers or panels **2146** are pressed inwardly toward the receiver axis B to allow for insertion of the entire insert **2014** into the receiver opening **2110**. As best shown in FIG. **152**, the outer cylindrical surface **2150** is sized and shaped so that once the insert **2014** is in an operational position, and the panels **2146** are frictionally mated about the shank upper portion **2008**, the outer cylindrical surface **2150** is in slidable engagement or slightly spaced from the receiver inner cylindrical wall **2096**. [0516] The location of the ledge or lip **2151** is designed such that the upper open retainer **2016** seats on the ledge **2151** when in an operational position as also shown in FIG. **152**, for example. As will be described in greater detail below, during early stages of assembly, the insert **2014** outer surface **2150** is gripped by the resilient retainer **2016** pre-assembled within the receiver **2010**, the retainer **2016** holding the insert **2014** in a desired stationary position in the receiver for ultimate assembly with the shank upper portion **2008**.

[0517] It is foreseen that in some embodiments of the invention the compression insert **2014** may further include upstanding arms that cradle the rod **2021** or other connecting member. Such arms may be located spaced from the closure top **2018** in some embodiments and may be sized and shaped to contact the closure top **2018** in other embodiments in order to provide locking of the polyaxial mechanism of the assembly with capture but without fixing of the rod **2021** or other longitudinal connecting member with respect to the closure top **2018**.

[0518] With particular reference to FIGS. **125** and **134-136**, the open upper resilient, ring-like retainer **2016** that operates to capture the compression inert **2014** within the receiver **2010** has a central axis that is operationally the same as the axis B associated with the receiver **2010** when the retainer **2016**, the insert **2014**, the shank upper portion **2008** and the retainer **2012** are installed within the receiver **2010**. The retainer **2016** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **2016** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The upper retainer **2016** has a central channel or hollow through bore, generally **2153**, that passes entirely through the structure **2016** from a top surface **2154** to a bottom surface **2156** thereof. The channel or bore **2153** is defined by a discontinuous inner cylindrical surface **2157** adjacent to both the top surface **2154** and the bottom surface **2156**. A discontinuous outer cylindrical surface **2158** is also adjacent to both the top surface **2154** and the bottom surface **2156**. In some embodiments of the invention spaced notches (not shown) may be formed in the cylindrical surfaces to receive a holding and manipulation tool (not shown) used for contraction and insertion of the retainer **2016** into the receiver **2010**. In some embodiments further notches may be made to evenly distribute stress across the entire retainer **2016** during contraction and expansion thereof. In other embodiments of the invention, such notches may be on the inside of the retainer **2016** ring. It is further noted that the geometry of the retainer **2016** (as well as that of the retainer **2012**) is not limited to the particular cylindrical or planar surface shapes shown in the drawings figures. The retainers **2016** and **2012** may be of a rounded ring-shape, for example, or include more or fewer planar surfaces. The resilient retainer **2016** further includes first and second end surfaces, **2159** and **2160** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **2159** and **2160** are disposed substantially perpendicular to the top surface **2154** and the bottom surface **2156**. A width X' between the surfaces **2159** and **2160** is determined by a desired amount of compressibility of the open retainer **2016** when loaded into the receiver **2010**. The space

X' shown in FIG. 134 provides adequate space between the surfaces 2159 and 2160 for the retainer 2016 to be pinched, with the surfaces 2159 and 2160 compressed toward one another (as shown in FIG. 139) to a closely spaced or even touching configuration, if necessary, to an extent that the compressed retainer 2016 is top loadable through the receiver channel opening 2066. After passing through the opening 2066 and along the channel 2064, the retainer 2016 is allowed to expand or spring back to an original uncompressed, rounded or collar-like configuration in the receiver arm recess formed in part by the cylindrical surface 2094, see, e.g., FIG. 140. The embodiment of the retainer 2016 shown in FIGS. 134-136 illustrates the surfaces 2159 and 2160 as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer 2016 into the receiver 2010.

[0519] With reference to FIGS. 115, 151 and 152, the illustrated elongate rod or longitudinal connecting member 2021 (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface 2022 of uniform diameter. The rod 2021 is the same or substantially similar to the rods previously described herein, such as the rods 21 and 1021. With reference to the '849 patent application, polyaxial bone screw assemblies 2001 according to the invention may be used with soft or dynamic stabilization longitudinal connecting member assemblies that may include, but are not limited to one or more sleeves with cooperating, spacers, bumpers and an inner tensioned cord.

[0520] With reference to FIGS. 115, 137 and 138, the closure structure or closure top 2018 shown with the assembly 2001 is rotatably received between the spaced arms 2062 of the receiver 2010. It is noted that the closure 2018 top could be a twist-in or slide-in closure structure. The illustrated closure structure 2018 is substantially cylindrical and includes a an outer helically wound guide and advancement structure 2162 in the form of a flange that operably joins with the guide and advancement structure 2072 disposed on the arms 2062 of the receiver 2010. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure 2018 downward between the arms 2062 and having such a nature as to resist splaying of the arms 2062 when the closure structure 2018 is advanced into the channel 2064, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the squared off U-shape channel of the illustrated receiver 2010 and reduced profile of the receiver 2010 that advantageously engages longitudinal connecting member components as will be further described below. The illustrated closure structure 2018 also includes a top surface 2164 with an internal drive 2166 in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive 2166 is used for both rotatable engagement and, if needed, disengagement of the closure 2018 from the receiver arms 2062. It is also foreseen that the closure structure 2018 may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface 2168 of the closure is planar and further includes a point 2169 and a rim 2170 for engagement and penetration into the surface 2022 of the rod 2021 in certain embodiments of the invention. The closure top 2018 may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure 2018 interior for a length of wire (not shown) inserted therein to provide a

guide for insertion of the closure top into the receiver arms **2062**.

[0521] Preferably, the receiver **2010**, the retainers **2012** and **2016** and the compression insert **2014** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainers **2012** and **2016** as well as compressing or expanding the insert **2014** panels **2146**. In some circumstances, the shank **2004** is also assembled with the receiver **2010**, the retainers **2012** and **2016** and the compression insert **2014** at the factory. In other instances, it is desirable to first implant the shank **2004**, followed by addition of the pre-assembled receiver, retainers and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **2004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized shank advantageously reduces inventory requirements, thus reducing overall cost.

[0522] Pre-assembly of the receiver **2010**, retainers **2012** and **2016** and compression insert **2014** is shown in FIGS. **139-144**. First, the retainer **2016** is top loaded into the receiver **2010** through the opening **2066** of the channel **2064**. The resilient open retainer **2016** is prepared for insertion into the receiver **2010** by squeezing or pressing the retainer end surfaces **2159** and **2160** toward one another as shown in FIG. **139**. The compressed retainer **2016** is inserted into the upper opening **2066** with the bottom surface **2156** facing the receiver cavity **2061**. However, in the present embodiment, as the top and bottom surfaces are identical, either surface **2154** or **2156** may serve as a bottom or top surface. The retainer **2016** is typically moved downwardly into the channel **2064** and past the cylindrical surface **2090** and allowed to expand to a neutral uncompressed state within the cylindrical surface **2094** of each of the arms **2062** as shown in FIG. **140**.

[0523] Then, the compression insert **2014** is uploaded into the receiver **2010** through the lower opening **2110** with the insert top surface **2139** facing the receiver bottom surface **2108**. The insert **2014** is slid upwardly toward the channel seat **2068** until the ledge **2151** nears the receiver bottom **2108**. Then, the insert panels **2146** are pressed radially inwardly toward the axis B to compress the insert slightly so that the outer lower cylindrical surface **2150** clears the receiver surface **2106** at the opening **2110**. With reference to FIG. **142**, the insert **2014** is pressed upwardly within the inner surface **2157** of the open retainer **2016**, expanding the retainer into the arm recesses formed by the cylindrical surfaces **2094**. The surface **2092** prohibits upward movement of the retainer **2016** as the insert **2014** is moved upwardly to a desired pre-assembly position with the insert bottom surface **2140** being substantially aligned with the annular receiver surface **2098**, the insert **2014** being located above the receiver cylindrical surface **2099** that functions as an expansion recess or chamber for the lower retainer **2012**. As the insert lower cylindrical surface **2150** has a diameter in a neutral state that is greater than an inner diameter of the retainer **2016**, also in a neutral state, the resilient retainer **2016** expands about and grips the insert **2014** within the receiver **2010**. The cylindrical surface **2096** of the receiver, now located about the insert **2014** is sized slightly larger than the outer diameter of the insert cylindrical surface **2150**, in a neutral state, so the surface **2096** does not function to compress or otherwise engage the insert panels **2146**, but the surface **2096** does block the retainer **2016** from moving in a downward direction. Also, the receiver upper stop **2088** abuts the insert top surface **2139**, prohibiting the pre-assembled insert from traveling any further up the receiver channel **2064**.

[0524] With reference to FIGS. **143** and **144**, the retainer **2012** is then prepared for insertion into the receiver **2010** by squeezing or pressing the retainer end surfaces **2134** and **2135** toward one another as shown in FIG. **143**. The compressed retainer **2012** is inserted into the lower opening **2110** with the planar top surface **2122** facing the receiver bottom surface **2108**. The retainer **2012** is typically moved upwardly into the receiver **2010** and past the cylindrical surface **2106** and allowed to expand to an almost neutral or slightly compressed state within the cylindrical surface **2101** as

shown in FIG. 144. Also as shown in FIG. 144, at this time, both the compression insert **2014** and the retainer **2012** are captured within the receiver **2010** in a manner that substantially prevents movement or loss of such parts out of the receiver **2010**. The receiver **2010**, compression insert **2014** (held by the retainer **2012**) and the retainer **2012** (held by the cylindrical surface **2101**) combination is now pre-assembled and ready for assembly with the shank **2004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **2004** as will be described herein.

[0525] As illustrated in FIG. 151, the bone screw shank **2004** or an entire assembly **2001** made up of the assembled shank **2004**, receiver **2010**, retainers **2012** and **2016** and compression insert **2014**, is screwed into a bone, such as the vertebra **2017**, by rotation of the shank **2004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **2006** by engagement thereof at the internal drive **2046**.

[0526] When the shank **2004** is driven into the vertebra **2017** without the remainder of the assembly **2001**, the shank **2004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer. With reference to FIGS. 145-150 the pre-assembled receiver, insert and retainers are placed above the shank upper portion **2008** until the shank upper portion is received within the opening **2110**. With particular reference to FIGS. 146 and 147, as the shank is moved into the interior of the receiver base, the shank upper portion **2008** presses the retainer **2012** upwardly into the recess or expansion chamber partially defined by the cylindrical surface **2099** (if the retainer is not already located within such recess). As the portion **2008** continues to move upwardly toward the channel **2064**, the top surface **2122** of the retainer **2012** abuts against the insert bottom surface **2140** as well as the annular rim stop **2098** of the receiver **2010**, stopping upward movement of the retainer **2012** and forcing outward movement of the retainer **2012** towards the cylindrical surface **2099** defining the receiver expansion recess as the spherical surface **2034** continues in an upward direction. The retainer **2012** begins to contract about the spherical surface **2034** as the center of the sphere passes beyond the center of the retainer expansion recess defined by the surface **2099** (see FIG. 148). At this time also (back to FIG. 147), the spherical surface **2034** moves into engagement with the insert **2014** spherical surface **2142** with the panels **2146** expanding slightly outwardly to receive the surface **2034** and pushing outwardly against the resilient upper retainer **2016**. The panels **2146** press outwardly against the surface **2096** that provides enough clearance for the spherical surface **2034** to enter into full frictional engagement with the panels **2146** as shown in FIG. 148. At this time, the insert **2014** and the surface **2034** are in a fairly tight friction fit, the surface **2034** being pivotable with respect to the insert **2014** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the insert **2014** and the shank upper portion **2008**.

[0527] With reference to FIG. 149, the retainer **2012** and attached insert **2014** are then moved down into a final operative position shown in FIGS. 149-152 by either an upward pull on the receiver **2010** or, in some cases, by driving the shank **2004** further into the vertebra **2017**. Also, in some embodiments, when the receiver **2010** is pre-assembled with the shank **2004**, the entire assembly **2001** may be implanted at this time by inserting the driving tool into the receiver and the shank drive **2046** and rotating and driving the shank **2004** into a desired location of the vertebra **2017**.

[0528] Also with reference to FIG. 149, at this time, the compression insert **2014** lower cylindrical surface **2150** is located below the open retainer **2016** and the retainer **2016** is disposed at or near the insert ledge **2151** and about the substantially non-compressible cylindrical surface **2148**. The insert **2014** is thus prohibited from moving upwardly at the ledge **2151** by the retainer **2016**, but the retainer **2016** is otherwise in a neutral state and does not place a compressive force on the insert **2014**. With reference to FIG. 150, at this time, the receiver **2010** may be articulated to a desired angular position with respect to the shank **2004**, that will be held, but not locked, by the frictional engagement between the insert **2014** and the shank upper portion **2008**.

[0529] With reference to FIGS. **151-152**, the rod **2021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **2001**. The closure structure **2018** is then inserted into and advanced between the arms **2062** of each of the receivers **2010**. The closure structure **2018** is rotated, using a tool engaged with the inner drive **2166** until a selected pressure is reached at which point the rod **2021** engages the flat top surface **2139** of the compression insert **2014**, further pressing the insert stepped surfaces **2143** against the shank spherical surface **2034**, the edges of the stepped surfaces penetrating into the spherical surface **2034** and also pressing the shank upper portion **2008** into locked frictional engagement with the retainer **2012**. Specifically, as the closure structure **2018** rotates and moves downwardly into the respective receiver **2010**, the point **2169** and rim **2170** engage and penetrate the rod surface **2022**, the closure structure **2018** pressing downwardly against and biasing the rod **2021** into compressive engagement with the insert **2014** that urges the shank upper portion **2008** toward the retainer **2012** and into locking engagement therewith, the retainer **2012** frictionally abutting the surface **2104** and expanding outwardly against the cylindrical seating surface **2101**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **2006** with respect to the receiver **2010**.

[0530] If removal of the rod **2021** from any of the bone screw assemblies **2001** is necessary, or if it is desired to release the rod **2021** at a particular location, disassembly is accomplished by using the driving tool (not shown) that mates with the internal drive **2166** on the closure structure **2018** to rotate and remove such closure structure from the cooperating receiver **2010**. Disassembly is then accomplished in reverse order to the procedure described previously herein for assembly.

[0531] With reference to FIGS. **157-187** the reference number **3001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **3001** includes a shank **3004**, that further includes a body **3006** integral with an upwardly extending upper portion or head-like capture structure **3008**; a receiver **3010**; a retainer structure illustrated as a resilient open ring **3012**, and a friction fit crown collet compression or pressure insert **3014**. The receiver **3010**, retainer **3012** and compression insert **3014** are initially assembled and may be further assembled with the shank **3004** either prior or subsequent to implantation of the shank body **3006** into a vertebra **3017**, as will be described in greater detail below. FIGS. **157** and **185-187** further show a rod **3021** and closure structure **3018** the same or similar to rods and closures previously described herein, for example, the rod **2021** and the closure **2018** described with reference to the assembly **2001**. As with other assemblies of the invention, the receiver **3010** and the shank **3004** cooperate in such a manner that the receiver **3010** and the shank **3004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **3010** with the shank **3004** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0532] The shank **3004**, best illustrated in FIGS. **157-159** is substantially similar to the shank **1004** previously described herein with respect to the assembly **1000**. Thus, the shank **3004** includes the shank body **3006**, upper portion or head **3008**, a shank thread **3024**, a neck **3026**, a tip **3028**, a top of thread **3032**, an upper portion spherical surface **3034** a top surface **3038**, an internal drive **3046** with a base surface **3045** and an cannulation bore **3050** the same or substantially similar to the respective body **1006**, upper portion or head **1008**, shank thread **1024**, neck **1026**, tip **1028**, top of thread **1032**, spherical surface **1034**, top surface **1038**, internal drive **1046** with base surface **1045** and cannulation bore **1050** previously described herein with respect to the shank **1004** of the assembly **1001**. To provide a biologically active interface with the bone, the threaded shank body **3006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**. The shank spherical surface **3034** has an outer radius configured for frictional, non-floppy, sliding cooperation with a discontinuous concave surface **3152** of the compression insert **3014** having a substantially similar or slightly smaller or

slightly larger radius, as well as ultimate frictional engagement and penetration by a stepped, gripping portion **3150** of the insert **3014**, as will be discussed more fully in the paragraphs below. The top surface **3038** is substantially perpendicular to a central axis A. The spherical surface **3034** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and ultimate frictional engagement with the compression insert **3014** as well as ultimate frictional engagement with the retainer **3012**. The shank spherical surface **3034** is locked into place exclusively by the insert **3014** and the retainer **3012** and not by inner surfaces defining the receiver cavity.

[0533] With particular reference to FIGS. **157** and **171-175**, the receiver **3010** has a generally U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **3010** has an axis of rotation B that is shown in FIG. **157** as being aligned with and the same as the axis of rotation A of the shank **3004**, such orientation being desirable, but not required during assembly of the receiver **3010** with the shank **3004** (see, e.g., FIG. **179** showing a receiver **3010** being “popped on” to a shank **3006** that is implanted in a vertebra **3017** and disposed at an angle with respect to the receiver). After the receiver **3010** is pivotally attached to the shank **3004**, either before or after the shank **3004** is implanted in a vertebra **3017**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **187**.

[0534] The receiver **10** includes a substantially cylindrical base **3060** defining a bore or inner cavity, generally **3061**, the base **3060** being integral with a pair of opposed upstanding arms **3062** forming a cradle and defining a channel **3064** between the arms **3062** with an upper opening, generally **3066**, and a U-shaped lower channel portion or seat **3068**, the channel **3064** having a width for operably snugly receiving the rod **3021** or portion of another longitudinal connector between the arms **3062**, the channel **3064** communicating with the base cavity **3061**. Outer front and rear opposed substantially planar arm surfaces **3069** partially define the channel **3064** directly above the seat **3068**, the surfaces **3069** advantageously reduce the run on the rod (i.e., provide a more narrow receiver portion that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and provide the planar surface **3069** for flush or close contact with other connecting member components in certain embodiments, such as for bumpers or spacers that surround a hard or deformable rod or provide support for cord-type connecting members, such as those shown in the '849 application, incorporated by reference herein.

[0535] Each of the arms **3062** has an interior surface, generally **3070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **3072** located adjacent top surfaces **3073** of each of the arms **3062**. In the illustrated embodiment, the guide and advancement structure **3072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **3018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **3072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **18** downward between the arms **3062**, as well as eventual torquing when the closure structure **3018** abuts against the rod **3021** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0536] An opposed pair of key-hole like shallow tool receiving and engaging grooves or apertures **3074**, each having a through bore **3075**, are formed on outer surfaces **3076** of the arms **3062**. Each through bore **3075** extends between the outer surface **3076** and the inner surface **3070** and is located between upper and lower shallow grooved or recessed portions that do not extend completely through the respective arm **3062**. In the present embodiment, part of the grooved portion directly below the through bore **3075** is defined by a thin wall **3077** that is crimped into the insert **3014** during assembly thereof with the receiver **3010** as will be described in greater detail below. In other embodiments of the invention, other surfaces forming the groove or aperture **3074**

may be inwardly crimped. Alternatively, spring tabs or other movable structure may be included on the receiver **3010** or the insert **3014** for retaining the insert **3014** in a desired position, with regard to rotation and axial movement (along the axis A) with respect to the receiver **3010**. Preferably the insert and/or receiver are configured with structure for blocking rotation of the insert with respect to the receiver, but allowing some up and down movement of the insert with respect to the receiver during the assembly and implant procedure. Two additional pair of tool receiving and engaging apertures **3078** are also formed in the front and rear surfaces **3069** of the receiver arms **3062**.

Transition base surfaces **3079** span between the planar surfaces **3069** at the U-shaped seat **3068** and the cylindrical base **3060**, the surfaces **3079** sloping downwardly toward the base **3060** at an angle with respect to the axis B. Some or all of the apertures **3074** and **3077** may be used for holding the receiver **3010** during assembly with the insert **3014**, the retainer **3012** and the shank **3004**; during the implantation of the shank body **3006** into a vertebra when the shank is pre-assembled with the receiver **3010**; during assembly of the bone anchor assembly **3001** with the rod **3021** and the closure structure **3018**; and during lock and release adjustment of the insert **3014** with respect to the receiver **3010**, either into or out of frictional engagement with the inner surfaces of the receiver **3010** as will be described in greater detail below. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **3062**.

[0537] Returning to the interior surface **3070** of the receiver arms **3062**, located below the guide and advancement structure **3072** is a discontinuous cylindrical surface **3082** partially defining a run-out feature for the guide and advancement structure **3072**. The cylindrical surface **3082** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **3072**. Moving downwardly, in a direction toward the base **3060**, adjacent the cylindrical surface **3082** of each arm is a run-out seat or surface **3084** that extends inwardly toward the axis B and runs perpendicular to the axis B. Adjacent to and located below the surface **3084** is another cylindrical surface **3086** having a diameter smaller than the diameter of the surface **3082**. The through bores **3075** extends through the arms at the surfaces **3086**. Located directly below each bore **3075** is a surface portion **3087** that engages the insert **3014** when the thin wall **3077** is crimped toward the insert **3014** during assembly of such insert in the receiver **3010** as will be described in greater detail below. A discontinuous annular surface **3088** is located below and adjacent to the cylindrical surface **3086**. The surface **3088** is disposed substantially perpendicular to the axis B. The inner surfaces **3070** of the arms **3062** include an additional partially discontinuous and partially continuous inner cylindrical surface **3090** adjacent the annular surface **3088** and extending downwardly into the receiver base **3060**. The surface **3090** is disposed parallel to the receiver axis B. The surface **3090** has a diameter greater than the diameter of the surface **3082**. The cylindrical surfaces **3086** and **3090** are sized to receive respective upper- and mid-portions of the insert **3014** as will be described in greater detail below.

[0538] Now, with respect to the base **3060** and more specifically, the base cavity **3061**, a lower portion of the surface **3090** that extends into the base and partially defines the base cavity **3061** terminates at an annular surface or ledge **3095**. The ledge **3095** extends toward the axis B and is substantially perpendicular thereto. Extending downwardly from the ledge **3095** is a cylindrical surface **3096** that partially defines the base cavity **3061**, the surface **3096** running parallel to the axis B and having a diameter smaller than the diameter of the surface **3090**. The surface **3096** is sized and shaped to initially closely receive a lower portion of the insert **3014** and later frictionally engage a tapered portion of the insert **3014**, providing and lock and release function that will be described in greater detail below. The surface **3096** terminates at an annular surface **98** of the base cavity **3061** that functions as an upper stop for the retainer **3012**, particularly when in an expanded state as shown in FIG. **181** and as will be described in greater detail below. Another cylindrical surface **3099** is located below and adjacent to the surface **3098**. The cylindrical surface **3099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer

3012. The surfaces **3098** and **3099** define a circumferential recess or expansion chamber that is sized and shaped give clearance to and to receive the retainer **3012** as it expands around the shank upper portion **3008** as the shank **8** moves upwardly toward the channel **3064** during assembly, as well as form a restriction to prevent the expanded retainer **3012** from moving upwardly with the shank portion **3008**, the surface **3098** and the insert **3014** preventing the retainer **3012** from passing upwardly out of the cavity **3061** whether the retainer **3012** is in a partially or fully expanded position or state, or in a neutral or original operative position or state. A cylindrical surface **3101** located below the cylindrical surface **3099** is sized and shaped to closely receive the retainer **3012** when the retainer is in a neutral or slightly compressed operative position as shown in FIGS. **184** and **185**, for example. Thus, the cylindrical surface **3101** has a diameter smaller than the diameter of the cylindrical surface **3099** that defines the expansion area for the retainer **3012**. The surface **3101** is joined or connected to the surface **3099** by one or more beveled, curved or conical surfaces **3102**. The surfaces **3102** allow for sliding gradual movement and/or contraction of the retainer **3012** into the final seating space defined by the surface **3101** and ultimate seating of the retainer **3012** on a lower annular surface **3104** located below and adjacent to the cylindrical surface **3101**. [0539] Located below and adjacent to the annular seating surface **3104** is another substantially cylindrical surface **3106** that communicates with a beveled or flared bottom opening surface **3107**, the surface **3107** communicating with an exterior base surface **3108** of the base **3060**, defining a lower opening, generally **3110**, into the base cavity **3061** of the receiver **3010**. The illustrated surface **3100** has a diameter requiring compression or squeezing of the retainer **3012** during uploading of the retainer **3012** through the lower opening **3110** (see FIG. **177**, for example). [0540] With particular reference to FIGS. **157**, **160-164** and **177**, the lower open retainer ring **3012** that operates to capture the shank upper portion **3008** and attached compression insert **3014** within the receiver **3010** has a central axis that is operationally the same as the axis B associated with the receiver **3010** when the shank upper portion **3008** and the retainer **3012** are installed within the receiver **3010**. The retainer ring **3012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **3012** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The retainer **3012** has a central channel or hollow through bore, generally **3121**, that passes entirely through the ring **3012** from a top surface **3122** to a bottom surface **3124** thereof. Surfaces that define the channel or bore **3121** include a discontinuous inner cylindrical surface **3125** adjacent the top surface **3122**, a discontinuous frusto-conical surface **3127** adjacent the surface **3125** and a beveled surface **3128**, all three surfaces coaxial when the retainer **3012** is in a neutral non-compressed, non-expanded orientation. The retainer **3012** further includes an outer cylindrical surface **3130** located adjacent the top surface **3122** and an outer beveled or frusto-conical surface **3132** adjacent the bottom surface **3124**. The surface **3130** is oriented parallel to the central axis of the retainer **3012**. In some embodiments of the invention, spaced notches (not shown) may be formed in the cylindrical surface **3130** to receive a holding and manipulation tool (not shown) used for contraction and insertion of the retainer **3012** into the receiver **3010**. In some embodiments further notches may be made to evenly distribute stress across the entire retainer **3012** during contraction and expansion thereof. In other embodiments of the invention, such notches may be on the inside of the retainer **3012** ring. The resilient retainer **3012** further includes first and second end surfaces, **3134** and **3135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **3134** and **3135** are disposed substantially perpendicular to the top surface **3122** and the bottom surface **3124**. A width X between the surfaces **3134** and **3135** is determined by a desired amount of compressibility of the open retainer **312** when loaded into the receiver **310** as shown in FIG. **177**. The space X shown in FIG. **160** provides adequate space between the surfaces **3134** and **3135** for the retainer **3012** to be pinched, with the surfaces **3134** and **3135** compressed toward one another (as shown in FIG. **177**) to a closely spaced or even touching configuration, if necessary, to an extent that the compressed retainer **3012** is up or bottom loadable through the receiver opening

3110. After passing through the opening **3110** and along a portion of the lower inner surface **3106**, the retainer **3012** expands or springs back to an original uncompressed, rounded or collar-like configuration of FIGS. **160-164**, see, e.g., FIG. **178**. The embodiment shown in FIGS. **160-164** illustrates the surfaces **3134** and **3135** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer **3012** into the receiver **3010**.

[0541] With particular reference to FIGS. **157** and **165-170**, the friction fit, lock and release crown compression insert **3014** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **3010** at the upper opening **3066**. The compression insert **3014** has an operational central axis that is the same as the central axis B of the receiver **3010**. In operation, the insert advantageously frictionally engages the bone screw shank upper portion **3008**, allowing for unlocked but non-floppy placement of the angle of the shank **3004** with respect to the receiver **3010** during surgery prior to locking of the shank with respect to the receiver near the end of the procedure. Furthermore, as will be described more fully below, an insert **3014** that has locked the shank **3004** in a desired angular position with respect to the receiver **3010**, by, for example, compression from the rod **3021** and closure top **3018**, is also wedged into engagement with the receiver **3010** at the inner surface **3096** and thus retains the shank **3006** in a locked position even if the rod **3021** and closure top **3018** are removed as shown in FIG. **186**. Such locked position may also be released by the surgeon if desired. The insert **3014** is thus preferably made from a resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be expanded about and then contracted, snapped or popped onto the shank upper portion **3008** as well as pinched and un-wedged from the receiver **3010**.

[0542] The lock- and-release crown collet compression insert **3014** includes a substantially cylindrical body **3136** integral with a pair of upstanding arms **3137** at an upper end thereof and integral with an opposed pair of crown collet extensions **3138** at a lower end thereof. A bore, generally **3140**, is disposed primarily within and through the body **3136** and communicates with a generally U-shaped through channel **3141** that is defined by the upstanding arms **3137**. The channel **3141** has a lower seat **3142** sized and shaped to closely, snugly engage the rod **3021**. It is foreseen that an alternative embodiment may be configured to include planar holding surfaces that closely hold a square or rectangular bar as well as hold a cylindrical rod-shaped, cord, or sleeved cord longitudinal connecting member. The arms **3137** disposed on either side of the channel **3141** extend upwardly from the body **3136**. The arms **3137** are sized and configured for ultimate placement near the cylindrical run-out surface **3082** below the receiver guide and advancement structure **3072**. It is foreseen that in some embodiments of the invention, the arms may be extended and the closure top configured such the arms ultimately directly engage the closure top **3018** for locking of the polyaxial mechanism, for example, when the rod **3021** is made from a deformable material. In such embodiments, the insert **3014** would include a rotation blocking structure or feature that abuts against cooperating structure located on an inner wall of the receiver **3010**, preventing rotation of the insert with respect to the receiver when the closure top is rotated into engagement with the insert. In the present embodiment, the arms **3137** include outer surfaces **3143** and top surfaces **3144** that are ultimately positioned in spaced relation with the closure top **3018**, so that the closure top **3018** frictionally engages the rod **3021** only, pressing the rod **3021** downwardly against the seating surface **3142**, the insert **3014** in turn pressing against the shank **3004** upper portion **3008** that presses against the retainer **3012** to lock the polyaxial mechanism of the bone screw assembly **3001** at a desired angle. As will be discussed in greater detail below, frictional engagement between the insert **3014** and the receiver **3010** maintains the upper portion **3008** in locked engagement with the retainer **3012** even if the closure top **3018** and/or rod **3021** are thereafter removed from the receiver **3010**.

[0543] The bore, generally **3140**, is substantially defined at the body **3136** by an inner cylindrical surface **3146** that communicates with a lower collet space that extends to discontinuous bottom

surfaces **3148** of the collet extensions **3138**. The body **3135** (and bore **3140**) is further defined by a shank gripping surface portion, generally **3150**, the gripping portion **3150** being adjacent to the cylindrical surface **3146**. Located below and adjacent to the gripping portion **3150** is an inner partially spherical surface **3152** that is continuous at the body **3136** and is discontinuous at the extensions **3138** wherein the surface **3152** extends downwardly, defining the inner shank holding portion of each of the collet extensions **3138** and terminating at the extension bottom surfaces **3148**. The gripping surface portion **3150** preferably includes two or more graduated cylindrical surfaces disposed substantially parallel to the axis B and adjacent perpendicular step surfaces that are disposed generally perpendicular to the axis B when the insert **3014** is mounted within the receiver **3010**. It is foreseen that the stepped surface portion **3150** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping surface portion **3150** and also the spherical surface **3152** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **3008**. The two collet extensions **3138** that generally extend in a direction opposite to the two arms **3137** and have the discontinuous inner spherical surface **3152**, also include through slits or slots **3153** running substantially vertically from adjacent the shank gripping surface portion **3150** through the bottom surfaces **3148**. The illustrated embodiment includes one slot **3153** centrally located in each extension **3138**. It is foreseen that other embodiments of the invention may include more or fewer slots **3153**. The slots **3153** substantially equally partition each of the extensions **3138**, forming four distinct resilient, partially spherical fingers, tab or panels **3154** that extend from the shank gripping portion **3150** to the bottom surface **3148**. In other words, the discontinuous inner spherical surface **3152** is further separated into four surface portions **3154**, each being partially spherical and sized and shaped to resiliently expand about the spherical surface **3034** of the shank upper portion **3008** and then snap on and frictionally grip the surface **3034**. The illustrated spherical surface **3152** is designed such that the gripping tabs or panels **3154** have a neutral or non-expanded radius that is the same or in some instances may be slightly smaller than a radius of the shank surface **3034** so that when the tabs or panels **3154** are gripping the surface **3034**, the insert **3014** collet extension portion **3138** is in a slightly expanded state. In other embodiments, as illustrated in the embodiment shown in FIG. 222, the non-expanded radius is the same or larger than a radius of the shank surface. The contacting surface area between the shank and the insert is sufficient to provide a non-floppy frictional fit in such instances. Furthermore, the shank surface **3034** and/or the spherical surface **3152** may include a roughened or grooved surface feature to provide for a frictional fit between the shank and the insert. In other embodiments, the resilient panels **3154** having a slightly larger pre-assembly radius than the shank surface **3034** may be bent inwardly to result in a tighter frictional fit with the shank surface. When the shank **3004** is locked into position by a rod **3021** or other connecting member being pressed downwardly on the insert seat **3142** by the closure top **3018**, the insert **3014** shank gripping portion **3150** that is initially slidable along the shank surface **3034** then digs or penetrates into the surface **3034** and thus securely fixes the shank upper portion **3008** to the insert at the portion **3150**.

[0544] The compression insert **3014** through bore **3140** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **3046** when the shank body **3006** is driven into bone with the receiver **3010** attached.

[0545] The illustrated insert **3014** further includes features that allow for a lock and release frictional fit between the insert **3014** and the receiver **3010**. These features include a shallow, substantially vertical or key-hole like slot **3155** disposed on the outer surface **3143** of each arm **3137**, the slot **3155** running substantially vertically from near the top surface **3144** through the body **3136** to near one of the collet extension through slots **3153**. In the illustrated embodiment, the slots **3155** and **3153** are substantially aligned and run substantially parallel to the axis B. Each slot **3155** further includes a through bore **3156** at or near a top thereof, the bore **156** running radially through each of the arms **3137** in a direction substantially perpendicular to the axis B. The through

bore and slots are directly opposed from one another and are sized and shaped to receive tools for manipulating the insert **3014** with respect to the receiver **3010** as will be described herein as well as for receiving tabs or crimped material from the receiver **3010** for maintaining alignment between the insert **3014** channel **3141** and the receiver **3010** channel **3064**. Directly below each arm **3137** and intersecting with a portion of each slot **3155** is a frusto-conical or otherwise outwardly flaring or tapered surface **3158** sized and shaped for engaging with the receiver **3010** at the surface **3096** as will be described more fully below. Each surface **3158** tapers inwardly toward the axis B as the surface runs toward the crown collet extensions **3138**. Below and adjacent to each surface **3158** is a cylindrical surface **3159** that partially defines an outer surface of a respective crown collet extension **3138**. Another frusto-conical surface **3160** is located below the surface **3159**, followed by a substantially cylindrical surface **3161** that defines a lower portion of each extension **3138**. The surface **3161** has a diameter smaller than a diameter of the surface **3159**. The surface **3161** is sized and shaped for being closely but slidably received by the receiver cavity **3061** at the cylindrical surface **3096**.

[0546] The insert body **3136** located between the arms **3137** and the collet extensions **3138** has an outer diameter slightly smaller than a diameter between crests of the guide and advancement structure **3072** of the receiver **3010**, allowing for top loading of the compression insert **3014** into the receiver opening **3066**, with the arms **3137** of the insert **3014** being located between the receiver arms **3062** during insertion of the insert **3014** into the receiver **3010**. Once the arms **3137** of the insert **3014** are generally located beneath the guide and advancement structure **3072**, the insert **3014** is rotated into place about the receiver axis B until the top surfaces **3144** are located directly below the guide and advancement structure **3072** as will be described in greater detail below.

[0547] With reference to FIGS. **157** and **185-187**, the illustrated elongate rod or longitudinal connecting member **3021** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **3022** of uniform diameter. The rod **3021** is substantially similar to the rods previously described herein, such as the rods **21**, **1021** and **2021** and may be a soft connecting member assembly as described, for example, in the '849 application incorporated by reference herein, and therefore shall not be discussed in any greater detail here.

[0548] With reference to FIGS. **157** and **185-187**, the closure structure or closure top **3018** shown with the assembly **3001** is rotatably received between the spaced arms of the receiver **3010**. It is noted that the closure **3018** top could be a twist-in or slide-in closure structure. The illustrated closure structure **3018** is substantially cylindrical and includes a an outer helically wound guide and advancement structure **3162** in the form of a flange that operably joins with the guide and advancement structure **3072** disposed on the arms **3062** of the receiver **3010**. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure **3018** downward between the arms **3062** and having such a nature as to resist splaying of the arms **3062** when the closure structure **3018** is advanced into the channel **3064**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the squared off U-shape channel of the illustrated receiver **3010** and reduced profile of the receiver **3010** that advantageously engages longitudinal connecting member components as will be further described below. The illustrated closure structure **3018** also includes a top surface **3164** with an internal drive **3166** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal

drive **3166** is used for both rotatable engagement and, if needed, disengagement of the closure **3018** from the receiver arms **3062**. It is also foreseen that the closure structure **3018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **3168** of the closure is planar and further includes a point **3169** and a rim **3170** for engagement and penetration into the surface **3022** of the rod **3021** in certain embodiments of the invention. The closure top **3018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **3018** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **3062**.

[0549] Preferably, the receiver **3010**, the retainer **3012** and the compression insert **3014** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **3012** as well as compressing or expanding the insert **3014** arms and collet extensions, if needed, as well as crimping a portion of the receiver **3010** toward the insert **3014**. In some circumstances, the shank **3004** is also assembled with the receiver **3010**, the retainer **3012** and the compression insert **3014** at the factory. In other instances, it is desirable to first implant the shank **3004**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **3004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **3008** and/or hydroxyapatite on the shank **3006**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **3004** advantageously reduces inventory requirements, thus reducing overall cost.

[0550] Pre-assembly of the receiver **3010**, retainer **3012** and compression insert **3014** is shown in FIGS. **175-177**. First, the compression insert **3014** is downloaded into the receiver **3010** through the upper opening **3066** with the crown collet extension bottom surfaces **3148** facing the receiver arm top surfaces **3073** and the insert arms **3137** as well as the insert collet extensions **3138** located between the opposed receiver arms **3062**. The insert **3014** is then lowered toward the channel seat **3068** until the insert **3014** arm upper surfaces **3144** are adjacent the run-out area below the guide and advancement structure **3072** defined in part by the cylindrical surface **3082**. Thereafter, the insert **3014** is rotated in a clockwise or counter-clockwise manner about the receiver axis B until the upper arm surfaces **3144** are directly below the guide and advancement structure **3072** as illustrated in FIG. **176** with the U-shaped channel **3141** of the insert **3014** aligned with the U-shaped channel **3064** of the receiver **3010**. In some embodiments, the insert arms **3137** and collet extensions **3138** may need to be compressed slightly during rotation to clear inner surfaces of the receiver arms **3062**. As shown in FIGS. **176** and **177**, the outer lower cylindrical surface **3161** of the insert **3014** is received with the cylindrical surface **3096** of the receiver. With reference to FIG. **177**, the receiver thin walls **3077** are then crimped inwardly toward the axis B by inserting a tool (not shown) through the receiver apertures **3074**, the tool pressing the walls **3077** until the wall surface **3087** engages the insert **3014** at the shallow central slot **3155** formed on the outer surface **3143** of each of the insert arms **3137**. The crimping of the wall surface **3087** into the slot **3155** keeps the insert **3014** U-shaped channel **3141** aligned with the receiver U-shaped channel **3064** and also retains the insert **3014** at the upward location shown in FIG. **177** with the insert arm top surfaces **3144** adjacent the guide and advancement structure **3072** until the insert **3014** is pushed downwardly toward the receiver base **3060** after assembly with the shank **3004**. Thus, the crimping of the receiver walls **3077** prohibits rotation of the insert **3014** about the receiver axis B but allows for limited axial movement of the insert **3014** with respect to the receiver **3010** along the axis B.

when some force is exerted to slide the crimped surface **3087** up or down along the groove **3155**. The insert **3014** is fully captured within the receiver **3010** by the guide and advancement structure **3072** prohibiting movement of the insert **3014** up and out through the receiver opening **3066** as well as by the frusto-conical surface **3158** of the insert **3014** that is sized to engage and wedge against the cylindrical surface **3096** of the receiver, preventing movement of the insert **3014** out of the lower receiver opening **3110**. In some embodiments of the invention, top or side surfaces of the insert **3014** may include a resilient projection or projections for temporarily frictionally engaging with an inner surface of the receiver **3010** to hold the insert **3014** in an upper portion of the receiver **3010** during some of the assembly steps, also providing a frictional but slidable fit between the insert **3014** and the receiver **3010**. In some embodiments, the insert **3014** may also be freely slidable in the upper portion of the receiver **3010** in an axial direction, but preferably kept above the receiver cylindrical surface **3099** that functions as an expansion recess or chamber for the retainer **3012**.

[0551] Also with reference to FIG. **177**, the retainer **3012** is then prepared for insertion into the receiver **3010** by squeezing or pressing the retainer end surfaces **3134** and **3135** toward one another. The compressed retainer **3012** is inserted into the lower opening **3110** with the planar top surface **3122** facing the receiver bottom surface **3108**. The retainer **3012** is typically moved upwardly into the receiver **3010** and past the cylindrical surface **3106** and allowed to expand to a substantially neutral state within the cylindrical surface **3101** as shown in FIG. **178**. Also as shown in FIG. **178**, at this time, both the compression insert **3014** and the retainer **3012** are captured within the receiver **3010** in a manner that substantially prevents movement or loss of such parts out of the receiver **3010**. The receiver **3010**, compression insert **3014** and the retainer **3012** combination is now pre-assembled and ready for assembly with the shank **3004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **3004** as shown, for example, in FIG. **179**, with the shank axis A and the receiver axis B either being aligned during assembly as shown in FIG. **178** and most of the drawings figures illustrating the assembly process, or the axes being at an angle with respect to one another as shown in FIG. **179**.

[0552] As illustrated in FIG. **179**, the bone screw shank **3004** or an entire assembly **3001** made up of the assembled shank **3004**, receiver **3010**, retainer **3012** and compression insert **3014**, is screwed into a bone, such as the vertebra **3017**, by rotation of the shank **3004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **3006** by engagement thereof at the internal drive **3046**.

[0553] With reference to FIGS. **178** and **179**, the pre-assembled receiver, insert and retainers are placed above the shank upper portion **3008** until the shank upper portion is received within the opening **3110**. With particular reference to FIGS. **180** and **181**, as the shank upper portion **3008** is moved into the interior **3061** of the receiver base, the shank upper portion **3008** presses the retainer **3012** upwardly into the recess partially defined by the cylindrical surface **3099** (if the retainer is not already located within such recess). As the portion **3008** continues to move upwardly toward the channel **3064**, the top surface **3122** of the retainer **3012** abuts against the insert bottom surfaces **3148** as well as the annular rim stop **3098** of the receiver **3010**, stopping upward movement of the retainer **3012** and forcing outward movement of the retainer **3012** towards the cylindrical surface **3099** defining the receiver expansion recess as the spherical surface **3034** continues in an upward direction. The retainer **3012** begins to contract about the spherical surface **3034** as the center of the sphere passes beyond the center of the retainer expansion recess defined by the surface **3099**. At this time also, the spherical surface **3034** moves into engagement with the insert **3014** spherical surface **3152** with the collet panels **3154** expanding slightly outwardly to receive the surface **3034**. The panels **3154** press outwardly against the surface **3096** that provides enough clearance for the spherical surface **3034** to enter into full frictional engagement with the panel inner surfaces **3152** as shown in FIG. **182**. At this time, the insert **3014** and the surface **3034** are in a fairly tight friction fit, the surface **3034** being pivotable with respect to the insert **3014** with some force. Thus, a tight,

non-floppy ball and socket joint is now created between the insert **3014** and the shank upper portion **3008**.

[0554] With reference to FIGS. **183** and **184**, the shank **3004** and attached insert **3014** are then moved downwardly into a desired position for receiving the rod **3021** or other longitudinal connecting member by either an upward pull on the receiver **3010** or, in some cases, by driving the shank **3004** further into the vertebra **3017**. Also, in some embodiments, when the receiver **3010** is pre-assembled with the shank **3004**, the entire assembly **3001** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **3046** and rotating and driving the shank **3004** into a desired location of the vertebra **3017**.

[0555] Also with reference to FIG. **184**, at this time, the compression insert **3014** cylindrical surface **3159** is located within the receiver cylindrical surface **3096** with the insert frusto-conical surface **3158** at or near the surface **3096** at an edge thereof defining a juncture of the surface **3096** and the annular seat **3095**. The insert **3014** is thus prohibited from moving any further downwardly at the ledge or seat **3095** unless forced downwardly by a tool or by the closure top pressing downwardly on the rod that in turn presses downwardly on the insert **3014** in a later stage of assembly as shown in FIG. **185**. With further reference to FIG. **184**, at this time, the receiver **310** may be articulated to a desired angular position with respect to the shank **3004**, that will be held, but not locked, by the frictional engagement between the insert **3014** and the shank upper portion **3008**.

[0556] With reference to FIGS. **185-187**, the rod **21** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **3001** (or combination of **1**, **1001**, **2001** and **3001**, for example). The closure structure **3018** is then inserted into and advanced between the arms **3062** of each of the receivers **3010**. The closure structure **3018** is rotated, using a tool engaged with the inner drive **3166** until a selected pressure is reached at which point the rod **3021** engages the U-shaped seating surface **3142** of the compression insert **3014**, further pressing the insert stepped shank gripping surfaces **3150** against the shank spherical surface **3034**, the edges of the stepped surfaces penetrating into the spherical surface **3034** and also pressing the shank upper portion **3008** into locked frictional engagement with the retainer **3012**. Specifically, as the closure structure **3018** rotates and moves downwardly into the respective receiver **3010**, the point **3169** and rim **3170** engage and penetrate the rod surface **3022**, the closure structure **3018** pressing downwardly against and biasing the rod **3021** into compressive engagement with the insert **3014** that urges the shank upper portion **3008** toward the retainer **3012** and into locking engagement therewith, the retainer **3012** frictionally abutting the surface **3104** and expanding outwardly against the cylindrical surface **3101**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **3006** with respect to the receiver **3010**.

[0557] Also, as the closure structure **3018** and the rod **3021** press the insert **3014** downwardly toward the base of the receiver **3010**, the insert frusto-conical surface **3158** is forced into the receiver cylindrical surface **3096**, wedging the insert **3014** into fixed frictional engagement with the receiver surface **3096**. With reference to FIG. **186**, at this time, the closure top **3018** may be loosened or removed and/or the rod **3021** may be adjusted and/or removed and the frictional engagement between the insert **3014** and the receiver **3010** at the receiver surface **3096** will remain locked in place, advantageously maintaining a locked angular position of the shank **3004** with respect to the receiver **3010**. If the user wishes to release the insert **3014** from the receiver **3010** and unlock the polyaxial mechanism, a tool (not shown) may be used that includes extensions or prongs that are received by and through the opposed through bores **3075** of the receiver **3010** and received into the through bores **3156** of the insert **3014**. Such tool is then pulled upwardly in a direction along the axis B away from the receiver base **3060**, thereby pulling the insert slightly upwardly and away from the receiver base **3060** and releasing the frusto-conical surface **3158** from the cylindrical surface **3096**. Alternatively, if both the closure top **3018** and the rod **3021** are already removed from

the receiver **3010**, another manipulation tool (not shown) may be used that is inserted into the receiver at the opening **3066** and into the insert channel **3141**, with prongs or extensions thereof extending outwardly into the insert through bores **3156**; a piston-like portion of the tool thereafter pushing directly on the shank upper portion **3008**, thereby pulling the insert **3014** surface **3158** away from the receiver surface **3096** and thus releasing the polyaxial mechanism. At such time, the shank **3004** may be articulated with respect to the receiver **3010**, but the desired friction fit remains or returns between the insert **3014** and the shank surface **3034**, so that an adjustable, but non-floppy relationship exists between the shank **3004** and the receiver **3010**. If further disassembly of the assembly **3001** is desired, such is accomplished in reverse order to the procedure described previously herein for assembly.

[0558] With reference to FIGS. **188-221** the reference number **3201** generally represents another embodiment of a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **3201** includes a shank **3204**, that further includes a body **3206** integral with an upwardly extending upper portion or head-like capture structure **3208**; a receiver **3210**; a retainer structure illustrated as a resilient open ring **3212**, and a friction fit crown collet compression or pressure insert **3214**. The receiver **3210**, retainer **3212** and compression insert **3214** are initially assembled and may be further assembled with the shank **3204** either prior or subsequent to implantation of the shank body **3206** into a vertebra **3217**, as will be described in greater detail below. FIGS. **188, 220** and **221** further show a closure structure **3218** for capturing a longitudinal connecting member, for example, a rod **3221** which in turn engages the compression insert **3214** that presses against the shank upper portion **3208** into fixed frictional contact with the retainer **3212**, so as to capture, and fix the longitudinal connecting member **3221** within the receiver **3210** and thus fix the member relative to the vertebra **3217**. The illustrated rod **3221** is substantially similar to the hard, stiff rod **3021** previously described herein, having an outer cylindrical surface **3222**. In other embodiments, the stiff rod **3221** may take other shapes and/or be made from other materials or be part of a longitudinal connecting member assembly that may include sleeves that are fixable to a core member or slidable with respect thereto, spacers (compressible or not) and cords, for example, all as previously described herein with respect to the rods **21, 1021, 2021** and **3021**, for example, and fully incorporated by reference herein with respect to the rod **3221**.

[0559] The receiver **3210** and the shank **3204** cooperate in such a manner that the receiver **3210** and the shank **3204** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **3210** with the shank **3204** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0560] The shank **3204**, best illustrated in FIGS. **188-190**, is the same or substantially similar to the shank **3004** previously described herein. Therefore, the shank **3204** includes the body **3206**, the upper portion or head **3208**, a thread **3224**, a neck **3226**, a shank body top **3232**, an upper portion spherical surface **3234**, an upper portion planar top surface **3238**, an aperture with a stepped base **3245** partially defining an internal drive feature **3246** and a cannulation bore **3250**, all the same or substantially similar to the respective body **3006**, upper portion or head **3008**, thread **3024**, neck **3026**, shank body top **3032**, upper portion spherical surface **3034**, upper portion planar top surface **3038**, aperture with stepped base **3045**, internal drive feature **3046** and cannulation bore **3050** of the shank **3004** previously described herein with respect to the assembly **3001**.

[0561] With particular reference to FIGS. **188** and **202-208**, the receiver **3210** has a generally squared-off, U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **3210** has an axis of rotation BB that is shown in FIG. **188** as being aligned with and the same as the axis of rotation AA of the shank **3204**, such orientation being desirable, but not required during assembly of the receiver **3210** with the shank **3204**. After the receiver **3210** is pivotally attached to the shank **3204**, either before or after the shank **3204** is implanted in a vertebra **3217**, the axis BB is typically disposed at an angle with respect to the axis

AA, as shown, for example, in FIG. 221.

[0562] The receiver **3210** includes a substantially cylindrical base **3260** defining a bore or inner cavity, generally **3261**, the base **3260** being integral with a pair of opposed upstanding arms **3262** forming a cradle and defining a channel **3264** between the arms **3262** with an upper opening, generally **3266**, and a substantially planar lower channel portion or seat **3268**, the channel **3264** having a width for operably snugly receiving the rod **3221** or portion of another longitudinal connector between the arms **3262**, the channel **3264** communicating with the base cavity **3261**. Directly below each channel seat **3268**, the cylindrical base **3260** is cut or truncated, forming opposed planar surfaces **3267**. Outer front and rear opposed substantially planar arm surfaces **3269** partially define the channel **3264** substantially directly above the seat **3268**, the arm surfaces **3269** as well as the base surfaces **3267** advantageously reducing the run on the rod (i.e., providing a more narrow receiver portion that in turn provides more space and thus more access between bone anchors along the rod or other connecting member) and providing planar contact surfaces for flush or close cooperation with other connecting member components in certain embodiments, such as for bumpers or spacers that surround a hard or deformable rod or provide support for elastic or cord-type connecting members. The squared-off geometry of the channel **3264** and lower seat **3268** allow for use with a variety of longitudinal connecting members, including, but not limited to those with circular, square and rectangular cross-sections. As compared to a U-shaped channel that includes a lower seat having a surface with a radius the same or slightly larger than a cooperating cylindrical rod or other connecting member, the squared-off seat **3268** provides improved stress management, moving stress risers outwardly toward the two arms **3262** rather than being focused primarily at a center base line of the radiused lower seat.

[0563] Each of the arms **3262** has an interior surface, generally **3270**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **3272** located adjacent top surfaces **3273** of each of the arms **3262**. In the illustrated embodiment, the guide and advancement structure **3272** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **3218**, the same or similar to the guide and advancement structure **3072** previously described herein with respect to the receiver **3010** of the assembly **3001**. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **3272** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **3218** downward between the arms **3262**, as well as eventual torquing when the closure structure **3218** abuts against the rod **3221** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0564] An opposed pair of substantially circular shallow tool receiving and engaging apertures **3274** are formed on outer surfaces **3276** of the arms **3262**. Two additional pair of tool receiving and engaging apertures **3278** are also formed in the front and rear surfaces **3269** of the receiver arms **3262**. Transition base surfaces **3279** span between the planar surfaces **3269** and the planar seating surface **3268** at either side of the planar base surfaces **3267**. Some or all of the apertures **3274** and **3278** may be used for holding the receiver **3210** during assembly with the insert **3214**, the retainer **3212** and the shank **3204**; during the implantation of the shank body **3206** into a vertebra when the shank is pre-assembled with the receiver **3210**; during assembly of the bone anchor assembly **3201** with the rod **3221** and the closure structure **3218**; and during disassembly of the component parts, when needed. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **3262**.

[0565] Returning to the interior surface **3270** of the receiver arms **3262**, located below the guide and advancement structure **3272** is a discontinuous cylindrical surface **3282** partially defining a run-out feature for the guide and advancement structure **3272**. The cylindrical surface **3282** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure

3272. Moving downwardly, in a direction toward the base 3260, adjacent the cylindrical surface 3282 of each arm is a run-out seat or surface 3284 that extends inwardly toward the axis BB and runs perpendicular to the axis BB. Adjacent to and located below the surface 3284 is another cylindrical surface 3286 having a diameter smaller than the diameter of the surface 3282. Four inner crimping structures, generally 3288, for keeping the insert 3214 in a desired alignment within the receiver 3210 are each cut or otherwise formed in the receiver 3210 below the surface 3286 and extend downwardly into the base cavity 3261. The crimping structures 3288 may be made, for example, by making electrical discharge machining (EDMO) cuts below the surface 3286, creating a pair of opposed discontinuous annular surfaces 3289 and 3290, the surfaces 3290 defining upper surfaces of the structures 3288. Substantially parallel vertical EDM cuts (parallel to the axis BB) located near each channel seat 3268 create outer surfaces 3292 of each of the structures 3288. A discontinuous cylindrical surface 3293 defines a contact surface of each of the crimping structures 3288. The surfaces 3293 have a diameter smaller than the diameter of the run-out surface 3282 but larger than the diameter of the discontinuous cylindrical surface 3286. Each structure 3288 terminates in the base cavity 3261 at a partially annular surface or lower rim 3294 disposed perpendicular to the axis BB. As best shown in FIGS. 206 and 213, the crimping structures 3288 are deployed by bending each structure toward the insert 3214 at the surface 3293 as will be described in greater detail below. It is foreseen that the structures 3288 may have other geometries for cooperating with the insert 3214 and/or other structure, such as spring tabs or thin crimped walls may alternatively be utilized to retain the insert 3214 in a desired position within the receiver 3210.

[0566] Each inner arm surface 3270 further includes a substantially centrally located recess or partial curvate groove, generally 3295, for receiving portions of the insert 3214 in a neutral state thereof as will be described in greater detail below. The recess 3295 is defined by an upper partially annular surface or stop 3296 disposed perpendicular to the axis BB and a curvate or partially cylindrical surface 3297 that cuts into the cylindrical surface 3293. Each recess 3295 generally terminates at or aligned with the discontinuous lower rim 3294.

[0567] The EDM cuts that form the outer surfaces 3292 of the crimping structures 3288 also form planar surfaces 3298 partially defining an upper portion of the cavity 3261 located below and at either side of each channel seat 3268. A pair of opposed centrally located curved or partially cylindrical surfaces 3299 also partially define an upper portion of the cavity 3261, each surface 3299 spanning between planar surfaces 3298. A diameter of each surface 3299 is the same as the un-crimped diameter of the crimping structure 3288 inner cylindrical surfaces 3293 as best illustrated in FIG. 206.

[0568] Further with respect to the base 3260 and more specifically, the base cavity 3261, located below and adjacent to the discontinuous lower rim 3294 is a cylindrical surface 3300 oriented substantially parallel to the axis BB and sized and shaped to receive an expanded retainer 3212. The surfaces 3294 and 3300 define a circumferential recess or chamber that is sized and shaped to receive the retainer 3212 as it expands around the shank upper portion 3208 as the shank 3204 moves upwardly toward the channel 3264 during assembly, the insert 3214 forming a restriction to prevent the neutral or expanding retainer 3212 from moving upwardly with the shank portion 3208. Prior to assembly of the insert 3214 with the receiver 3210, the discontinuous rim 3294 aids in keeping the retainer 3212 within the receiver cavity 3261. A cylindrical surface 3301 located below the cylindrical surface 3300 is sized and shaped to closely receive the retainer 3212 when the retainer is in a neutral or operative position as shown in FIGS. 219 and 220, for example. Thus, the cylindrical surface 3301 has a diameter smaller than the diameter of the cylindrical surface 3300 that defines the expansion area for the retainer 3212. The surface 3301 is joined or connected to the surface 3300 by one or more beveled, curved or conical surfaces 3302. The surfaces 3302 allow for sliding gradual movement and/or contraction of the retainer 3212 into the space defined by the surface 3301 and ultimate seating of the retainer 3212 on a lower annular surface 3304 located

below and adjacent to the cylindrical surface **3301**. Located below and adjacent to the annular seating surface **3304** is another substantially cylindrical surface **3306** that communicates with a beveled or flared bottom opening surface **3307**, the surface **3307** communicating with an exterior base surface **3308** of the base **3260**, defining a lower opening, generally **3310**, into the base cavity **3261** of the receiver **3210**.

[0569] With particular reference to FIGS. **188**, **191-195** and **207**, the open retainer ring **3212** that operates to capture the shank upper portion **3208** and attached compression insert **3214** within the receiver **3210** has a central axis that is operationally the same as the axis BB associated with the receiver **3210** when the shank upper portion **3208** and the retainer **3212** are installed within the receiver **3210**. The retainer ring **3212** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **3212** may be expanded during assembly as will be described in greater detail below. The retainer **3212** has a central channel or hollow through bore, generally **3321**, that passes entirely through the ring **3212** from a top surface **3322** to a bottom surface **3324** thereof. Surfaces that define the channel or bore **3321** include a discontinuous inner cylindrical surface **3325** adjacent the top surface **3322**, a discontinuous frusto-conical surface **3327** adjacent the surface **3325** and a beveled surface **3328**, all three surfaces coaxial when the retainer **3212** is in a neutral, non-expanded orientation. The retainer **3212** further includes an outer cylindrical surface **3330** located adjacent the top surface **3322** and an outer beveled or frusto-conical surface **3332** adjacent the bottom surface **3324**. The surface **3330** is oriented parallel to the central axis of the retainer **3212**. The resilient retainer **3212** further includes first and second end surfaces, **3334** and **3335** disposed in spaced relation to one another (they may also be touching) when the retainer is in a neutral state. Both end surfaces **3334** and **3335** are disposed substantially perpendicular to the top surface **3322** and the bottom surface **3324**. A width XX between the surfaces **3334** and **3335** is very narrow as compared to the width X between the surfaces **3034** and **3035** of the retainer **3012** of the assembly **3001**. Unlike the retainer **3012** and receiver **3010** of the assembly **3001**, the retainer **3212** and the receiver **3210** are sized and shaped for top loading of the insert **3212** into the receiver **3210** which does not require compressing or pinching of the surfaces and **3335** toward one another during the loading step. Therefore, the gap between the surfaces **3334** and **3335** functions only in expansion to allow the retainer **3212** to expand about the shank upper portion **3208** and ultimately against the receiver when finally locked in place. This results in a stronger retainer that provides more surface contact with the shank upper portion **3208**, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **3212** is only expanded and not compressed and expanded like the retainer **3012**, the retainer **3212** does not undergo the mechanical stress that typically is placed on the retainer **3012**.

[0570] With particular reference to FIGS. **188** and **196-201**, the friction fit crown compression insert **3214** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **3210** at the upper opening **3266**. The compression insert **3214** has an operational central axis that is the same as the central axis BB of the receiver **3210**. In operation, the insert **3214** advantageously frictionally engages the bone screw shank upper portion **3208**, allowing for unlocked but non-floppy placement of the angle of the shank **3204** with respect to the receiver **3210** during surgery prior to locking of the shank **3204** with respect to the receiver **3210** near the end of the procedure. The insert **3214** is thus preferably made from a resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be expanded about and then contracted, snapped or popped onto the shank upper portion **3208** as well as pinched and un-wedged from the receiver **3210**.

[0571] The crown collet compression insert **3214** includes a partially cylindrical body **3336** having a planar top surface **3337** and being integral with an opposed pair of crown collet extensions, generally **3338** at a lower end thereof opposite the planar top surface **3337**. Furthermore, extending opposite the extensions **3338** are two upwardly and outwardly extending resilient structures or prongs **3339** that engage the receiver **3210** during certain steps of the assembly process as will be

described in greater detail below. A bore **3340** extends through the body **3336** with the collet extensions **3338** and the prongs **3339** being located generally on opposed sides of the bore **3340**. The prongs **3339** may be formed in a variety of ways. In the illustrated embodiment, the top surface **3337** is initially substantially rectangular and opposed side surfaces taper outwardly (e.g., frusto-conical). EDM cut-outs are made near each of the tapered side surfaces, forming the prong inner curved surfaces **3342** and facing substantially planar body surfaces **3343** as well as a groove or base surface **3344** that spans between each surface **3342** and **3343**. Each prong **3339** further includes an outer tapered or frusto-conical (or otherwise curved) surface **3345** and a top surface **3346**. In a neutral state, the top surface **3346** is level or flush with the top surface **3337** of the body **3336**. It is noted that in other embodiments of the invention, the prongs **3339** may be made into an insert having outer cylindrical surfaces, for example, by making a straight cut into the surface **3337** and then bending each prong **3339** outwardly from the body **3336** into a position wherein the prong top surfaces **3346** are spaced from the body top surface **3337** similar to what is shown in the drawings. However formed, the prongs **3339** are sized, shaped and positioned to extend outwardly from the body **3336** when in a neutral state and to fit within the central recesses **3295** of the receiver **3210** when in a deployed or operating position as shown, for example in FIGS. **219** and **220**, with each prong top surface **3346** located directly beneath a respective receiver surface **3296** and each prong outer surface **3345** engaging or extending outwardly near the receiver surface **3297**. The prongs **3339** have adequate resilience to be pinched or squeezed toward the insert body **3336** at the receiver arm surfaces **3293** when being rotated into an initial position in the receiver **3210** as shown, for example, in FIG. **210** in preparation for assembly with the shank upper portion **3208** as described in greater detail below.

[0572] The bore **3340** is disposed generally centrally through the body **3336** and is further defined by an inner cylindrical surface **3347** that communicates with a lower collet space that extends to discontinuous bottom surfaces **3348** of the collet extensions **3338**. The body **3335** (and bore **3340**) is further defined by a shank gripping surface portion, generally **3350**, the gripping portion **3350** being adjacent to the cylindrical surface **3347**. Located below and adjacent to the gripping portion **3350** is an inner partially spherical surface **3352** that is continuous at the body **3336** and is discontinuous at the extensions **3338** wherein the surface **3352** extends downwardly, defining the inner shank holding portion of each of the collet extensions **3338** and terminating at the extension bottom surfaces **3348**. The gripping surface portion **3350** preferably includes two or more graduated cylindrical surfaces disposed substantially parallel to the axis BB and adjacent perpendicular step surfaces that are disposed generally perpendicular to the axis BB when the insert **3214** is mounted within the receiver **3210**. It is foreseen that the stepped surface portion **3350** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping surface portion **3350** and also the spherical surface **3352** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **3208**. The two collet extensions **3338** that generally extend in a direction opposite to the prongs **3339** and have the discontinuous inner spherical surface **3352**, also include through slits or slots **3353** running substantially vertically from adjacent the shank gripping surface portion **3350** through the bottom surfaces **3348**. The illustrated embodiment includes two substantially equally spaced slots **3353** located in each extension **3338**. It is foreseen that other embodiments of the invention may include more or fewer slots **3353**. The slots **3353** substantially equally partition each of the extensions **3338**, forming a total of six distinct resilient, partially spherical fingers, tab or panels **3354** that extend from the shank gripping portion **3350** to the bottom surface **3348**. In other words, the discontinuous inner spherical surface **3352** is further separated into two opposed pairs of three surface portions **3354** each, with each portion **3354** being partially spherical and sized and shaped to resiliently expand about the spherical surface **3234** of the shank upper portion **3208** and then snap on and frictionally grip the surface **3234**. Preferably, the spherical surface **3352** is designed such that the gripping tabs

or panels **3354** have a neutral or non-expanded radius that is slightly smaller than a radius of the shank surface **3234** so that when the tabs or panels **3354** are gripping the surface **3234**, the insert **3214** collet extension portion **3338** is in a slightly expanded state. When the shank **3204** is locked into position by a rod **3221** or other connecting member being pressed downwardly on the insert seat **3342** by the closure top **3218**, the insert **3214** shank gripping portion **3350** that is initially slidable along the shank surface **3234** then digs or penetrates into the surface **3234** and thus securely fixes the shank upper portion **3208** to the insert at the portion **3350**.

[0573] The compression insert **3214** through bore **3340** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **3246** when the shank body **3206** is driven into bone with the receiver **3210** attached. The illustrated insert **3214** further includes surface features located primarily on the insert body **3336** that cooperate with the receiver crimping structures **3288**. Specifically, the insert body **3336** includes a centrally located substantially cylindrical portion **3356** located about and co-axial with the bore **3340**. The portion **3356** includes opposed cylindrical surfaces **3357** that run from the top surface **3337** to a lower or bottom surface **3358** located substantially centrally between the collet extensions **3338**. Located on either side of each surface **3357** is a curvate transition surface **3359** followed by a substantially planar surface **3360** that extends substantially to the body surface **3343** and is substantially perpendicular thereto. Each surface **3359** runs from the planar top surface **3337** to the respective bottom surface **3358**. The portion of the bottom surface **3358** that is adjacent to the surface **3359** curves downwardly in a direction toward the respective collet extension bottom surface **3348**. Each surface **3360** runs from the planar top surface **3337** and along each crown collet extension **3338** to bottom surfaces **3348** thereof. With particular reference to FIGS. **211-213** and as will be described in greater detail below, the four inner crimping structures **3288** of the receive **3210** are crimped or bent to a location at or near the four surfaces **3360** of the insert **3214**, the structures **3288** being crimped or bent toward the insert **3214** spanning from a location near the insert top surface **3337** to a location near the collet bottom surfaces **3348**, substantially limiting or prohibiting rotation of the insert **3214** with respect to the receiver **3210** about the axis BB (only a 2.5 degree collet rotation possible in the illustrated embodiment). However, the crimping structures **3288** advantageously allow for up and down movement of the insert **3214** with respect to the receiver **3210** along the axis BB.

[0574] The opposed cylindrical insert body surfaces **3357** have an outer diameter slightly smaller than a diameter between crests of the guide and advancement structure **3272** of the receiver **3210**, allowing for top loading of the compression insert **3214** into the receiver opening **3266**, with the collet extensions **3338** and the prongs **3339** of the insert **3214** being located between the receiver arms **3262** during insertion of the insert **3214** into the receiver **3210**. Once the upper prongs **3339** the insert **3214** are generally located below the discontinuous annular surface **3289**, the insert **3214** is rotated into place about the receiver axis BB until the prong top surfaces **3346** are located directly beneath the surfaces **3289**, the prong outer surfaces **3345** engaging the receiver discontinuous cylindrical surface **3293** during rotation of the insert **3214**, the prongs **3339** being pressed inwardly toward the axis BB as will be described in greater detail below. The frictional engagement between the prong surfaces **3345** and the receiver arm surfaces **3293** advantageously maintains the insert **3214** in an upper portion of the receiver cavity **3261** prior to and during assembly with the shank **3204**.

[0575] The closure top **3218** illustrated in FIGS. **188**, **220** and **221** is the same or substantially similar to the closure top **3018** previously described herein. Therefore, the closure top **3218** includes a guide and advancement structure **3362**, a top surface **3364**, an internal drive feature **3366**, and a bottom surface **3368** further having a point **3369** and a rim **3370** the same or similar to the respective guide and advancement structure **3162**, top surface **3164**, internal drive feature **3166**, and bottom surface **3168** with point **3169** and rim **3170** of the closure top **3018** of the assembly **3001** previously described herein.

[0576] Preferably, the receiver **3210**, the retainer **3212** and the compression insert **3214** are

assembled at a factory setting that includes tooling for holding, alignment, compression and expansion of the component pieces, if needed, as well as crimping the structures **3288** of the receiver **3210** toward the insert **3214**. In some circumstances, the shank **3204** is also assembled with the receiver **3210**, the retainer **3212** and the compression insert **3214** at the factory. In other instances, it is desirable to first implant the shank **3204**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **3204**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **3208** and/or hydroxyapatite on the shank **3206**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **3204** advantageously reduces inventory requirements, thus reducing overall cost.

[0577] Pre-assembly of the receiver **3210**, retainer **3212** and compression insert **3214** is shown in FIGS. **207-214**. With particular reference to FIGS. **207** and **208**, first the retainer **3212** is inserted into the upper receiver opening **3266** with the planar top surface **3322** facing one of the receiver guide and advancement structures **3272** (not shown), the retainer **3212** lowered into the channel **3264** and partially into the receiver cavity **3261**, followed by turning the retainer **3212** such that the top surface **3322** is moved into a position within the cavity facing upwardly toward the receiver channel opening **3266**. The retainer **3212** may then be pressed downwardly into a lower portion of the receiver cavity **3261**, preferably to a position wherein the retainer **3212** bottom surface **3324** engages the receiver annular surface **3304**, the retainer **3212** being slightly compressed with the outer cylindrical surface **3330** frictionally engaging the receiver cylindrical surface **3301**.

[0578] With particular reference to FIGS. **208-210**, the compression insert **3214** is downloaded into the receiver **3210** through the upper opening **3266** with the crown collet extension bottom surfaces **3348** facing the receiver arm top surfaces **3273** and the insert upper prongs **3339** as well as the insert collet extensions **3338** being located between the opposed receiver arms **3262**. The insert **3214** is then lowered toward the receiver base **3260** until the insert **3214** body top surface **3337** is substantially adjacent to and located slightly below the receiver arm surfaces **3289** that are located directly below the arm cylindrical surfaces **3286**. Thereafter, the insert **3214** is rotated in a clockwise or counter-clockwise manner about the receiver axis BB until the prong upper surfaces **3346** are each directly below the surfaces **3289**. As the insert **3214** is rotated, the prongs **3339** are squeezed toward one another so that each prong outer surface **3345** slidably frictionally engages the receiver surfaces **3293**. With reference to FIG. **210**, the insert **3213** is rotated until the prongs **3339** and the collet extensions **3338** are centrally located beneath the surfaces **3289** and centrally aligned with each of the receiver arms **3262**. With particular reference to FIGS. **211-213**, each of the four receiver crimping structures **3288** are then crimped or bent towards the insert surfaces **3360**, limiting any further rotation of the insert **3214** with respect to the receiver **3210** about the axis BB to no more than a few degrees. At this time, the surfaces **3289** prevent upward movement of the insert **3214**, but with some force, the insert **3214** may be moved downwardly toward the receiver base **3260**. However, it is desirable at this time to keep the insert **3214** wedged at the arm surfaces **3293** and the retainer **3212** engaged with the cavity surface **3301**.

[0579] The receiver **3210**, compression insert **3214** and the retainer **3212** combination is now pre-assembled and ready for assembly with the shank **3204** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **3204**, with the shank axis AA and the receiver axis BB either being aligned during assembly as shown in FIG. **214** and most of the drawings figures illustrating the assembly process, or the axes being at an angle with respect to one another as shown, for example, in FIG. **221**.

[0580] The bone screw shank **3204** or an entire assembly **3201** made up of the assembled shank **3204**, receiver **3210**, retainer **3212** and compression insert **3214**, is screwed into a bone, such as the

vertebra **3217**, by rotation of the shank **3204** using a suitable driving tool (not shown) that operably drives and rotates the shank body **3206** by engagement thereof at the internal drive **3246**. Specifically, the vertebra **3217** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **3204** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank **3204** or the entire assembly **3201** is threaded onto the guide wire utilizing the cannulation bore **3250** by first threading the wire into the opening at the bottom **3228** and then out of the top opening at the drive feature **3246**. The shank **3204** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **3221** (also having a central lumen in some embodiments) and the closure top **3218** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires. When the shank **3204** is driven into the vertebra **3217** without the remainder of the assembly **3201**, the shank **3204** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer.

[0581] With reference to FIGS. **214-219**, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **3208** until the shank upper portion is received within the opening **3310**. With particular reference to FIG. **215**, as the shank upper portion **3208** is moved into the interior **3261** of the receiver base, the shank upper portion **3208** presses the retainer **3212** upwardly into the recess partially defined by the cylindrical surface **3300** (if the retainer is not already located within such recess). With particular reference to FIG. **216**, as the portion **3208** continues to move upwardly toward the channel **3264**, the top surface **3322** of the retainer **3212** abuts against the insert bottom surfaces **3348**, stopping upward movement of the retainer **3212** and forcing outward movement of the retainer **3212** towards the cylindrical surface **3300** as the shank spherical surface **3234** continues in an upward direction. With further reference to FIG. **216**, the retainer **3212** contracts about the spherical surface **3234** as the center of the sphere passes beyond the center of the retainer expansion recess. At this time also, the spherical surface **3234** moves into engagement with the insert **3214** spherical surface **3352** with the collet panels **3354** expanding slightly outwardly to receive the surface **3234**. The spherical surface **3352** then enters frictional engagement with the panel inner surfaces **3352** as shown in FIG. **217**. At this time, the insert **3214** and the surface **3234** are in a fairly tight friction fit, the surface **3234** being pivotable with respect to the insert **3214** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the insert **3214** and the shank upper portion **3208**. At this time the retainer **3212** has returned to a neutral position and is typically located within the receiver transition surface or surfaces **3302**.

[0582] With reference to FIGS. **218** and **219**, the shank upper portion **3208** and attached insert **3214** are then moved downwardly into a desired position for receiving the rod **3221** or other longitudinal connecting member by either an upward pull on the receiver **3210** or, in some cases, by driving the shank **3204** further into the vertebra **3217**. Also with reference to FIGS. **217-219**, as the shank **3204** is moved downwardly toward the receiver base **3260**, the insert prongs **3339** slide along the surfaces **3293** until top surfaces **3346** thereof clear the surfaces **3293**, allowing the prongs **3330** to snap outwardly to a neutral position located below the surfaces **3293** and within the central recesses **3295** of each of the receiver arms **3262**. The prong top surfaces **3346** now being located beneath surfaces **3296** of the central recesses **3295**, thus retaining the insert **3214** in a desired location within the receiver cavity **3261**, the shank upper portion **3208** pressing downwardly on the retainer **3212** and the retainer seated on the receiver surface **3304**. In some embodiments, when the receiver **3210** is pre-assembled with the shank **3204**, the entire assembly **3201** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **3246** and rotating and driving the shank **3204** into a desired location of the vertebra **3217**.

[0583] With reference to FIG. **220**, the rod **3221** is eventually positioned in an open or

percutaneous manner in cooperation with the at least two bone screw assemblies **3201** (or with other bone screws of the invention). The closure structure **3218** is then inserted into and advanced between the arms **3262** of each of the receivers **3210**. The closure structure **3218** is rotated, using a tool engaged with the inner drive **3366** until a selected pressure is reached at which point the rod **3221** engages the planar surface **3337** of the compression insert **3214**, further pressing the insert stepped shank gripping surfaces **3350** against the shank top **3238** and/or spherical surface **3234**, the edges of the stepped surfaces penetrating into the spherical surface **3234** and also pressing the shank upper portion **3208** into locked frictional engagement with the retainer **3212**. Specifically, as the closure structure **3218** rotates and moves downwardly into the respective receiver **3210**, the point **3369** and rim **3370** engage and penetrate the rod surface **3222**, the closure structure **3218** pressing downwardly against and biasing the rod **3221** into compressive engagement with the insert **3214** that urges the shank upper portion **3208** toward the retainer **3212** and into locking engagement therewith, the retainer **3212** frictionally abutting the surface **3304** and expanding outwardly against the cylindrical surface **3301**. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **3206** with respect to the receiver **3210**.

[0584] With reference to FIGS. **222-234** the reference number **3401** generally represents another embodiment of a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **3401** includes a shank **3404**, that further includes a body **3406** integral with an upwardly extending upper portion or head-like capture structure **3408**; a receiver **3410**; a retainer structure illustrated as a resilient open ring **3412**, and a lock and release friction fit compression or pressure insert **3414**. The receiver **3410**, retainer **3412** and compression insert **3414** are initially assembled and may be further assembled with the shank **3404** either prior or subsequent to implantation of the shank body **3406** into a vertebra, such as the vertebra **3017** or **3217** previously shown herein. FIG. **222** also shows a closure structure **3418** for capturing a longitudinal connecting member, for example, a rod **3421** which in turn engages the compression insert **3414** that presses against the shank upper portion **3408** into fixed frictional contact with the retainer **3412**, so as to capture, and fix the longitudinal connecting member **3421** within the receiver **3410** and thus fix the member **3421** relative to the vertebra. The illustrated rod **3421** is substantially similar to the hard, stiff rod **3021** previously described herein, having an outer cylindrical surface. In other embodiments, the stiff rod **3421** may take other shapes and/or be made from other materials or be part of a longitudinal connecting member assembly that may include rigid sleeves that are fixable to a core member or slidable with respect thereto, spacers (compressible or not) and cords, for example, all as previously described herein with respect to the rods **21**, **1021**, **2021** and **3021**, for example, and fully incorporated by reference herein with respect to the rod **3421**.

[0585] The receiver **3410** and the shank **3404** cooperate in such a manner that the receiver **3410** and the shank **3404** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **3410** with the shank **3404** until both are locked or fixed relative to each other near the end of an implantation procedure. [0586] The shank **3404** is identical or substantially similar to the shank **3004** previously described herein. The receiver **3410** is identical or substantially similar to the receiver **3010** previously described herein, thus having an inner bore or cavity **3461** with an insert locking inner cylindrical surface **3496** and inner arm surfaces **3470**, as well as other features that are the same or similar to the cavity **3061**, cylindrical surface **3096** and inner arm surfaces **3070** previously described herein with respect to the receiver **3010**.

[0587] The retainer **3412** is the same or substantially the same as the top loading retainer **3212** previously described herein. Thus, with reference to FIG. **223**, the retainer **3412** is open, but includes a very narrow slit, providing a strong retainer that is not compressed during assembly with the receiver because it is top loaded and is only expanded about the shank head **3408** during

assembly as shown in FIGS. 226 and 227 and also expands slightly within the receiver upon final locking with the shank 3404 as shown, for example in FIG. 228.

[0588] The insert 3414 is substantially similar or identical to the insert 3014 previously described herein with respect to the assembly 3001. Thus, the insert 3414 includes a pair of upstanding arms 3537 and a pair of crown collet extensions 3538 the same or similar to the respective arms 3137 and extensions 3138 of the insert 3014 previously described herein. The insert 3414 extension 3538 further include panels 3554 having inner spherical gripping surfaces as well as an outer frusto-conical surface 3558 for frictional locking with the inner cylindrical surface 3496 of the receiver 3410, such features being the same or similar to the respective panels 3154, with spherical surfaces 3152 and the outer frusto-conical surface 3158 of the insert 3014. The inner spherical surfaces 3552 preferably have a pre-assembly radius that is the same or slightly larger than a radius of the shank upper portion or head 3408. As discussed above with respect to the insert 3014, the insert 3414 and/or the shank head 3408 may include surface treatments or the panels may be crimped or bent to result in a desired frictional fit between the insert 3414 and the shank head 3408 during the temporary steps of manipulating the bone screws 3401 during assembly with the rod. As with the assembly 3001, the final locking of the assembly 3401 is accomplished by frictional contact between the bone screw shank upper portion 3408 and the retainer ring 3412 when the insert 3414 presses down upon the shank upper portion 3408.

[0589] Thus, the assembly 3401 includes features of both the assembly 3001 and the assembly 3201 to result in a polyaxial assembly wherein the shank 3404 may be snapped or popped onto the receiver 3410 either before or after implantation of the shank 3404 into a vertebra. As shown in FIG. 223, the retainer 3412 is initially top loaded into the receiver 3410 followed by top loading of the insert 3414 with the insert arms 3437 located between arms of the receiver 3410. With reference to FIGS. 224 and 225, the insert 3414 is rotated into place. With reference to FIGS. 226 and 227, the shank 3404 is bottom loaded by pressing the shank head or upper portion 3408 through the open retainer 3412 within an expansion area of the receiver 3410 followed by non-floppy frictional engagement between the shank upper portion 3408 and the flexible tabs or panels 3554. FIGS. 228-230 illustrate the locking of the insert 3414 frusto-conical surface 3558 against the receiver surface 3496, the same or similar to what was described above with respect to the assembly 3001. Thus, as shown in FIG. 229, the closure top 3418 and/or the hard rod 3421 may be removed without unlocking the polyaxial mechanism of the screw 3401. With reference to FIG. 230, a deformable rod 3421', such as a PEEK rod with a cooperating alternative closure top 3418' may be placed in the receiver 3410 and secured therein. The deformable rod 3421' does not compromise the secure lock of a desired angle between the shank 3404 and the receiver 3410 provided by the wedging of the insert 3414 into the receiver 3410 at the respective surfaces 3496 and 3558.

[0590] With reference to FIGS. 231-233, an alternative insert 3414' for use in the assembly 3001 or the assembly 3401 in lieu of the insert 3014 or the insert 3414 is shown. The insert 3414' is substantially similar or identical to the inserts 3014 and 3414 previously described above with the exception that the substantially curved or spherical surfaces 3152 or 3552 and been replaced by planar surfaces 3552'. Thus, the insert 3414' includes four planar surfaces 3552' that frictionally grip the shank upper portion 3008 or the shank upper portion 3408 to provide a friction fit between the respective shank and the insert 3414'. It is foreseen that more or fewer planar surfaces may be included to grip the shank upper portion 3008 or the shank upper portion 3408.

[0591] With reference to FIG. 234, another alternative non-locking insert 3414'' is shown that is identical to the insert 3414' with the exception that a frusto-conical outer surface 3558' of the insert 3414' has been replaced by a cylindrical surface 3558''. Thus, when used with the receiver 3410, for example, the cylindrical surface 3558' is slidably received by the inner cylindrical surface 3496 and does not wedge or lock into the receiver 3410.

[0592] With reference to FIGS. 235-244 an alternative receiver 4010, open retainer 4012 and

compression insert **4014** are shown that may be used with the shank **3404**, rod **3421** and closure top **3418** previously described herein with respect to the assembly **3401**. Furthermore, the open, top loadable retainer **4012** is identical to the retainer **3412** of the assembly **3401**. The alternative receiver **4010** and cooperating insert **4014** differ only slightly from the receiver **3410** and cooperating insert **3414** of the assembly **3401**. The receiver **4010** includes features of both the receiver **3410** and the receiver **1210** of the assembly **1201** previously described herein. In particular, the receiver **4010** has spring tabs **4290** with insert engaging surfaces **4311** substantially similar to the respective spring tabs **1290** with inner engaging surfaces **1311** of the receiver **1210**. However, the receiver **4010** also includes an inner cylindrical surface **4096** located at a base of the spring tabs **1290**, the surface **4096** sized and shaped to frictionally engage and lock a tapered or frusto-conical surface **4158** of the insert **4014** so that the insert **4014** locks against the receiver, substantially similar to the cooperation between the insert **3414** and the receiver **3410** previously described herein. The insert further includes grooves **4159** in the arms thereof for receiving the receiver spring tabs **4290** at the surfaces **4311**. As shown in FIGS. **238** and **239**, when the insert **4014** is dropped down into the receiver **4010** and rotated into position, the spring tabs **4290** are pushed outward and away from the insert **4014**. However, once the insert **4014** completes its rotation when the insert U-shaped channel is aligned with the receiver U-shaped channel, the spring tabs **4290** snap into grooves **4159** of the insert **4014**, capturing the insert **4014** in the receiver **4010**, keeping the insert in alignment with the receiver and allowing only some upward and downward movement of the insert with respect to the receiver **4010**. The tabs **4290** frictionally retain the insert in an upper portion of the receiver during the “pop-on” attachment to the bone screw shank **3404**. As shown in FIG. **241**, a downward force upon the insert **4014**, such as by the rod **3421** and closure top **3418** causes the insert tapered surface **4158** to wedge up against the cylindrical surface **4096** of the receiver **4010**, locking the polyaxial mechanism.

[0593] With reference to FIGS. **242-244** a two-piece tool **4600** is illustrated for releasing the insert **4014** from the receiver **4010**. The tool **4600** includes an inner flexible tube-like structure with opposed inwardly facing prongs **4312** located on either side of a through-channel **4616**. The channel **4616** may terminate at a location spaced from the prongs **4312** or may extend further upwardly through the tool, resulting in a two-piece tool **4610**. The tool **4600** includes an outer, more rigid tubular member **4620** having a smaller through channel **4622**. The member **4620** slidably fits over the tube **4610** after the flexible member **4610** prongs **4612** are fitted within opposed apertures **4074** of the receiver **4010** and aligned opposed apertures **4156** located on arms of the insert **4014**. In FIG. **242**, the tool **4600** is shown having unlocked the insert **4014** from the receiver **4010** with the outer member **4620** surrounding the inner member **4610** and holding the prongs **4612** within the receiver and insert apertures while the tool **4600** is pulled upwardly away from the shank **3404**. It is foreseen that another tube within a tube type tool may be used for locking the lower pressure insert downward into the receiver **4010** wherein prongs of an inner flexible tubular member that are larger than the prongs **4612** drive the insert **4014** downwardly into locking engagement with the receiver **4010** as the prongs enter the larger receiver apertures **4074** and then the insert apertures **4156**.

[0594] With reference to FIGS. **245-283** the reference number **5001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **5001** includes a shank **5004**, that further includes a body **5006** integral with an upwardly extending upper portion or head structure **5008**; a receiver **5010**; a friction fit retainer **5012**, and a crown-like compression or pressure insert **5014**. The receiver **5010**, retainer **5012** and compression insert **5014** are initially assembled and may be further assembled with the shank **5004** either prior or subsequent to implantation of the shank body **5006** into a vertebra **5017**, as will be described in greater detail below. FIGS. **1** and **278-280** further show a closure structure **5018** for capturing a longitudinal connecting member, for example, a rod **5021** which in turn engages the compression insert **5014** that presses against the shank upper portion **5008** into fixed frictional contact with the

retainer **5012**, so as to capture, and fix the longitudinal connecting member **5021** within the receiver **5010** and thus fix the member **5021** relative to the vertebra **5017**. The closure top **5018** and the rod **2010** are identical or substantially similar to many of the closure tops and rods previously described herein, for example, the closure top **4018** and the rod **4021** of the assembly **4001** having the same form and function and therefore shall not be discussed any further here.

[0595] The shank **5004**, best illustrated in FIGS. **245-247**, is substantially similar to the shank **3004** previously described herein with respect to the assembly **3001**. Thus, the shank **5004** includes the shank body **5006**, upper portion or head **5008**, a shank thread **5024**, a neck **5026**, a tip **5028**, a top of thread **5032**, an upper portion spherical surface **5034** a top surface **5038**, an internal drive **5046** with a base surface **5045** and an cannulation bore **5050** the same or substantially similar to the respective body **3006**, upper portion or head **3008**, shank thread **3024**, neck **3026**, tip **3028**, top of thread **3032**, spherical surface **3034**, top surface **3038**, internal drive **3046** with base surface **3045** and cannulation bore **3050** previously described herein with respect to the shank **3004** of the assembly **3001**.

[0596] To provide a biologically active interface with the bone, the threaded shank body **5006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**. In the illustrated embodiment, a frusto-conical surface **5039** extends from the spherical surface **5034** to the top surface **5038**, providing additional clearance during shank angulation as best shown in FIG. **279**. The spherical surface **5034** has an outer radius configured for temporary frictional, non-floppy, sliding cooperation with panels of the retainer **5012** having concave or flat surfaces, as well as ultimate frictional engagement with the insert **5014** at an inner partially spherical surface thereof, as will be discussed more fully in the paragraphs below. The top surface **5038** is substantially perpendicular to the axis A. The spherical surface **5034** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and ultimate frictional engagement with the compression insert **5014** as well as ultimate frictional engagement with a lower ring-like portion of the retainer **5012**. The shank spherical surface **5034** is locked into place exclusively by the insert **5014** and the retainer **5012** lower portion and not by inner surfaces defining the receiver cavity. The illustrated internal drive feature **5046** differs from the feature **3046** of the shank **3004** in that the feature **5046** is an aperture formed in the top surface **5038** that has a star shape designed to receive a tool (not shown) of an Allen wrench type, into the aperture for rotating and driving the bone screw shank **5004**. As illustrated in FIGS. **246** and **247**, the drive seat **5045** may include beveled or stepped surfaces that may further enhance gripping with the driving tool. In operation, a driving tool (not shown) is received in the internal drive feature **5046**, being seated at the base **5045** and engaging the faces of the drive feature **5046** for both driving and rotating the shank body **5006** into the vertebra **5017**, either before the shank **5004** is attached to the receiver **5010** or after the shank **5004** is attached to the receiver **5010**, with the shank body **5006** being driven into the vertebra **5017** with the driving tool extending into the receiver **5010**.

[0597] With particular reference to FIGS. **245** and **248-253**, the receiver **5010** has a generally U-shaped appearance with partially discontinuous and partially cylindrical inner and outer profiles. The receiver **5010** has an axis of rotation B that is shown in FIG. **245** as being aligned with and the same as an axis of rotation A of the shank **5004**, such orientation being desirable, but not required during assembly of the receiver **5010** with the shank **5004**. After the receiver **5010** is pivotally attached to the shank **5004**, either before or after the shank **5004** is implanted in a vertebra **5017**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **279**.

[0598] The receiver **5010** includes a substantially cylindrical base **5060** defining a bore or inner cavity, generally **5061**, the base **5060** being integral with a pair of opposed upstanding arms **5062** forming a cradle and defining a channel **5064** between the arms **5062** with an upper opening, generally **5066**, and a U-shaped lower channel portion or seat **5068**, the channel **5064** having a

width for operably snugly receiving the rod **5021** or portion of another longitudinal connector between the arms **5062**, the channel **5064** communicating with the base cavity **5061**. Inner opposed substantially planar arm surfaces **5069** partially define the channel **5064** directly above the seat **5068** and are located on either side of each arm interior surface generally **5070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **5072** located adjacent top surfaces **5073** of each of the arms **62**. In the illustrated embodiment, the guide and advancement structure **5072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **5018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **5072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-thread-like helically wound discontinuous advancement structures, for operably guiding under rotation and advancing the closure structure **5018** downward between the arms **5062**, as well as eventual torquing when the closure structure **5018** abuts against the rod **5021** or other longitudinal connecting member. It is foreseen that the arms **5062** could have break-off extensions.

[0599] An opposed pair of key-hole like tool receiving and engaging grooves or apertures, generally **5074**, each having an upper arched through bore **5075**, are formed on outer surfaces **5076** of the arms **5062**. Each through bore **5075** extends between the outer surface **5076** and the inner surface **5070** and is located above a rectangular shaped shallow recessed arm portion or crimp wall **5077** that defines the portion of the aperture **5074** that does not extend completely through the respective arm **5062**. The thin walled portion **5077** is pressed or crimped into the insert **5014** to prohibit rotation and misalignment of the insert **5014** with respect to the receiver **5010** as will be described in greater detail below. In other embodiments of the invention, other surfaces forming the groove or aperture **5074** may be inwardly crimped. Alternatively, spring tabs or other movable structure may be included on the receiver **5010** or the insert **5014** for retaining the insert **5014** in a desired position, with regard to rotation and axial movement (along the axis A) with respect to the receiver **5010**. Preferably the insert and/or receiver are configured with structure for blocking rotation of the insert with respect to the receiver, but allowing some up and down movement of the insert with respect to the receiver during the assembly and implant procedure. Two additional rectangular shaped through bores **5078** are also formed in the arms **5062** and located directly below the apertures **5074**. The through bores **5078** are sized and shaped for receiving portions of the retainer **5012** during top loading of the retainer **5012** into the receiver **5010** as will be described more fully below and as shown, for example, in FIG. 266. An upper surface **5079** defining each bore **5078** functions as an upper stop for a portion of the retainer **5012**, during shipping and during assembly, as shown, for example, in FIG. 272, and as will be described in greater detail below. Also formed in each outer arm surface **5076** near the top surface **5073** is an undercut tool receiving and engaging groove **5081**. Some or all of the apertures **5074** and **5078** and the groove **5081** may be used for holding the receiver **5010** during assembly with the insert **5014**, the retainer **5012** and the shank **5004**; during the implantation of the shank body **5006** into a vertebra when the shank is pre-assembled with the receiver **5010**; during assembly of the bone anchor assembly **5001** with the rod **5021** and the closure structure **5018**; and during lock and release adjustment of the insert **5014** with respect to the receiver **5010**, either into or out of frictional engagement with the inner surfaces of the receiver **5010** as will be described in greater detail below. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **5062**.

[0600] Returning to the interior surface **5070** of the receiver arms **5062**, located below the guide and advancement structure **5072** is a discontinuous cylindrical surface **5082** partially defining a run-out feature for the guide and advancement structure **5072**. The cylindrical surface **5082** has a diameter equal to or slightly greater than a greater diameter of the guide and advancement structure **5072**. Moving downwardly, in a direction toward the base **5060**, following the cylindrical surface

5082 of each arm is a cylindrical surface **5084** partially defined by a run-out seat or surface **5085** that extends inwardly toward the axis B and runs perpendicular to the axis B. The surface **5084** has a diameter smaller than the diameter of the surface **5082**. The surface **5084** is sized and shaped to initially closely receive a lower portion of the insert **5014** and later frictionally engage a tapered or frusto-conical upper portion of the insert **5014**, providing a lock and release function that will be described in greater detail below. A discontinuous annular surface **5086** is located below and adjacent to the surface **5084**. The surface **5086** is substantially perpendicular to the axis B. Another discontinuous cylindrical surface **5088** is located below and adjacent to the surface **5086**. The surface **5088** has a diameter slightly larger than the diameter of the surface **5084**. A discontinuous annular surface or narrow ledge **5089** is located below the surface **5088** and is substantially perpendicular to the axis B. A partially discontinuous cylindrical surface **5090** is located one each arm below and adjacent to the surface **5089**. The surface **5090** also defines an upper cylindrical surface of the base cavity **5061**. The surface **5090** has a diameter slightly smaller than the diameter of the surface **5088** but larger than the diameter of the surface **5084**. It is noted that in some embodiments of the invention, the surfaces **5088** and **5090** are combined and form a single smooth cylindrical surface.

[0601] The through bores **5075** each extend through the arms at the surfaces **5082**, **5084** and **5088**. The crimping wall **5077** is located in an inner recessed surface area **5092** that is formed in both the surfaces **5088** and **5090**. In the illustrated embodiment, the crimping wall **5077** has an inner surface **5093** that is primarily located at the portion of the area **5092** that is formed in the cylindrical surface **5088**. Each through bore **5078** is located directly below the area **5092**.

[0602] An annular surface **5098** partially defining the base cavity **5061** is located below and adjacent to the cylindrical surface **5090**. The surface **5098** is disposed substantially perpendicular to the axis B. Another cylindrical surface **5099** is located below and adjacent to the surface **5098**. The cylindrical surface **5099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded portion of retainer **5012**. The surfaces **5098** and **5099** define a circumferential recess or expansion chamber that is sized and shaped to receive the retainer **5012** as it expands around the shank upper portion **5008** as the shank **5008** moves upwardly toward the channel **5064** during assembly. A cylindrical surface **5101** located below the cylindrical surface **5099** is sized and shaped to closely receive a lower portion of the retainer **5012** when the retainer is in a substantially neutral position as shown in FIG. 267, for example. Thus, the cylindrical surface **5101** has a diameter smaller than the diameter of the cylindrical surface **5099** that defines the expansion area for the retainer **5012**. The surface **5101** is joined or connected to the surface **5099** by one or more beveled, curved or conical surfaces **5102**. The surfaces **5102** allow for sliding gradual movement and/or contraction of the retainer **5012** into the space defined by the surface **5101** and ultimate seating of the retainer **5012** on a lower annular surface **5104** located below and adjacent to the cylindrical surface **5101**.

[0603] Located below and adjacent to the annular seating surface **5104** is another substantially cylindrical surface **5106** that communicates with a beveled or flared bottom opening surface **5107**, the surface **5107** communicating with an exterior base surface **5108** of the base **5060**, defining a lower opening, generally **5110**, into the base cavity **5061** of the receiver **5010**.

[0604] With particular reference to FIGS. 245 and 254-259, the lower open friction fit retainer **5012** that operates to capture the shank upper portion **5008** and attached compression insert **5014** within the receiver **5010** has a central axis that is operationally the same as the axis B associated with the receiver **5010** when the shank upper portion **5008** and the retainer **5012** are installed within the receiver **5010**. The retainer **5012** includes a substantially cylindrical discontinuous lower body **5116**, a plurality of flex fingers or panels, **5117** extending upwardly from the body **5116** and a pair of opposed spring arms or tabs **5118**, also extending upwardly from the body **5116**. The retainer ring **5012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **5012** body **5116** may be expanded and the fingers and tabs (**5117** and **5118**) of the

retainer may be manipulated during various steps of assembly as will be described in greater detail below. The retainer **5012** has a central channel or hollow through bore, generally **5121**, that passes entirely through the retainer **5012** from tab **5118** top surfaces **5122** to a bottom surface **5124** of the retainer body **5116**. Surfaces that define the channel or bore **5121** include an inner lower frusto-conical surface **5128** adjacent to the retainer body bottom surface **5124**, a substantially cylindrical surface **5130** adjacent the frusto-conical surface **5128**, a narrow frusto-conical or beveled surface **5131** adjacent the cylindrical surface **5130** and a partially continuous partially discontinuous substantially spherical surface **5132** adjacent the surface **5131**, the surface **5132** being substantially continuous near the cylindrical surface **5130** with the exception of the opposed spring tabs **5118** and a through slot or slit, generally **5134**. The surface **5132** is in a plurality of segments or pieces at the flex fingers **5117** wherein a plurality of substantially evenly spaced slots **5136** running outwardly and upwardly through an upper surface **5137** separate the surface **5132** into the individual flex fingers **5117**. In the illustrated embodiment, the slots **5136** and the through slit **5134** form the six substantially uniform flex fingers or tabs **5117** as well as partially define the two spring tabs **5118**, each finger having the inner spherical surface **5132**. It is foreseen that more or fewer flex fingers may be made by the forming of more or fewer slots **5136**. The discontinuous spherical surface **5132** is sized and shaped to closely fit about and snap onto the shank surface **5034** during assembly as will be described in greater detail below. Preferably the surface **5132** has a radius the same, slightly smaller or slightly larger than the radius of the spherical shank surface **5034**. In operation, the discontinuous surface **5132** advantageously frictionally engages the bone screw shank upper portion **5008**, allowing for un-locked but non-floppy placement of the angle of the shank **5004** with respect to the receiver **5010** during surgery prior to locking of the shank **5004** with respect to the receiver **5010** near the end of the procedure. At the time of locking engagement, as shown in FIG. 278, for example, downward and outward force placed on the retainer **5012** by the shank upper portion **5008** expands the retainer body **5116** at the slit **5134** and the individual flex fingers **5117** no longer frictionally grip the spherical surface **5034** of the upper portion **5008**. To aid in bending flexibility and resiliency, certain flex fingers **5117** may have sloping outer surfaces or other geometry to gain the level of resiliency desired for expansion and gripping of the fingers **5117** about the shank upper portion **5008**. The spherical surfaces **5132** may include a surface treatment or roughening to provide a desired friction fit. It is noted that the surfaces **5132** need not be spherical and may be planar or faceted or include other surface geometries that resiliently grip the shank upper portion or head **5008**. In some embodiments, the flexible tabs **5117** may be bent to further enhance frictional engagement. It is noted that the fingers **5117** that are directed generally upwardly toward the receiver channel **5064** advantageously sufficiently snap about and then grip the shank surface **5034** to an extent to provide the friction fit desired for non-floppy placement of the shank body **5006** at a desired angle with respect to the receiver **5010** during manipulation of the bone screws **5001** and the rod **5021** or other longitudinal connecting member during surgery. However, as compared to bone screw inserts such as collets known in the art that include downwardly directed portions or panels that are ultimately wedged between a receiver surface and a shank surface upon final locking of the shank to the receiver, the thin upwardly directed fingers **5117** that extend away from the shank locking surface that are not as strong as the retainer body **5116** or the insert **5014**, do not participate or cooperate with the final locking of the insert **5014** to the shank upper portion **5008**, the shank upper portion **5008** to the retainer **5012**, and the retainer **5012** to the receiver inner surfaces **5101** and **5104**. For such purpose, the more substantial retainer body **5116** having only the very narrow slit **5134**, used for expansion purposes only, is the component that locks the shank upper portion **5008** between the receiver **5010**, the insert **5014** and the rod **5021** or other longitudinal connecting member.

[0605] The retainer body **5116**, the flex fingers **5117** and a portion of each of the spring tabs **5118** have an outer substantially cylindrical profile, sized and shaped to closely and slidingly fit within the receiver cavity **5061** with the exception of outward extensions or wings, generally **5140**, of the

spring tabs **5118** that are located adjacent to the upper surfaces **5122**, each wing extending outwardly away from the respective tab body **5118** and having a projected outward surface **5142** spaced from each top surface **5122** that is sized and shaped to closely cooperate and frictionally engage upper surfaces **5079** defining the through bores **5078**. Outer surfaces **5143** located directly beneath each upper surface **5122** and above the surfaces **5142** are sized and shaped to cooperate with and frictionally engage the cylindrical surface **5090** during assembly and shipping as shown, for example, in FIG. **270**. The tab wings **5140** may include more or fewer projections or notches as needed for tooling to resiliently hold the retainer in an upper portion of the cavity **5061** when desired, but readily release the retainer **5012** into a lower portion of the receiver cavity **5061** once the retainer flex tabs **5117** engage the shank head **5008**. The illustrated spring tabs **5118** each includes one or more planar or curved inner surfaces **5144** running from the top surface **5122** to a tab base surface or seat **5145** located adjacent to the surface **5131**. The surfaces **5144** extend both outwardly and upward from the base surface **5145**. It is foreseen that in other embodiments of the invention, fewer or greater number of planar or other surfaces with other geometries may extend between the top surface **5122** and the inner surfaces defining the body **5116** of the retainer **5012**.

[0606] The through slit **5134** of the resilient retainer **5012** is defined by first and second end surfaces, **5146** and **5147** disposed in spaced relation to one another (they may also be touching) when the retainer is in a neutral state. Both end surfaces **5146** and **5147** are disposed substantially perpendicular to the bottom surface **5124**. A width X between the surfaces **5146** and **5147** is very narrow (slit may be made by EDM process) to provide stability to the retainer **5012** during operation. Because the retainer **5012** is top loadable in a neutral state and the retainer **5012** does not need to be compressed to fit within the receiver cavity **5061**, the width X may be much smaller than might be required for a bottom loaded compressible retainer ring. The gap X functions only in expansion to allow the retainer **5012** to expand about the shank upper portion **5008**. This results in a stronger retainer that provides more surface contact with the shank upper portion **5008** upon locking, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **5012** body **5116** is only expanded and not compressed, the retainer **5012** does not undergo the mechanical stress that typically is placed on spring ring type retainers that are both compressed and expanded during assembly.

[0607] It is foreseen that in some embodiments of the invention, the retainer **5012** inner surfaces may include a roughening or additional material to increase the friction fit against the shank upper portion **5008** prior to lock down by the rod **5021** or other longitudinal connecting member. Also, the embodiment shown in FIGS. **254-259** illustrates the surfaces **5146** and **5147** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle.

[0608] With particular reference to FIGS. **245** and **260-265**, the lock and release crown compression insert **5014** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **5010** at the upper opening **5066**. The compression insert **5014** has an operational central axis that is the same as the central axis B of the receiver **5010**. In operation, the insert advantageously frictionally engages the bone screw shank upper portion **5008**. Furthermore, as will be described more fully below, an insert **5014** that has locked the shank **5004** in a desired angular position with respect to the receiver **5010**, by, for example, compression from the rod **5021** and closure top **5018**, is also wedged into engagement with the receiver **5010** at the inner surface **5084** and thus retains the shank **5006** in a locked position even if the rod **5021** and closure top **5018** are removed as shown in FIG. **280**. Such locked position may also be released by the surgeon if desired. The insert **5014** is thus preferably made from a resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be expanded about and then contracted, snapped or popped onto the shank upper portion **5008** as well as pinched and un-wedged from the receiver **5010**.

[0609] The lock- and-release compression insert **5014** includes a substantially cylindrical body

5156 integral with a pair of upstanding arms **5157**. A bore, generally **5160**, is disposed primarily within and through the body **5156** and communicates with a generally U-shaped through channel **5161** that is defined by the upstanding arms **5157**. The channel **5161** has a lower seat **5162** sized and shaped to closely, snugly engage the rod **5021**. It is foreseen that an alternative embodiment may be configured to include planar holding surfaces that closely hold a square or rectangular bar as well as hold a cylindrical rod-shaped, cord, or sleeved cord longitudinal connecting member. The arms **5157** disposed on either side of the channel **5141** extend upwardly from the body **5156**. The arms **5157** are sized and configured for ultimate placement beneath the cylindrical run-out surface **5082** located below the receiver guide and advancement structure **5072**. It is foreseen that in some embodiments of the invention, for example, when the insert is non-locking as the insert **5014**" shown in FIGS. **282** and **283**, the arms may be extended and the closure top configured such the arms ultimately directly engage the closure top **5018** for locking of the polyaxial mechanism, for example, when the rod **5021** is made from a deformable material. In such embodiments, the insert **5014** would include a rotation blocking structure or feature that abuts against cooperating structure located on an inner wall of the receiver **5010**, preventing rotation of the insert with respect to the receiver when the closure top is rotated into engagement with the insert. In the present embodiment, the arms **5157** include outer upper flared or frusto-conical surfaces **5163** and top surfaces **5164** that are ultimately positioned in spaced relation with the closure top **5018**, so that the closure top **5018** frictionally engages the rod **5021** only, pressing the rod **5021** downwardly against the seating surface **5162**, the insert **5014** in turn pressing against the shank **5004** upper portion **5008** that presses against the retainer **5012** to lock the polyaxial mechanism of the bone screw assembly **5001** at a desired angle. As will be discussed in greater detail below, frictional engagement between the insert **5014** and the receiver **5010**, more particularly, the wedging of the tapered surfaces **5163** into the cylindrical surfaces **5084**, provides independent locking of the polyaxial mechanism of the assembly **5001**, maintaining the upper shank portion **5008** in locked engagement by and between the retainer **5012** and the insert **5014** even if the closure top **5018** and/or rod **5021** are thereafter removed from the receiver **5010**.

[0610] The bore, generally **5160**, is substantially defined at the body **5156** by an inner cylindrical surface **5166** that communicates with the seat **5162** and a lower concave substantially spherical surface **5168** having a radius the same or substantially similar to a radius of the surface **5034** of the shank upper portion **5008**. The surface **5168** terminates at an annular and substantially planar base surface **5169** of the body **5156**. In some embodiments of the invention, located between the cylindrical surface **5166** and the spherical surface **5168** or located along the spherical surface **5168** is a shank gripping surface portion, generally **5170**, illustrated in FIG. **265** on an alternative insert **5014'** that is otherwise identical to the insert **5014**. The gripping surface portion **5170** includes one or more stepped surfaces or ridges sized and shaped to grip and penetrate into the shank head **5008** when the insert **5014'** is locked against the head surface **5034**. It is foreseen that the stepped surface portion **5170** may include greater or fewer number of stepped surfaces. It is foreseen that the shank gripping surface portion **5170** and also the spherical surface **5168** may additionally or alternatively include a roughened or textured surface or surface finish, or may be scored, knurled, or the like, for enhancing frictional engagement with the shank upper portion **5008**.

[0611] The compression insert **5014** through bore **5160** is sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature **5046** when the shank body **5006** is driven into bone with the receiver **5010** attached. Also, the bore **5160** receives a manipulation tool (not shown) used for releasing the insert **5014** from a locked position with the receiver, the tool pressing down on the shank and also gripping the insert **5014** at through bores **5172** located in the arms **5157** or with other tool engaging features. A manipulation tool for un-wedging the insert **5014** from the receiver **5010** may also access the bores **5172** from the receiver through bores **5075** (see, e.g., FIG. **242244**)

[0612] The illustrated insert **5014** further includes other features for manipulating and holding the

insert **5014** within the receiver **5010**. Each insert arm **5157** includes an outer surface **5174** having a substantially vertical groove **5175** formed thereon, the groove **5175** located below the through bore **5172**. The grooves **5175** cooperate with the receiver crimp wall **5077** to aid in alignment of the insert channel **5161** with the receiver channel **5064**. Located beneath each groove **5175** is a recessed area or portion **5178** sized and shaped to receive the upper surface **5122** of the retainer wings **5140**, as shown, for example, in FIG. **270**, during assembly and shipping of the pre-assembled receiver **5010**, retainer **5012** and insert **5014**.

[0613] The insert body **5156** has an outer diameter slightly smaller than a diameter between crests of the guide and advancement structure **5072** of the receiver **5010**, allowing for top loading of the compression insert **5014** into the receiver opening **5066**, with the arms **5157** of the insert **5014** being located between the receiver arms **5062** during insertion of the insert **5014** into the receiver **5010**. Once the arms **5157** of the insert **5014** are generally located beneath the guide and advancement structure **5072**, the insert is rotated into place about the receiver axis B until the top surfaces **5164** are located directly below the guide and advancement structure **5072** as will be described in greater detail below.

[0614] With reference to FIGS. **282** and **283**, an alternative non-locking insert **5014''** is identical or substantially similar to the insert **5014** with the exception of outer arm surfaces **5174''** that are substantially cylindrical and extend from a top surface **5164''** to near a bottom surface **5169''** of the insert **5014''**. In other words, the insert **5014''** does not include the tapered surfaces **5163** of the insert **5014**. The arm surfaces **5174''** are fully and slidably received by the receiver surfaces **5084** as well as the other receiver **5010** inner arm surfaces and thus the insert **5014''** cannot be wedged into the receiver **5010** to independently lock the polyaxial mechanism of the assembly **5001**. In all other respects, the insert **5014''** functions the same as the insert **5014**.

[0615] With reference to FIGS. **245** and **278-280**, the closure structure or closure top **5018** shown with the assembly **5001** includes a guide and advancement structure **5182** that is a flange form as described in Applicant's U.S. Pat. No. 6,726,689, which is incorporated herein by reference. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure **5018** downward between the arms **5062** and having such a nature as to resist splaying of the arms **5062** when the closure structure **5018** is advanced into the channel **5064**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the any reduced profile of the receiver **5010** that may more advantageously engage longitudinal connecting member components. The illustrated closure structure **5018** also includes a top surface **5184** with an internal drive **5186** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **5186** is used for both rotatable engagement and, if needed, disengagement of the closure **5018** from the receiver arms **5062**. It is also foreseen that the closure structure **5018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **5188** of the closure is planar and further includes a point **5189** and a rim **5190** for engagement and penetration into the surface **5022** of the rod **5021** in certain embodiments of the invention.

[0616] An alternative closure top **5018'** for use with a deformable rod **5021'**, such as a PEEK rod, is shown in FIG. **281**. The top **5018'** is identical to the top **5018** with the exception that a point **5189'** is located on a domed surface **5190'** in lieu of the planar bottom with point and rim of the closure

top **5018**.

[0617] Preferably, the receiver **5010**, the retainer **5012** and the compression insert **5014** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **5012** spring tabs **5118** and rotating and otherwise manipulating the insert **5014** arms, as well as crimping a portion of the receiver **5010** toward the insert **5014**. In some circumstances, the shank **5004** is also assembled with the receiver **5010**, the retainer **5012** and the compression insert **5014** at the factory. In other instances, it is desirable to first implant the shank **5004**, followed by addition of the pre-assembled receiver, retainer and compression insert at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **5004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **5008** and/or hydroxyapatite on the shank **5006**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **5004** advantageously reduces inventory requirements, thus reducing overall cost.

[0618] Pre-assembly of the receiver **5010**, retainer **5012** and compression insert **5014** is shown in FIGS. **266-272**. With particular reference to FIG. **266**, first the retainer **5012** is inserted into the upper receiver opening **5066**, leading with one of the spring tabs **5118** with both of the spring tab top surfaces **5122** facing one arm **5062** and the retainer bottom surface **5124** facing the opposing arm **5062** (shown in phantom). The retainer **5012** is then lowered in such sideways manner into the channel **5064** and partially into the receiver cavity **5061**, followed by tilting the retainer **5012** such that the top surface **5122** and thereafter the outer tab or wing **5140** of the leading spring tab **5118** is moved into a nearby receiver arm through bore **5078**. With reference to FIG. **267**, the retainer **5012** is then further tilted or turned and manipulated within the receiver to a position within the cavity until the retainer **5012** bottom surface **5124** is directed toward the receiver cavity **5061** and the spring tab upper surfaces **5122** are facing upwardly toward the receiver channel opening **5066**. To accomplish such tilting and turning of the retainer **5012**, the spring tab arm **5118** located within the receiver bore **5078** is manipulated downwardly and then upwardly within the bore **5078** and finally shifted out of the bore **5078** when the opposed spring tab arm **5118** outer tab or wing **5140** moves past and clears the cylindrical surface **5084** of the receiver **5010**. Once the retainer bottom surface **5124** seats on the receiver surface **5104**, both of the spring tab wings **5140** are partially located in opposed receiver bores **5078**.

[0619] With reference to FIGS. **267** and **268**, the compression insert **5014** is then downloaded into the receiver **5010** through the upper opening **5066** with the bottom surface **5169** facing the receiver arm top surfaces **5073** and the insert arms **5157** located between the opposed receiver arms **5062**. The insert **5014** is then lowered toward the channel seat **5068** until the insert **5014** arm upper surfaces **5164** are adjacent the run-out area below the guide and advancement structure **5072** defined in part by the cylindrical surface **5082**. Thereafter, the insert **5014** is rotated in a clockwise or counter-clockwise manner about the receiver axis B until the upper arm surfaces **5164** are directly below the guide and advancement structure **5072** as illustrated in FIG. **268** with the U-shaped channel **5161** of the insert **5014** aligned with the U-shaped channel **5064** of the receiver **5010**. In some embodiments, the insert arms **5157** may need to be compressed slightly during rotation to clear inner surfaces of the receiver arms **5062**. As shown in FIGS. **269** and **270**, the outer lower cylindrical surface **5174** of the insert **5014** is received within the cylindrical surface **5090** of the receiver.

[0620] With further reference to FIGS. **268** and **269**, a tool (not shown) is then used to grip the retainer spring tab arms **5118** at outer surfaces thereof and squeeze or press the tabs **5118** toward one another while moving the retainer **5012** in an upward direction away from the surface **5104**. With reference to FIG. **270**, when the spring tab wing surface projections **5142** abut against the

surface **5079**, the tool (not shown) is released and a portion or portions **5143** of each spring tab **5118** spring out to engage the surface portion **5092** formed in the receiver cylindrical surface **5090**. With reference to FIGS. **270-272**, the retainer **5012** is now in a desired position for shipping and with assembly with the shank **5004**. The insert **5014** recessed areas **5178** are located adjacent to the retainer spring tab top surfaces **5122**.

[0621] With reference to FIGS. **271** and **272**, the receiver thin walls **5077** are then crimped inwardly toward the axis B by inserting a tool (not shown) through the receiver apertures **5074**, the tool pressing the walls **5077** until the wall surface **5087** engages the insert **5014** at the shallow central grooves **5175** formed on the outer surface **5174** of each of the insert arms **5157**. The crimping of the wall surface **5093** into the groove **5175** keeps the insert **5014** U-shaped channel **5161** aligned with the receiver U-shaped channel **5064** and also helps retain the insert **5014** at the upward location shown in FIG. **270** with the insert arm top surfaces **5164** adjacent the guide and advancement structure **5072** until the insert **5014** is pushed downwardly toward the receiver base **5060** after assembly with the shank **5004**. Thus, the crimping of the receiver walls **5077** prohibits rotation of the insert **5014** about the receiver axis B but allows for limited axial movement of the insert **5014** with respect to the receiver **5010** along the axis B when some force is exerted to slide the crimped surface **5093** up or down along the groove **5175**. The insert **5014** is fully captured within the receiver **5010** by the guide and advancement structure **5072** prohibiting movement of the insert **5014** up and out through the receiver opening **5066** as well as by retainer **5012** located below the insert.

[0622] Typically, the receiver and retainer combination are shipped or otherwise provided to the end user with the spring tab outer wings **5140** wedged against the receiver as shown in FIG. **270**. The receiver **5010**, retainer **5012** and insert **5014** combination is now pre-assembled and ready for assembly with the shank **5004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **5004** as will be described herein.

[0623] As illustrated in FIG. **273**, the bone screw shank **5004** or an entire assembly **5001** made up of the assembled shank **5004**, receiver **5010**, retainer **5012** and compression insert **5014**, is screwed into a bone, such as the vertebra **5017** (shown in phantom), by rotation of the shank **5004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **5006** by engagement thereof at the internal drive **5046**.

[0624] With further reference to FIG. **273**, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **5008** until the shank upper portion is received within the opening **5110**. With particular reference to FIGS. **274** and **275**, as the shank upper portion **5008** is moved into the interior **5061** of the receiver base, the shank upper portion **5008** presses upwardly against the retainer **5012** in the recess partially defined by the cylindrical surface **5099**. As the portion **5008** continues to move upwardly toward the channel **5064**, the surface **5034** forces outward movement of the retainer **5012** towards the cylindrical surface **5099** defining the receiver expansion recess or chamber. The retainer **5012** begins to contract about the spherical surface **5034** as the center of the sphere (shown in dotted lines) passes beyond the center of the retainer expansion recess. At this time also, the spherical surface **5034** moves into engagement with the surfaces **5132** of the retainer flex tabs **5117**, the tabs **5117** expanding slightly outwardly to receive the surface **5034**. With reference to FIG. **276**, the spherical surface **5034** then enters into full frictional engagement with the panel inner surfaces **5132**. At this time, the retainer **5012** panels and the surface **5034** are in a fairly tight friction fit, the surface **5034** being pivotable with respect to the retainer **5012** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the retainer **5012** and the shank upper portion **5008**.

[0625] With reference to FIG. **277**, the shank **5004** and attached retainer **5012** are then moved downwardly into a desired position with the retainer seated on the surface **5104**. This may be accomplished by either an upward pull on the receiver **5010** or, in some cases, by driving the shank **5004** further into the vertebra **5017**. The insert **5014** may be pressed downwardly by a tool or by a

rod and closure top as shown in FIG. 278. Also, in some embodiments, when the receiver 5010 is pre-assembled with the shank 5004, the entire assembly 5001 may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive 5046 and rotating and driving the shank 5004 into a desired location of the vertebra 5017.

[0626] Also with reference to FIGS. 277 and 278, prior to assembly with the rod 5021 and the closure top 5018, the compression insert 5014 frusto-conical surface 5163 is near the surface 5084. The insert 5014 is prohibited from moving any further downwardly at the beginning of the surface 5084 unless forced downwardly by a tool or by the closure top pressing downwardly on the rod that in turn presses downwardly on the insert 5014 as shown in FIG. 278. With further reference to FIG. 277 and also to FIG. 279, at this time, the receiver 5010 may be articulated to a desired angular position with respect to the shank 5004, such as that shown in FIG. 279, that will be held, but not locked, by the frictional engagement between the retainer 5012 and the shank upper portion 5008.

[0627] The rod 5021 is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies 5001. The closure structure 5018 is then inserted into and advanced between the arms 5062 of each of the receivers 5010. The closure structure 5018 is rotated, using a tool engaged with the inner drive 5186 until a selected pressure is reached at which point the rod 5021 engages the U-shaped seating surface 5162 of the compression insert 5014, further pressing the insert spherical surface 5168 (or stepped shank gripping surfaces 5170 of the insert 5014') against the shank spherical surface 5034, (the edges of the stepped surfaces 5170 penetrating into the spherical surface 5034), pressing the shank upper portion 5008 into locked frictional engagement with the retainer 5012. Specifically, as the closure structure 5018 rotates and moves downwardly into the respective receiver 5010, the point 5189 and rim 5190 engage and penetrate the rod surface 5022, the closure structure 5018 pressing downwardly against and biasing the rod 5021 into compressive engagement with the insert 5014 that urges the shank upper portion 5008 toward the retainer 5012 and into locking engagement therewith, the retainer 5012 frictionally abutting the surface 5104 and expanding outwardly against the cylindrical surface 5101. For example, about 80 to about 120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank 5006 with respect to the receiver 5010.

[0628] Also, as the closure structure 5018 and the rod 5021 press the insert 5014 downwardly toward the base of the receiver 5010, the insert frusto-conical surface 5163 is forced into the receiver cylindrical surface 5084, wedging the insert 5014 into fixed frictional engagement with the receiver surface 5084. With reference to FIG. 280, at this time, the closure top 5018 may be loosened or removed and/or the rod 5021 may be adjusted and/or removed and the frictional engagement between the insert 5014 and the receiver 5010 at the receiver surface 5084 will remain locked in place, advantageously maintaining a locked angular position of the shank 5004 with respect to the receiver 5010. If the user wishes to release the insert 5014 from the receiver 5010 and unlock the polyaxial mechanism, a tool (not shown) may be used that includes extensions or prongs that are received by and through the opposed through bores 5075 of the receiver 5010 and received into the through bores 5172 of the insert 5014. Such tool is then pulled upwardly in a direction along the axis B away from the receiver base 5060, thereby pulling the insert slightly upwardly and away from the receiver base 5060 and releasing the frusto-conical surface 5163 from the cylindrical surface 5084. Alternatively, if both the closure top 5018 and the rod 5021 are already removed from the receiver 5010, another manipulation tool (not shown) may be used that is inserted into the receiver at the opening 5066 and into the insert channel 5161, with prongs or extensions thereof extending outwardly into the insert through bores 5172; a piston-like portion of the tool thereafter pushing directly on the shank upper portion 5008, thereby pulling the insert 5014 surface 5163 away from the receiver surface 5084 and thus releasing the polyaxial mechanism. At such time, the shank 5004 may be articulated with respect to the receiver 5010, and the desired friction fit returns between the retainer 5012 and the shank surface 5034, so that an adjustable, but non-floppy relationship still exists between the shank 5004 and the receiver 5010. If further disassembly if the

assembly is desired, such is accomplished in reverse order to the procedure described previously herein for assembly.

[0629] With reference to FIG. **281**, an alternative assembly **5001'** is shown in which the rod **5021** and closure top **5018** of the assembly **5001** of FIG. **280** are replaced with a deformable rod **5018'** and alternative closure top **5018'**. Because of the lock between the insert **5014** and the receiver **5010**, any loosening of the rod **5021'** from the receiver **5010** that may occur due to rod deformation does not compromise the locked polyaxial mechanism formed by the wedged in insert **5014**, the shank upper portion **5008**, the retainer **5012** and the receiver **5010**.

[0630] With reference to FIGS. **284-306** the reference number **6001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **6001** includes a shank **6004**, that further includes a body **6006** integral with an upwardly extending upper portion or head-like capture structure **6008**; a receiver **6010**; and a lower retainer structure illustrated as a resilient open ring **6012**. The receiver **6010** and retainer structure **6012** are initially assembled and may be further assembled with the shank **6004** either prior or subsequent to implantation of the shank body **6006** into a vertebra **6017**, as will be described in greater detail below. FIG. **284** further shows a closure structure **6018** for capturing a longitudinal connecting member, for example, a rod **6021** which in turn presses against the shank upper portion **6008** into fixed frictional contact with the lower retainer **6012**, so as to capture, and fix the longitudinal connecting member **6021** within the receiver **6010** and thus fix the member **6021** relative to the vertebra **6017**. The illustrated rod **6021** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **6022**. The rod **6021** is the same or substantially similar to the rods previously discussed herein, such as the rods **21**, **1021**, **2021**, **3021**, **4021**, **5021** and **6021**. It is foreseen that in other embodiments, the rod **6021** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **6010** and the shank **6004** cooperate in such a manner that the receiver **6010** and the shank **6004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **6010** with the shank **6004** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0631] The shank **6004**, best illustrated in FIGS. **284-286**, is elongate, with the shank body **6006** having a helically wound bone implantable thread **6024** (single or dual lead thread form) extending from near a neck **6026** located adjacent to the upper portion or head **6008**, to a tip **6028** of the body **6006** and extending radially outwardly therefrom. During use, the body **6006** utilizing the thread **6024** for gripping and advancement is implanted into the vertebra **6017** leading with the tip **6028** and driven down into the vertebra with an installation or driving tool (not shown), so as to be implanted in the vertebra to a location at or near the neck **6026**, as more fully described in the paragraphs below. The shank **6004** has an elongate axis of rotation generally identified by the reference letter A.

[0632] The neck **6026** extends axially upward from the shank body **6006**. The neck **6026** may be of the same or is typically of a slightly reduced radius as compared to an adjacent upper end or top **6032** of the body **6006** where the thread **6024** terminates. Further extending axially and outwardly from the neck **6026** is the shank upper portion or head **6008** that provides a connective or capture apparatus disposed at a distance from the upper end **6032** and thus at a distance from the vertebra **6017** when the body **6006** is implanted in such vertebra.

[0633] The shank upper portion **6008** is configured for a pivotable connection between the shank **6004** and the retainer **6012** and receiver **6010** prior to fixing of the shank **6004** in a desired position with respect to the receiver **6010**. The shank upper portion **6008** has an outer, convex and substantially spherical surface **6034** that extends outwardly and upwardly from the neck **6026** and terminates at an outer annular rim surface **6038**. The spherical surface **6034** has an outer radius configured for frictional, sliding cooperation with the retainer **6012** as will be described in greater detail below. The top surface **6038** is substantially perpendicular to the axis A. The spherical

surface **6034** shown in the present embodiment is substantially smooth, but in some embodiments may include a roughening or other surface treatment and is sized and shaped for cooperation and ultimate frictional engagement with the lower retainer **12**. The shank spherical surface **6034** is locked into place exclusively by the rod **6021** and the retainer **6012** and not by inner surfaces defining the receiver cavity.

[0634] The shank upper portion **6008** further includes a substantially spherical, curved or domed top surface **6040**. In operation, the surface **6040** directly engages the rod **6021** within the channel of the receiver **6010**. The domed surface **6040** is located above and is spaced from the annular rim **6038** where the surface **6034** terminates. A counter sunk feature, generally **6042** separates the domed surface **6040** from the annular rim **6038**. The feature **6042** is further defined by a discontinuous cylindrical surface **6043** located about the domed surface **6040** and a frusto-conical surface **6044** extending from the rim **6038** downwardly and inwardly toward the surface **6043**. The surface **6043** runs parallel to the axis A. The surface **6044** terminates at a narrow annular track **6046** that encircles the cylindrical surface **6043**. Six evenly spaced cylindrical cut-outs, **6048** are formed primarily into the cylindrical surface **6043** and also partially into the frusto-conical surface **6044**, each of the cutouts **6048** running parallel to the shank axis A. Cylindrical surfaces created by the cutouts **6048** and the surfaces **6043** and **6044** provide a partially external and partially internal drive feature for receiving a driving tool (not shown) for rotating and driving the bone screw shank body **6006** into the vertebra **6017**. It is foreseen that the shank driving feature may take on other various shapes and forms, as for example, will be described herein with respect to the alternative bone screw shanks **6004'** and **6004''** shown in FIGS. **305** and **306**, respectively. Other forms may include more or fewer apertures of various shapes. As illustrated in FIGS. **285** and **386**, the external and internal portions of the drive may also include beveled or stepped surfaces that may further enhance gripping with the driving tool. In operation, a driving tool (not shown) is received in the drive feature **6042**, being seated at the frusto-conical surface **6044** and engaging the various curved faces at and about the cylindrical surface **6043** for both driving and rotating the shank body **6006** into the vertebra **6017**, either before the shank **6004** is attached to the receiver **6010** or after the shank **6004** is attached to the receiver **6010**, with the shank body **6006** being driven into the vertebra **6017** with the driving tool extending into the receiver **6010**.

[0635] The shank **6004** shown in the drawings is cannulated, having a small central bore **6050** extending an entire length of the shank **6004** along the axis A. The bore **6050** is defined by an inner cylindrical wall of the shank **6004** and has a circular opening at the shank tip **6028** and an upper opening communicating with the top surface **6040**. The bore **6050** is coaxial with the threaded body **6006** and the upper portion **6008**. The bore **6050** provides a passage through the shank **6004** interior for a length of wire (not shown) inserted into the vertebra **6013** prior to the insertion of the shank body **6006**, the wire providing a guide for insertion of the shank body **6006** into the vertebra **6017**.

[0636] To provide a biologically active interface with the bone, the threaded shank body **606** may be coated, perforated, made porous or otherwise treated. The treatment may include, but is not limited to a plasma spray coating or other type of coating of a metal or, for example, a calcium phosphate; or a roughening, perforation or indentation in the shank surface, such as by sputtering, sand blasting or acid etching, that allows for bony ingrowth or ongrowth. Certain metal coatings act as a scaffold for bone ingrowth. Bio-ceramic calcium phosphate coatings include, but are not limited to: alpha-tri-calcium phosphate and beta-tri-calcium phosphate ($\text{Ca.sub.3(PO.sub.4).sub.2}$), tetra-calcium phosphate ($\text{Ca.sub.4P.sub.2O.sub.9}$), amorphous calcium phosphate and hydroxyapatite ($\text{Ca.sub.o(PO.sub.4).sub.6(OH).sub.2}$). Coating with hydroxyapatite, for example, is desirable as hydroxyapatite is chemically similar to bone with respect to mineral content and has been identified as being bioactive and thus not only supportive of bone ingrowth, but actively taking part in bone bonding.

[0637] With particular reference to FIGS. **284** and **290-295**, the receiver **6010** has a generally

cylindrical and U-shaped appearance. The receiver **6010** has an axis of rotation B that is shown in FIG. **284** as being aligned with and the same as the axis of rotation A of the shank **6004**, such orientation being desirable, but not required during assembly of the receiver **6010** with the shank **6004**. After the receiver **6010** is pivotally attached to the shank **6004**, either before or after the shank **6004** is implanted in a vertebra **6017**, the axis B is typically disposed at an angle with respect to the axis A.

[0638] The receiver **6010** includes a substantially cylindrical base **6060** defining a bore or inner cavity, generally **6061**, the base **6060** being integral with a pair of opposed upstanding arms **6062** forming a cradle and defining a channel **6064** between the arms **6062** with an upper opening, generally **6066**, and a lower channel portion including a partially planar and partially U-shaped lower seat **6068**, the channel **6064** having a width for operably snugly receiving the rod **6021** or portion of another longitudinal connector between the arms **6062**, the channel **6064** communicating with the base cavity **6061**.

[0639] Each of the arms **6062** has an interior surface, generally **6070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **6072** located adjacent top surfaces **6073** of each of the arms **6062**. In the illustrated embodiment, the guide and advancement structure **6072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **18**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **6072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structures, such as a flange form, for operably guiding under rotation and advancing the closure structure **6018** downward between the arms **6062**, as well as eventual torquing when the closure structure **6018** abuts against the rod **6021** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0640] An opposed pair of tool receiving and engaging apertures **6074** are formed on outer surfaces **6076** of the arms **62**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **6062**.

[0641] Returning to the interior surface **6070** of the receiver arms **6062**, located below the guide and advancement structure **6072** is a run-out feature for the guide and advancement structure **6072** partially defined by a discontinuous cylindrical surface **6082** having a diameter approximately the same or slightly greater than a greater diameter of the guide and advancement structure **6072**.

Below the surface **6082**, moving in a direction toward the base **6060**, is another cylindrical surface **6084** having a diameter smaller than the diameter of the surface **6082** and illustrated as substantially the same as an inner or lesser diameter of the guide and advancement structure **6072**. The surface **6084** is initially discontinuous (at the arms **6062**) and transitions into a continuous surface at the channel seat **6068**. Directly above the seat **6068**, located in each of the arms **6062** is a discontinuous, radially extending, rounded lip **6086** extending inwardly toward the axis B from each of the surfaces **6084**. As will be described in greater detail below, the lip **6086** provides a frictional stop for the shank upper portion **6008** during pop-on assembly with the retainer **6012** within the receiver **6010**. A continuous, beveled annular upper rim or stop surface **6098** is located below and adjacent to the cylindrical surface **6084**. The surface **6098** is disposed in the base **6060**, partially forming the base cavity **6061** and forms an abutment stop for the resilient retainer **6012**, prohibiting the retainer **6012** (when in an uncompressed configuration) from moving upwardly into the space defined by the cylindrical surface **6084** and the channel **6064**. Another cylindrical surface **6099** is located below and adjacent to the surface **6098**. The cylindrical surface **6099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **6012**. The surfaces **6098** and **6099** define a circumferential recess that is sized and shaped to receive the retainer **6012** as it expands around the shank upper portion **6008** as the shank **6008** moves upwardly toward the channel **6064** during assembly, as well as form a restriction to prevent the

expanded retainer **6012** from moving upwardly with the shank portion **6008**, the surface **6098** preventing the retainer **6012** from passing upwardly out of the cavity **6061** whether the retainer **6012** is in a partially or fully expanded position or state, or in a neutral or original operative position or state (see, e.g., FIGS. **299** and **300**). A cylindrical surface **6101** located below the cylindrical surface **6099** is sized and shaped to closely receive the retainer **6012** when the retainer is in a neutral or slightly compressed operative position as shown in FIGS. **297** and **301**, for example. Thus, the cylindrical surface **6101** has a diameter smaller than the diameter of the cylindrical surface **6099** that defines the expansion area for the retainer **6012**. The surface **6101** is joined or connected to the surface **6099** by one or more beveled, curved or conical surfaces **6102**. The surfaces **6102** allow for sliding gradual movement and/or contraction of the retainer **6012** into the space defined by the surface **6101** and ultimate seating of the retainer **6012** on a lower annular surface **6104** located below and adjacent to the cylindrical surface **6101**.

[0642] Located below and adjacent to the annular seating surface **6104** is another substantially cylindrical surface **6106** that communicates with a beveled or flared bottom opening surface **6107**, the surface **6107** communicating with an exterior base surface **6108** of the base **6060**, defining a lower opening, generally **6110**, into the base cavity **6061** of the receiver **6010**. The illustrated surface **6106** has a diameter requiring compression or squeezing of the retainer **6012** during uploading of the retainer **6012** through the lower opening **6110** (see FIG. **296**, for example).

[0643] With particular reference to FIGS. **284** and **287-289**, the lower open retainer ring **6012** that operates to capture the shank upper portion **6008** within the receiver **6010** has a central axis that is operationally the same as the axis B associated with the receiver **6010** when the shank upper portion **6008** and the retainer **6012** are installed within the receiver **6010**. The retainer ring **6012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **6012** may be both compressed and expanded during various steps of assembly as will be described in greater detail below. The lower retainer **6012** has a central channel or hollow through bore, generally **6121**, that passes entirely through the ring **6012** from a top surface **6122** to a bottom surface **6124** thereof. Surfaces that define the channel or bore **6121** include a discontinuous inner cylindrical surface **6125** near or adjacent the top surface **6122**, a first discontinuous frusto-conical or curved surface **6127** adjacent the surface **6125** and a second frusto-conical or beveled surface **6128** adjacent the surface **6127** and also adjacent to the bottom **6124**, all three surfaces **6125**, **6127** and **6128** being coaxial when the retainer **6012** is in a neutral non-compressed, non-expanded orientation. The retainer **6012** further includes an outer cylindrical surface **6130** located adjacent the top surface **6122** and an outer beveled or frusto-conical surface **6132** adjacent the bottom surface **6124**. The surface **6130** is oriented parallel to the central axis of the retainer **6012**. In some embodiments of the invention spaced notches (not shown) may be formed in the cylindrical surface **6130** to receive a holding and manipulation tool (not shown) used for contraction and insertion of the retainer **6012** into the receiver **6010**. In some embodiments further notches may be made to evenly distribute stress across the entire retainer **6012** during contraction and expansion thereof. In other embodiments of the invention, such notches may be on the inside of the retainer **6012** ring. It is also foreseen that in some embodiments of the invention, the retainer **6012** inner surfaces may include a roughening or additional material to provide a friction fit against the shank upper portion **6008** prior to lock down by the rod **6021** or other longitudinal connecting member. The resilient retainer **6012** further includes first and second end surfaces, **6134** and **6135** disposed in spaced relation to one another when the retainer is in a neutral non-compressed state. Both end surfaces **6134** and **6135** are disposed substantially perpendicular to the top surface **6122** and the bottom surface **6124**. A width X between the surfaces **6134** and **6135** is determined by a desired amount of compressibility of the open retainer **6012** when loaded into the receiver **6010**. The space X shown in FIG. **287** provides adequate space between the surfaces **6134** and **6135** for the retainer **6012** to be pinched, with the surfaces **6134** and **6135** compressed toward one another (as shown in FIG. **296**) to a closely spaced or even touching configuration, if necessary, to an extent that the

compressed retainer **6012** is up or bottom loadable (as illustrated) through the receiver opening **6110** or alternatively top loaded through the channel opening **6066** (not shown). After passing through the opening **6110** and along a portion of the lower inner surface **6106**, the retainer **6012** expands or springs back to an original uncompressed, rounded or collar-like configuration of FIGS. **287-289**, see, e.g., FIG. **297**. The embodiment shown in FIGS. **287-289** illustrates the surfaces **6134** and **6135** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle depending upon the amount of compression desired during loading of the retainer **6012** into the receiver **6010**. It is further noted that the geometry of the retainer **6012** is not limited to the particular cylindrical or planar surface shapes shown in the drawings figures. The retainer **6012** may be of a rounded ring-shape, for example, or include more or fewer planar, conical or curved surfaces.

[0644] With reference to FIGS. **284**, **303** and **304**, the illustrated elongate rod or longitudinal connecting member **6021** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **6022** of uniform diameter. The rod **6021** may be made from a variety of metals, metal alloys and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials.

[0645] Longitudinal connecting members for use with the assembly **6001** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the receiver **6010** may be modified so as to closely hold, and if desired, fix or slidingly capture the longitudinal connecting member to the assembly **6001**. Some embodiments of the assembly **6001** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the receiver **6010** or alternative receiver having a U-shaped, rectangular- or other-shaped channel, for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **6001**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **6001**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from deformable plastics to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

[0646] With reference to FIG. **284**, the closure structure or closure top **6018** shown with the assembly **6001** is rotatably received between the spaced arms **6062** of the receiver **6010**. It is noted that the closure **6018** top could be a twist-in or slide-in closure structure. The illustrated closure structure **6018** is substantially cylindrical and includes a an outer helically wound guide and advancement structure **6162** in the form of a flange that operably joins with the guide and advancement structure **6072** disposed on the arms **6062** of the receiver **6010**. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure

structure **6018** downward between the arms **6062** and having such a nature as to resist splaying of the arms **6062** when the closure structure **6018** is advanced into the channel **6064**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the reduced profile of the receiver **6010** that advantageously engages longitudinal connecting member components as will be further described below. The illustrated closure structure **6018** also includes a top surface **6164** with an internal drive **6166** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **6166** is used for both rotatable engagement and, if needed, disengagement of the closure **6018** from the receiver arms **6062**. It is also foreseen that the closure structure **6018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 6070 to 6140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **6168** of the closure is planar and further includes a point **6169** and a rim **6170** for engagement and penetration into the surface **6022** of the rod **6021** in certain embodiments of the invention. The closure top **6018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **6018** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **6062**.

[0647] Preferably, the receiver **6010** and the retainers **6012** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **6012**. In some circumstances, the shank **6004** is also assembled with the receiver **6010** and the retainer **6012** at the factory. In other instances, it is desirable to first implant the shank **6004**, followed by addition of the pre-assembled receiver and retainer at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **6004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size with the already assembled receiver and retainer. Allowing the surgeon to choose the appropriately sized shank advantageously reduces inventory requirements, thus reducing overall cost.

[0648] Pre-assembly of the receiver **6010** and the retainer **6012** is shown in FIGS. 296-297. The retainer **6012** is prepared for insertion into the receiver **6010** by squeezing or pressing the retainer end surfaces **6134** and **6135** toward one another as shown in FIG. 296. The compressed retainer **6012** is inserted into the lower opening **6110** with the planar top surface **6122** facing the receiver bottom surface **6108**. The retainer **6012** is typically moved upwardly into the receiver **6010** and past the cylindrical surface **106** and allowed to expand to an almost neutral or slightly compressed state within the cylindrical surface **101** as shown in FIG. 297. The receiver **6010** and the retainer **6012** (held by the cylindrical surface **6101**) combination is now pre-assembled and ready for assembly with the shank **6004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **6004** as will be described herein.

[0649] As illustrated in FIG. 298, the bone screw shank **6004** alone (or an entire assembly **6001** made up of the assembled shank **6004**, receiver **6010** and retainer **6012**) is screwed into a bone, such as the vertebra **6017**, by rotation of the shank **6004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **6006** by engagement thereof at the drive feature **6042**. Specifically, the vertebra **6017** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **6004** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as

a guide. Then, the bone screw shank **6004** or the assembly **6001** is threaded onto the guide wire utilizing the cannulation bore **6050** by first threading the wire into the opening at the bottom **6028** and then out of the top opening at the top surface **6040**. The shank **6004** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **6021** (also having a central lumen in some embodiments) and the closure top **18** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires.

[0650] When the shank **6004** is driven into the vertebra **6017** without the remainder of the assembly **6001**, the shank **6004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer. With reference to FIG. **298**, the pre-assembled receiver and retainer is placed above the shank upper portion **6008** until the shank upper portion is received within the opening **6110**. With particular reference to FIGS. **299-300**, as the shank is moved into the interior of the receiver base, the shank upper portion **6008** presses the retainer **6012** upwardly into the recess partially defined by the cylindrical surface **6099** (if the retainer is not already located within such recess). As the portion continues to move upwardly toward the channel **6064**, the top surface **6122** of the retainer **6012** abuts against the receiver surface **6098**. At this time, the shank upper portion moves upwardly into the channel **6064** until the outer surface **6034** frictionally engages the lip **6086** on the receiver cylindrical surface **6084**, stopping upward movement of the shank upper portion **6008**. As the retainer **6012** presses up against the surface **6098**, the shank upper portion **6008** forces outward movement of the retainer **6012** towards the cylindrical surface **6099** defining the receiver expansion recess as the spherical surface **6034** continues in an upward direction. The retainer **6012** begins to contract about the spherical surface **6034** as the center of the sphere passes beyond the center of the retainer expansion recess defined by the surface **6099**. At this time also, the spherical surface **6034** is in engagement with the receiver lip **6086**, prohibiting further upward movement of the shank **6004** into the channel **6064**.

[0651] With reference to FIG. **302**, the retainer **6012** and attached insert **6014** are ultimately moved down into a final operative position by either an upward pull on the receiver **6010** or, in some cases, by driving the shank **6004** further into the vertebra **6017**. Also, in some embodiments, when the receiver **6010** is pre-assembled with the shank **6004**, the entire assembly **6001** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **6042** and rotating and driving the shank **4** into a desired location of the vertebra **6017**.

[0652] With reference to FIG. **303-304**, the rod **6021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **6001**. The closure structure **6018** is then inserted into and advanced between the arms **6062** of each of the receivers **6010**. The closure structure **6018** is rotated, using a tool engaged with the inner drive **6166** until a selected pressure is reached at which point the rod **6021** engages the curved top surface **6040** of the shank **6004**, pressing the shank upper portion **6008** into locked frictional engagement with the retainer **6012**. Specifically, as the closure structure **6018** rotates and moves downwardly into the respective receiver **6010**, the point **6169** and rim **6170** engage and penetrate the rod surface **6022**, the closure structure **6018** pressing downwardly against and biasing the rod **6021** into compressive engagement with the shank upper surface **6040** that urges the shank upper portion **6008** toward the retainer **6012** and into locking engagement therewith, the retainer **6012** frictionally abutting the surface **6104** and expanding outwardly against the cylindrical surface **6101**. For example, about 6080 to about 6120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **6006** with respect to the receiver **6010**.

[0653] If removal of the rod **6021** from any of the bone screw assemblies **6001** is necessary, or if it is desired to release the rod **6021** at a particular location, disassembly is accomplished by using the driving tool (not shown) that mates with the internal drive **6166** on the closure structure **6018** to rotate and remove such closure structure from the cooperating receiver **6010**. Disassembly is then

accomplished in reverse order to the procedure described previously herein for assembly.

[0654] With reference to FIGS. 305 and 306, alternative bone screw shanks **6004'** and **6004''** according to the invention may include alternative drive features **6042'** and **6042''**, respectively. The bone screws **6004'** and **6004''** may be used in lieu of a screw **6004** in the assembly **1** described above. The bone screw **6004'** and **6004''** include respective outer lower spherical surface **6034'** and **6034''** and respective upper or top domed shaped surfaces **6040'** and **6040''**, that are the same or substantially similar to the respective spherical surface **6034** and top domed surface **6040** previously described herein with respect to the shank **6004** of the assembly **6001**. The bone screw **6004'** is identical to the screw **6004** with the exception that the six cylindrical cutouts **6048** are replaced by six partially cylindrical grooves **6048'** that in addition to forming cylindrical surfaces in an upstanding surface **6043'** (that is otherwise identical to the surface **6043** of the shank **6004**), also carve a groove into the frusto-conical surface **6044'** (that is otherwise identical to the surface **6044**) and through the outer spherical surface **6034'**, the grooves **6048'** each having a substantially planar bottom surface **6049'** that extends from the surface **6043'** radially outwardly and through the spherical surface **6034'**. With respect to the bone screw **6004''**, the cylindrical surface **6043** of the bone screw **6004** is replaced by a faceted surface **6043''** and a portion of the spherical surface **6034** is completely removed to result in an annular planar tool seating surface **6049''**. In the illustrated embodiment, the faceted surface **6043''** includes six surfaces sized and shaped to be received in a hex shaped socket type driving tool (not shown), the tool seatable on the planar surface **6049''**.

[0655] With reference to the '849 patent application incorporated by reference herein, polyaxial bone screws **6001** according to the invention may be used with dynamic stabilization longitudinal connecting member assemblies that include one or more sleeves with cooperating, spacers, bumpers and an inner tensioned cord.

[0656] With reference to FIGS. 307-339 the reference number **7001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **7001** includes a shank **7004**, that further includes a body **7006** integral with an upwardly extending upper portion or head-like capture structure **7008**; a receiver **7010**; and a lower retainer structure illustrated as a resilient open ring-like structure **7012**. The receiver **7010** and retainer structure **7012** are initially assembled and may be further assembled with the shank **7004** either prior or subsequent to implantation of the shank body **7006** into a vertebra **7017**, as will be described in greater detail below. FIG. 307 further shows a closure structure **7018** for capturing a longitudinal connecting member, for example, a rod **7021** which in turn presses against the shank upper portion **7008** into fixed frictional contact with the lower retainer **7012**, so as to capture, and fix the longitudinal connecting member **7021** within the receiver **7010** and thus fix the member **7021** relative to the vertebra **7017**. The illustrated rod **7021** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **7022**. It is foreseen that in other embodiments, the rod **7021** may be elastic, deformable and/or of a different cross-sectional geometry. The receiver **7010** and the shank **7004** cooperate in such a manner that the receiver **7010** and the shank **7004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **7010** with the shank **7004** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0657] The shank **7004**, best illustrated in FIGS. 307-309, is substantially similar to the shank **6004** previously described herein with respect to the assembly **6001**. Thus, the shank **7004** includes the shank body **7006**, upper portion or head **7008**, a shank thread **7024**, a neck **7026**, a tip **7028**, a top of thread **7032**, an upper portion spherical surface **7034** a top surface **7040**, a drive feature **7042** and a cannulation bore **7050** the same or substantially similar to the respective body **6006**, upper portion or head **6008**, shank thread **6024**, neck **6026**, tip **6028**, top of thread **6032**, spherical surface **6034**, domed top surface **6040**, drive feature **6042** and cannulation bore **6050** previously described herein with respect to the shank **6004** of the assembly **6001**. To provide a biologically active

interface with the bone, the threaded shank body **7006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**.

[0658] With particular reference to FIGS. **307** and **316-320**, the receiver **7010** has a generally cylindrical and U-shaped appearance. The receiver **7010** has an axis of rotation B that is shown in FIG. **307** as being aligned with and the same as the axis of rotation A of the shank **7004**, such orientation being desirable, but not required during assembly of the receiver **7010** with the shank **7004**, as shown, for example, in FIG. **337**. After the receiver **7010** is pivotally attached to the shank **7004**, either before or after the shank **7004** is implanted in a vertebra **7017**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **338**.

[0659] The receiver **7010** includes a substantially cylindrical base **7060** defining a bore or inner cavity, generally **7061**, the base **7060** being integral with a pair of opposed upstanding arms **7062** forming a cradle and defining a channel **7064** between the arms **7062** with an upper opening, generally **7066**, and a lower channel portion including a partially planar and partially U-shaped lower seat **7068**, the channel **7064** having a width for operably snugly receiving the rod **7021** or portion of another longitudinal connector between the arms **7062**, the channel **7064** communicating with the base cavity **7061**.

[0660] Each of the arms **7062** has an interior surface, generally **7070**, that includes various inner cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **7072** located adjacent top surfaces **7073** of each of the arms **7062**. In the illustrated embodiment, the guide and advancement structure **7072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **7018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **7072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structures, such as a flange form, for operably guiding under rotation and advancing the closure structure **7018** downward between the arms **7062**, as well as eventual torquing when the closure structure **7018** abuts against the rod **7021** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0661] An opposed pair of upper tool receiving and engaging apertures or grooves **7074** are formed on outer surfaces **7076** of the arms **7062**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **7062**. Located directly below the apertures are another pair of tool receiving and engaging apertures or through bores, generally **7078**, that extend from the surfaces **7076** to the inner surfaces **7070**. The through bores **7078** each have a substantially planar bottom surface **7079** and arched or U-shaped upper and side surfaces **7080**. It is foreseen that other geometries are possible. As will be described in greater detail below, the through bores **7078** are sized and shaped to provide clearance within the receiver **7010** for down-loading the retainer **7012** from the receiver upper opening **7066** and between the interior surfaces **7070** of the arms **7062** and into the receiver cavity **7061**. The bores **7078** also provide access into the receiver **7010** for manipulating the retainer **7012** after loading and during assembly with the shank **7004**.

[0662] Returning to the interior surface **7070** of the receiver arms **7062**, located below each guide and advancement structure **7072** is a run-out feature for the guide and advancement structure **7072** partially defined by a discontinuous cylindrical surface **7082** having a diameter approximately the same or slightly greater than a greater diameter of the guide and advancement structure **7072**. Below the surface **7082**, moving in a direction toward the base **7060**, is another cylindrical surface **7084** having a diameter smaller than the diameter of the surface **7082** and illustrated as slightly greater than an inner or lesser diameter of the guide and advancement structure **7072**. The surface **7084** is also discontinuous, being formed only at the arms **7062**. Located between each of the surfaces **7082** and **7084** is a discontinuous annular surface **7085** running substantially

perpendicular to the axis B. Formed in each of the surfaces **7084** is a curved recess or aperture, generally **7088**, that is located adjacent to and directly above the respective through bore **7078**, an upper portion of the bore **7078** also formed in and through the surface **7084**. Each recess **7088** is partially defined by a substantially cylindrical surface **7089** and also by an arched or upside-down U-shaped surface **7090** that runs from the surface **7084** to the surface **7089**. The recesses **7088** cooperate with the retainer **7012** during assembly with the receiver **7010** and the shank **7004** as will be described in greater detail below.

[0663] The surface **7084** terminates at a lower ledge **7098** that runs radially outwardly from the surface **7084** to another cylindrical surface **7099**. The ledge **7098** is substantially perpendicular to the axis B. The cylindrical surface **7099** is partially discontinuous at the arms **7062** and also extends downwardly into the base **7060**, defining a continuous upper cylindrical portion of the base cavity **7061**. Each bore **7078** is substantially formed in the surface **7099** and extends outwardly to the arm surface **7076**. The cylindrical surface **7099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **7012**. The surfaces **7098** and **7099** define a circumferential recess that is sized and shaped to receive a portion of the retainer **7012** as it expands around the shank upper portion **7008** at the surface **7034** as the shank **8** moves upwardly toward the channel **7064** during assembly. A cylindrical surface **7101** located below the cylindrical surface **7099** is sized and shaped to closely receive the retainer **7012** when the retainer is in a neutral or slightly expanded position as will be described in greater detail below. Thus, the cylindrical surface **7101** has a diameter smaller than the diameter of the cylindrical surface **7099** that defines the expansion area for the retainer **7012**. The surface **7101** is joined or connected to the surface **7099** by one or more beveled, curved or conical surfaces **7102**. The surfaces **7102** allow for sliding gradual movement of the retainer **7012** into the space defined by the surface **6101** and ultimate seating of the retainer **7012** on a lower annular surface **7104** located below and adjacent to the cylindrical surface **7101**. Located below and adjacent to the annular seating surface **7104** is another substantially cylindrical surface **7106** that communicates with a beveled or flared bottom opening surface **7107**, the surface **7107** communicating with an exterior base surface **7108** of the base **7060**, defining a lower opening, generally **7110**, into the base cavity **7061** of the receiver **7010**.

[0664] With particular reference to FIGS. **307** and **310-315**, the open, friction fit retainer **7012** that operates to capture and frictionally engage the shank upper portion **7008** within the receiver **7010** has a central axis that is operationally the same as the axis B associated with the receiver **7010** when the shank upper portion **7008** and the retainer **7012** are installed within the receiver **7010**. The retainer **7012** includes a substantially cylindrical discontinuous lower body **7116**, a plurality of flex fingers or panels, **7117** extending upwardly from the body **7116** and a pair of opposed spring arms or tabs **7118**, also extending upwardly from the body **7116**. The retainer ring **7012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **7012** body **7116** may be expanded and the fingers and tabs (**7117** and **7118**) of the retainer may be manipulated during various steps of assembly as will be described in greater detail below. The retainer **7012** has a central channel or hollow through bore, generally **7121**, that passes entirely through the retainer **7012** from curved retainer arm or tab **7118** top surfaces **7122** to a bottom surface **7124** of the retainer body **7116**. Surfaces that define the channel or bore **7121** include an inner lower frusto-conical surface **7128** adjacent to the retainer body bottom surface **7124**, a substantially cylindrical surface **7130** adjacent the frusto-conical surface **7128**, a narrow frusto-conical or beveled surface **7131** adjacent the cylindrical surface **7130** and a partially discontinuous substantially spherical surface **7132** adjacent the surface **7131**, the surface **7132** being continuous near the cylindrical surface **7130** with the exception of a through slot or slit, generally **7134**. The surface **7132** is in a plurality of segments or pieces at the flex fingers **7117** wherein a plurality of substantially evenly spaced slots **7136** running outwardly and upwardly through an upper surface **7137** separate the surface **7132** into the individual flex fingers **7117**. In the illustrated embodiment, the slots **7136** and

the through slit **7134** form six substantially uniform flex fingers or tabs **7117** as well as partially define the two spring tabs **7118**, each finger and tab having the inner spherical surface **7132**. It is foreseen that more or fewer flex fingers may be made by the forming of more or fewer slots **7136**. The discontinuous spherical surface **7132** is sized and shaped to closely fit about and snap onto the shank surface **7034** during assembly as will be described in greater detail below. Preferably the surface **7132** has a radius the same or slightly smaller than the radius of the spherical shank surface **7034**. In operation, the discontinuous surface **7132** advantageously frictionally engages the bone screw shank upper portion **7008**, allowing for un-locked but non-floppy placement of the angle of the shank **7004** with respect to the receiver **7010** during surgery prior to locking of the shank **7004** with respect to the receiver **7010** near the end of the procedure. At the time of locking engagement, as shown in FIG. **336**, for example, downward and outward force placed on the retainer **7012** by the shank upper portion **7008** expands the retainer body **7116** at the slit **7134** and the individual flex fingers **7117** no longer frictionally grip the spherical surface **7034** of the upper portion **7008**. To aid in bending flexibility and resiliency, certain flex fingers **7117** have sloping outer surfaces **7138**, reducing a width of, or, as illustrated, substantially eliminating, the top planar surface **7137**, resulting in four of the fingers **7117** having a combination of a narrow top edge surface **7137** with an outwardly and downwardly sloping frusto-conical surface **7138**. It is foreseen that in other embodiments of the invention other surface geometries may be used to gain the level of resiliency desired for expansion and gripping of the fingers **7117** about the shank upper portion **7008**. It is noted that the fingers **7117** that are directed generally upwardly toward the receiver channel **7064**, some of which that include narrow top edges, advantageously sufficiently snap about and then grip the shank surface **7034** to an extent to provide the friction fit desired for non-floppy placement of the shank body **7006** at a desired angle with respect to the receiver **7010** during manipulation of the bone screws **7001** and the rod **7021** or other longitudinal connecting member during surgery. However, as compared to bone screw inserts such as collets known in the art that include downwardly directed portions or panels that are ultimately wedged between a receiver surface and a shank surface upon final locking of the shank to the receiver, the thin upwardly directed fingers **7117** that extend away from the shank locking surface that are not as strong as the retainer body **7116** do not participate or cooperate with the final locking of the shank upper portion **7008** to the retainer **7012** and the retainer **7012** to the receiver inner surfaces **7101** and **7104**. For such purpose, the more substantial retainer body **7116** having only the very narrow slit **7134** used for expansion purposes only is the component that locks the shank upper portion **7008** between the receiver **7010** and the rod **7021** or other longitudinal connecting member.

[0665] The retainer body **7116**, the flex fingers **7117** and a substantial part of each of the spring tabs **7118** have an outer substantially cylindrical profile, sized and shaped to closely and slidingly fit within the receiver cavity **7061** with the exception of outward extensions or wings, generally **7140**, of the spring tabs **7118** that are located adjacent to the upper surfaces **7122**, each extending outwardly away from the respective tab and having a curved outward surface **7142** that is substantially cylindrical, being sized and shaped to closely cooperate and frictionally engage the cylindrical surface **7089** of the receiver recess **7088**. Each spring tab **7118** further includes an inner planar surface **7144** that runs from the curved top surface **7122** to the inner cylindrical surface **7132**.

[0666] The through slit **7134** of the resilient retainer **7012** is defined by first and second end surfaces, **7146** and **7147** disposed in spaced relation to one another (they may also be touching) when the retainer is in a neutral state. Both end surfaces **7146** and **7147** are disposed substantially perpendicular to the bottom surface **7124**. A width X between the surfaces **7146** and **7147** is very narrow, in some embodiments of about or less than 0.004 inches, the narrow slit functioning to provide stability to the retainer **7012** during operation, specifically retention of the shank upper portion **7008** within the receiver **7010** that must withstand extreme pressure both during assembly and subsequent patient movement. The slit **7134** may be made, for example, by an electrical

discharge machining (EDM) process with the resulting surfaces **7146** and **7147** almost touching. Because the retainer **7012** is top loadable in a neutral state and the retainer **7012** does not need to be compressed to fit within the receiver cavity **7061**, the width X may be much smaller than what is often required for a bottom loaded compressible retainer ring. The gap X functions only in expansion to allow the retainer **7012** to expand about the shank upper portion **8** both during assembly and during locking of the polyaxial mechanism. The narrow gap X provides for a stronger retainer that has more surface contact with the shank upper portion **7008** upon locking, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **7012** body **7116** is only expanded and not compressed, the retainer **7012** does not undergo the mechanical stress that typically is placed on spring ring type retainers that may be both compressed and expanded more than once during assembly and locking. [0667] It is foreseen that in some embodiments of the invention, the retainer **7012** inner surfaces may include a roughening or additional material to increase the friction fit against the shank upper portion **7008** prior to lock down by the rod **7021** or other longitudinal connecting member. Also, the embodiment shown in FIGS. **310-315** illustrates the surfaces **7146** and **7147** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle.

[0668] With reference to FIGS. **307**, **336** and **339**, the illustrated elongate rod or longitudinal connecting member **7021** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **7022** of uniform diameter. The rod **7021** may be made from a variety of metals, metal alloys and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials.

[0669] Longitudinal connecting members for use with the assembly **7001** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the receiver **7010** may be modified so as to closely hold, and if desired, fix or slidingly capture the longitudinal connecting member to the assembly **7001**. Some embodiments of the assembly **7001** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the receiver **7010** of the receiver having a U-shaped, rectangular- or other-shaped channel, for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **7001**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **7001**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from deformable plastics to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

[0670] With reference to FIGS. **307** and **336**, the closure structure or closure top **7018** shown with the assembly **7001** is rotatably received between the spaced arms **7062** of the receiver **7010**. It is noted that the closure **7018** top could be a twist-in or slide-in closure structure. The illustrated

closure structure **7018** is substantially cylindrical and includes a an outer helically wound guide and advancement structure **7162** in the form of a flange that operably joins with the guide and advancement structure **7072** disposed on the arms **7062** of the receiver **7010**. The flange form utilized in accordance with the present invention may take a variety of forms, including those described in Applicant's U.S. Pat. No. 6,726,689, which is incorporated herein by reference. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure **7018** downward between the arms **7062** and having such a nature as to resist splaying of the arms **7062** when the closure structure **7018** is advanced into the channel **7064**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred in some embodiments due to the added strength provided by such flange form that beneficially cooperates with and counters any reduction in receiver strength that may occur in some embodiments that have a receiver of reduced profile designed for closely fitting with sleeves or other longitudinal connecting member components. The illustrated closure structure **7018** also includes a top surface **7164** with an internal drive **7166** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **7166** is used for both rotatable engagement and, if needed, disengagement of the closure **7018** from the receiver arms **7062**. It is also foreseen that the closure structure **7018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 70 to 140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **7168** of the closure is planar and further includes a point **7169** and a rim **7170** for engagement and penetration into the surface **7022** of the rod **7021** in certain embodiments of the invention. The closure top **7018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **7018** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **7062**.

[0671] Preferably, the receiver **7010** and the retainer **7012** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer spring tabs **7118** toward one another. In some circumstances, the shank **7004** is also assembled with the receiver **7010** and the retainer **7012** at the factory. In other instances, it is desirable to first implant the shank **7004**, followed by addition of the pre-assembled receiver and retainer at the insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **7004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized shank advantageously reduces inventory requirements, thus reducing overall cost.

[0672] Pre-assembly of the receiver **10** and the retainer **7012** is shown in FIGS. **321-329**. With particular reference to FIG. **321**, first the retainer **7012** is inserted into the upper receiver opening **7066**, leading with one of the spring tabs **7118** with both of the spring tab top surfaces **7122** facing one arm **7062** and the retainer bottom surface **7124** facing the opposing arm **7062**. The retainer **7012** is then lowered in such sideways manner into the channel **7064** and partially into the receiver cavity **7061**, followed by tilting the retainer **7212** such that the top surface **7122** and thereafter the outer tab or wing **7140** of the leading spring tab **7118** is moved into a nearby receiver arm through bore **7078** as shown in FIGS. **322** and **323**. With reference to FIGS. **323-326**, the retainer **7012** is then further tilted or turned and manipulated within the receiver to a position within the cavity until

the retainer **7012** bottom surface **7124** is directed toward the receiver cavity **7061** and the spring tab upper surfaces **7122** are facing upwardly toward the receiver channel opening **7066**. To accomplish such tilting and turning of the retainer **7012**, the spring tab arm **7118** located within the receiver bore **7078** is manipulated downwardly and then upwardly within the bore **7078** and finally shifted out of the bore **7078** when the opposed spring tab arm **7118** outer tab or wing **7140** moves past and clears the cylindrical surface **7084** of the receiver **7010** as shown in FIG. **326**. Once the retainer bottom surface **7124** seats on the receiver surface **7104**, both of the spring tab wings **7140** are partially located in opposed receiver bores **7078**. With reference to FIGS. **328** and **329**, a tool (not shown) is then used to grip the spring tab arms **7118** at outer surfaces thereof and squeeze or press the tabs **7118** toward one another while moving the retainer **7012** in an upward direction away from the surface **7104**. When the spring tab wing surfaces **7122** abut against the surface **7090**, the tool (not shown) is released and a portion or portions of each spring tab **7118** curved outer surface **7142** spring out to engage the cylindrical surface **7089** that defines a portion of the receiver recess or aperture **7088**. With reference to FIG. **329**, the retainer **7012** is now in a desired position for assembly with the shank **7004** with the retainer body **7116** located near and centrally within the cylindrical surface **7099**. Typically, the receiver and retainer combination are shipped or otherwise provided to the end user with the spring tab outer wings **7140** wedged against the receiver as shown in FIG. **329**. The receiver **7010** and the retainer **7012** combination is now pre-assembled and ready for assembly with the shank **7004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **7004** as will be described herein.

[0673] As illustrated in FIG. **337**, the bone screw shank **7004** or an entire assembly **7001** made up of the assembled shank **7004**, receiver **7010** and retainer **7012** is screwed into a bone, such as the vertebra **7017**, by rotation of the shank **7004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **7006** by engagement thereof at the drive feature **7042**.

Specifically, the vertebra **7017** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of the shank **7004** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank **7004** or the assembly **7001** is threaded onto the guide wire utilizing the cannulation bore **7050** by first threading the wire into the opening at the bottom **7028** and then out of the top opening at the drive feature **7042**. The shank **7004** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **7021** (also having a central lumen in some embodiments) and the closure top **7018** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires.

[0674] When the shank **7004** is driven into the vertebra **7017** without the remainder of the assembly **7001**, the shank **7004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer. With reference to FIGS. **330** and **337**, the pre-assembled receiver and retainer is placed above the shank upper portion **708** until the shank upper portion is received within the opening **7110**. As shown in these two figures, the receiver may be snapped or popped on to the shank with the shank and receiver axes aligned or at an angle with respect to one another. With particular reference to FIGS. **331-332**, as the shank is moved into the interior of the receiver base, the shank upper portion **7008** presses upwardly against the retainer **7012**, the engagement of the retainer spring tabs with the receiver surfaces **7090** keeping the retainer body **7116** in the space defined by the cylindrical surface **7099**. As the retainer **7012** presses up against the surface **7090**, the shank upper portion **7008** forces outward movement of the retainer body **7116** towards the cylindrical surface **7099** defining the receiver expansion recess as the spherical surface **7034** continues in an upward direction. With reference to FIG. **331**, the spring tabs **7118** may bow outwardly as the retainer body **7116** expands. The retainer **7012** body **7116** then begins to contract about the spherical surface **7034** as the center of the sphere passes beyond the center of the retainer

expansion recess defined by the surface **7099**. At this time also, the spherical surface **7034** engages the spherical surfaces **7132** of the retainer flex fingers **7117**, the fingers **7117** also prohibiting further upward movement of the shank **7004** into the channel **7064**. The frictional engagement between the surface **7034** and the surfaces **7132** provide for a desired friction fit between such components that is snug or close, but not locked. Furthermore, the position of the spring tab outer wings **7140** within the receiver recesses **7088** prohibits rotation of the now coupled retainer **7012** and shank **7004** about the receiver axis B which might otherwise occur if the retainer **7012** was equipped with flex fingers **7117** but not the upwardly and outwardly extending spring tabs **7118**. [0675] With reference to FIGS. **333-335**, the shank **7004** and attached retainer **7012** are then moved down into a final operative position by either an upward pull on the receiver **7010** or a downward pull on the shank, and/or, in some cases, by driving the shank **7004** further into the vertebra **7017**. As best shown in FIG. **335**, such movements snaps the retainer **7012** into place with the wings **7140** moving outwardly and being ultimately located in opposed bores **7078** of the receiver **7010** directly beneath the arched surfaces **7080**, the spring tabs **7118** now in a neutral position with the receiver surface **7080** prohibiting upward movement of the retainer **7012** and attached shank **7004** within the receiver **7010**. Furthermore, capture of the spring tab portions **7140** within the opposed receiver bores **7078** prevent rotation (about the axis B) of the retainer **7012** and shank **7004** combination with respect to the receiver **7010**. The shank body **7006** may now only be manipulated (pivoted and rotated) in a non-floppy manner with respect to the receiver **7010**. As also illustrated in FIG. **335**, the retainer body **7116** is now seated on the receiver surface **7104**. However, there is still space between the outer surface of the retainer body and the cylindrical surface **7101** of the receiver to allow for expansion locking of the retainer **7012** with respect to the receiver **7010** surface **7101** when a downward force is placed upon a rod or other captured connecting member as shown, for example, in FIG. **336**. In some embodiments, when the receiver **7010** is pre-assembled with the shank **7004**, the entire assembly **7001** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **7042** and rotating and driving the shank **7004** into a desired location of the vertebra **7017**.

[0676] With reference to FIGS. **336, 338** and **339**, the rod **7021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **7001**. The closure structure **7018** is then inserted into and advanced between the arms **7062** of each of the receivers **7010**. The closure structure **7018** is rotated, using a tool engaged with the inner drive **7166** until a selected pressure is reached at which point the rod **7021** engages the curved top surface **7040** of the shank **7004**, pressing the shank upper portion **7008** into locked frictional engagement with the retainer **7012**. Specifically, as the closure structure **7018** rotates and moves downwardly into the respective receiver **7010**, the point **7169** and rim **7170** engage and penetrate the rod surface **7022**, the closure structure **7018** pressing downwardly against and biasing the rod **7021** into compressive engagement with the shank upper surface that urges the shank upper portion **7008** toward the retainer **7012** and into locking engagement therewith, the retainer **7012** frictionally abutting the surface **7104** and expanding outwardly against the cylindrical surface **7101**. For example, about 7080 to about 7120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **7006** with respect to the receiver **7010**. It is noted that at this time, the retainer flex finger **7117** inner spherical surfaces **7132** may pull away from the shank spherical surface **7034** as shown in FIG. **336**. As the final locking of the shank **7004** with respect to the receiver **7010** has now been accomplished, such a pulling away of the retainer fingers from the shank upper portion **7008** is of no consequence. The non-floppy, friction fit relationship between the retainer flex fingers **7117** and the shank surface **7034** is a temporary, advantageous engagement providing bone anchor stability and maneuverability during the bone anchor implantation and rod placement process.

[0677] If removal of the rod **7021** from any of the bone screw assemblies **7001** is necessary, or if it is desired to release the rod **7021** at a particular location, disassembly is accomplished by using the

driving tool (not shown) that mates with the internal drive **7166** on the closure structure **7018** to rotate and remove such closure structure from the cooperating receiver **7010**. Disassembly is then accomplished in reverse order to the procedure described previously herein for assembly.

[0678] With particular reference to FIGS. **340-385** the reference number **8001** generally represents a polyaxial bone screw apparatus or assembly according to the present invention. The assembly **8001** includes a shank **8004**, that further includes a body **8006** integral with an upwardly extending upper portion or head-like capture structure **8008**; a receiver **8010**; and a lower retainer structure illustrated as a resilient open ring-like structure **8012**. The receiver **8010** and retainer structure **8012** are initially assembled and may be further assembled with the shank **8004** either prior or subsequent to implantation of the shank body **8006** into a vertebra **8017**, as will be described in greater detail below. FIG. **340** further shows a closure structure **8018** for capturing a longitudinal connecting member, for example, a rod **8021** which in turn presses against the shank upper portion **8008** into fixed frictional contact with the lower retainer **8012**, so as to capture, and fix the longitudinal connecting member **8021** within the receiver **8010** and thus fix the member **8021** relative to the vertebra **8017**. The illustrated rod **8021** is hard, stiff, non-elastic and cylindrical, having an outer cylindrical surface **8022**. It is foreseen that in other embodiments, the rod **8021** may be elastic, deformable and/or of a different cross-sectional geometry. Furthermore, the assembly **8001** may cooperate with longitudinal connecting members that include sleeves. The receiver **8010** and the shank **8004** cooperate in such a manner that the receiver **8010** and the shank **8004** can be secured at any of a plurality of angles, articulations or rotational alignments relative to one another and within a selected range of angles both from side to side and from front to rear, to enable flexible or articulated engagement of the receiver **8010** with the shank **8004** until both are locked or fixed relative to each other near the end of an implantation procedure.

[0679] The shank **8004** is substantially similar to the shank **6004** previously described herein with respect to the assembly **6001**. Thus, the shank **8004** includes the shank body **8006**, upper portion or head **8008**, a shank thread **8024**, a neck **8026**, a tip **8028**, a top of thread **8032**, an upper portion spherical surface **8034** a top surface **8040**, a drive feature **8042** and a cannulation bore **8050** the same or substantially similar to the respective body **6006**, upper portion or head **6008**, shank thread **6024**, neck **6026**, tip **6028**, top of thread **6032**, spherical surface **6034**, domed top surface **6040**, drive feature **6042** and cannulation bore **6050** previously described herein with respect to the shank **6004** of the assembly **6001**. To provide a biologically active interface with the bone, the threaded shank body **8006** may be coated, perforated, made porous or otherwise treated as previously discussed herein with respect to the shank body **6** of the assembly **1**.

[0680] With particular reference to FIGS. **340** and **349-353**, the receiver **8010** has a generally cylindrical and U-shaped appearance. The receiver **8010** has an axis of rotation B that is shown in FIG. **340** as being aligned with and the same as the axis of rotation A of the shank **8004**, such orientation being desirable, but not required during assembly of the receiver **8010** with the shank **8004**, as shown, for example, in FIG. **370**. After the receiver **8010** is pivotally attached to the shank **8004**, either before or after the shank **8004** is implanted in a vertebra **8017**, the axis B is typically disposed at an angle with respect to the axis A, as shown, for example, in FIG. **382**.

[0681] The receiver **8010** includes a substantially cylindrical base **8060** defining a bore or inner cavity, generally **8061**, the base **8060** being integral with a pair of opposed upstanding arms **8062** forming a cradle and defining a channel **8064** between the arms **8062** with an upper opening, generally **8066**, and a lower channel portion including a partially planar and partially U-shaped lower seat **8068**, as well as pairs of opposed, facing substantially planar perimeter surfaces extending upwardly from either side of the u-shaped seat **8068**, the channel **8064** having a width between the opposed surfaces **8069** for operably snugly receiving the insert **8014** and the rod **8021** or portion of another longitudinal connector between the arms **8062**, the channel **8064** communicating with the base cavity **8061**.

[0682] Each of the arms **8062** has a pair of perimeter surfaces **8069** and an interior surface,

generally **8070** located therebetween, the surface **8070** including various inner concave and substantially cylindrical profiles, an upper one of which is a partial helically wound guide and advancement structure **8072** located adjacent top surfaces **8073** of each of the arms **8062**. In the illustrated embodiment, the guide and advancement structure **8072** is a partial helically wound interlocking flangeform configured to mate under rotation with a similar structure on the closure structure **8018**, as described more fully below. However, it is foreseen that for certain embodiments of the invention, the guide and advancement structure **8072** could alternatively be a square-shaped thread, a buttress thread, a reverse angle thread or other thread-like or non-threadlike helically wound discontinuous advancement structures, such as a flange form, for operably guiding under rotation and advancing the closure structure **8018** downward between the arms **8062**, as well as eventual torquing when the closure structure **8018** abuts against the rod **8021** or other longitudinal connecting member. It is foreseen that the arms could have break-off extensions.

[0683] An opposed pair of upper tool receiving and engaging apertures or grooves **8074** are formed on outer surfaces **8076** of the arms **8062**. It is foreseen that tool receiving grooves or apertures may be configured in a variety of shapes and sizes and be disposed at other locations on the receiver arms **8062**. Located directly below the apertures are another pair of tool receiving and engaging apertures or through bores, generally **8078**, that are illustrated as having a keyhole shape, and extend from the surfaces **8076** to the inner surfaces **8070**. The through bores **8078** each have a substantially planar bottom surface **8079** and keyhole-like curved side surfaces **8080** and an upper arched surface **8081**. It is foreseen that other geometries are possible. As will be described in greater detail below, the through bores **8078** are sized and shaped to provide clearance within the receiver **8010** for down-loading the retainer **8012** from the receiver upper opening **8066** and between the interior surfaces **8070** of the arms **8062** and into the receiver cavity **8061**. The bores **8078** also provide access into the receiver **8010** for manipulating the retainer **8012** and/or the insert **8014** during and after assembly.

[0684] Returning to the interior surface **8070** of the receiver arms **8062**, located below each guide and advancement structure **8072** is a run-out feature for the guide and advancement structure **8072** partially defined by a discontinuous cylindrical surface **8082** having a diameter approximately the same or slightly greater than a greater diameter of the guide and advancement structure **8072**. Below the surface **8082**, moving in a direction toward the base **8060**, is another cylindrical surface **8084** having a diameter smaller than the diameter of the surface **8082** and illustrated as slightly greater than an inner or lesser diameter of the guide and advancement structure **8072**. The surface **8084** is also discontinuous, being formed only at the arms **8062**. Located between each of the surfaces **8082** and **8084** is a discontinuous annular surface **8085** running substantially perpendicular to the axis B. Formed in each of the surfaces **8084** are curved recesses or apertures, generally **8088**, each partially formed by a cylindrical surface **8089** and arched surfaces **8090** extending from the surface **8089** to the surface **8084**. The recesses **8088** are located adjacent to and at either side of the respective through bore **8078**, the upper arched portion **8081** of the bore **8078** also being formed in and through the surface **8084**. The surfaces forming the recesses **8088** cooperate with the retainer **8012** during assembly with the receiver **8010** and the shank **8004**, allowing the resilient retainer **8012** to be temporarily retained in an upper portion of the receiver as will be described in greater detail below.

[0685] Returning to the substantially planar peripheral surfaces **8069**, each arm **8062** includes a pair of projecting ridges or stops **8092**, located on each surface **8069**, for a total of four stops **8092** that are located near the annular surface **8085** and are spaced from the cylindrical surface **8084**. The stops **8092** of one arm **8062** face the opposing pair of stops **8092** on the other arm **8062**, each stop **8092** projecting outwardly from the respective planar surface **8069**. The illustrated stops **8092** are elongate, running from arm outside or edge surfaces **8094** toward the respective cylindrical surface **8084** in a direction perpendicular to the axis B. As will be described in greater detail below, the stops **8092** cooperate with surfaces of the insert **8014** to retain the insert **8014** within the

channel **8064** of the receiver **8010**. In the illustrated embodiment, at a location below the stops **8092**, each arm includes a curved surface **8095** connecting each substantially planar surface **8069** with the U-shaped seat **8068**, an edge **8096** forming a juncture of the curved surface **8095** and the U-shaped seat **8068**. As will be described in greater detail below and is shown in FIGS. **374-379**, for example, when the insert **8014** is positioned in the receiver channel **8064** between the surfaces **8069** and below the stops **8092**, the insert **8014** initially typically rests or seats at or near the surface **8095** and the edge **8096** and is later pressed along and below the edge **8096** and into the seat **8068** into frictional, locking engagement with the receiver **8010**.

[0686] Returning to FIGS. **349-353**, the surface **8084** terminates at a lower ledge **8098** that runs radially outwardly from the surface **8084** to another cylindrical surface **8099**. The ledge **8098** is substantially perpendicular to the axis B and located adjacent the curved surfaces **8095** that partially form the edges **8096**. The cylindrical surface **8099** is partially discontinuous at the arms **8062** and also extends downwardly into the base **8060**, defining a continuous upper cylindrical portion of the base cavity **8061**. Each bore **8078** is substantially formed in the surface **8099** (with the exception of the upper arched portion **8081** that is formed in the surface **8084**), the bore **8078** extending outwardly to the arm surface **8076**. The cylindrical surface **8099** is oriented substantially parallel to the axis B and is sized and shaped to receive an expanded retainer **8012**. The surfaces **8098** and **8099** define a circumferential recess that is sized and shaped to receive a portion of the retainer **8012** as it expands around the shank upper portion **8008** at the surface **8034** as the shank **8008** moves upwardly toward the channel **8064** during assembly. A cylindrical surface **8101** located below the cylindrical surface **8099** is sized and shaped to closely receive the retainer **8012** when the retainer is in a neutral or slightly expanded position as will be described in greater detail below. Thus, the cylindrical surface **8101** has a diameter smaller than the diameter of the cylindrical surface **8099** that defines the expansion area for receiving the retainer **8012**. The surface **8101** is joined or connected to the surface **8099** by one or more beveled, curved or frusto-conical surfaces **8102**. The surfaces **8102** allow for sliding gradual movement of the retainer **8012** into the space defined by the surface **8101** and ultimate seating of the retainer **8012** on a lower annular surface **8104** located below and adjacent to the cylindrical surface **8101**. Located below and adjacent to the annular seating surface **8104** is a circular edge or narrow substantially cylindrical surface **8106** that communicates with a beveled or flared bottom opening surface **8107**, the surface **8107** communicating with an exterior base surface **8108** of the base **8060**, defining a lower opening, generally **8110**, into the base cavity **8061** of the receiver **8010**.

[0687] With particular reference to FIGS. **340** and **343-348**, the open, friction fit retainer **8012** that operates to capture and frictionally engage the shank upper portion **8008** within the receiver **8010** has a central axis that is operationally the same as the axis B associated with the receiver **8010** when the shank upper portion **8008** and the retainer **8012** are installed within the receiver **8010**. The retainer **8012** includes a substantially cylindrical discontinuous lower body **8116**, a plurality of flex fingers or panels, **8117** extending upwardly from the body **116** and a pair of opposed spring arms or tabs **8118**, also extending upwardly from the body **8116**. The retainer ring **8012** is made from a resilient material, such as a stainless steel or titanium alloy, so that the retainer **8012** body **8116** may be expanded and the fingers and tabs (**8117** and **8118**) of the retainer may be manipulated during various steps of assembly as will be described in greater detail below. The retainer **8012** has a central channel or hollow through bore, generally **8121**, that passes entirely through the retainer **8012** from curved retainer arm or tab **8118** top surfaces **8122** to a bottom surface **8124** of the retainer body **8116**. Surfaces that define the channel or bore **8121** include an inner lower frusto-conical surface **8128** adjacent to the retainer body bottom surface **8124**, a substantially cylindrical surface **8130** adjacent the frusto-conical surface **8128** and a partially discontinuous substantially spherical surface **8132** adjacent the surface **8130**, the surface **8132** being continuous near the cylindrical surface **8130** with the exception of a through slot or slit, generally **8134**. The surface **8132** is in a plurality of segments or pieces at the flex fingers **8117** wherein a plurality of

substantially evenly spaced slots **8136** running outwardly and upwardly through an upper surface **8137** separate the surface **8132** into the individual flex fingers **8117**. In the illustrated embodiment, the slots **8136** and the through slit **8134** form six substantially uniform flex fingers or tabs **8117** as well as partially define the two spring tabs **8118**, each finger and tab having the inner spherical surface **8132**. It is foreseen that more or fewer flex fingers may be made by the forming of more or fewer slots **8136**. The discontinuous spherical surface **8132** is sized and shaped to closely fit about and snap onto the shank surface **8034** during assembly as will be described in greater detail below. The surface **8132** may have a radius the same, slightly larger or slightly smaller than the radius of the spherical shank surface **8034**. The surface **8132** and/or the shank surface **8034** may include a surface treatment for enhancing friction between such surfaces. In some embodiments, the flexible tabs **8117** may be bent to further enhance frictional engagement. In other embodiments, some or all of the spherical surface **8132** may be replaced by planar or faceted surfaces. In operation, the discontinuous surface **8132** advantageously frictionally engages the bone screw shank upper portion **8008**, allowing for un-locked but non-floppy placement of the angle of the shank **8004** with respect to the receiver **8010** during surgery prior to locking of the shank **8004** with respect to the receiver **8010** near the end of the procedure. At the time of locking engagement, downward and outward force placed on the retainer **8012** by the shank upper portion **8008** expands the retainer body **8116** at the slit **8134** and the individual flex fingers **8117** no longer frictionally grip the spherical surface **8034** of the upper portion **8008**. In some embodiments, to aid in bending flexibility and resiliency, certain flex fingers **8117** may have sloping outer surfaces (not shown) that reduce a width of, or substantially eliminate the top planar surface **8137**. It is foreseen that in other embodiments of the invention other surface geometries may be used to gain a level of resiliency desired for expansion and gripping of the fingers **8117** about the shank upper portion **8008**. It is noted that the fingers **8117** that are directed generally upwardly toward the receiver channel **8064** advantageously sufficiently snap about and then grip the shank surface **8034** to an extent to provide the friction fit desired for non-floppy placement of the shank body **8006** at a desired angle with respect to the receiver **8010** during manipulation of the bone screws **8001** and the rod **8021** or other longitudinal connecting member during surgery. However, as compared to bone screw inserts such as collets known in the art that include downwardly directed portions or panels that are ultimately wedged between a receiver surface and a shank surface upon final locking of the shank to the receiver, the thin upwardly directed fingers **8117** that extend away from the shank locking surface that are not as strong as the retainer body **8116**, do not participate or cooperate with the final locking of the shank upper portion **8008** to the retainer **8012** and the retainer **8012** to the receiver inner surfaces **8101** and **8104**. For such purpose, the more substantial retainer body **8116** having only the very narrow slit **8134**, used for expansion purposes only, is the component that locks the shank upper portion **8008** between the receiver **8010** and the rod **8021** or other longitudinal connecting member.

[0688] The retainer body **8116**, the flex fingers **8117** and a substantial part of each of the spring tabs **8118** have an outer substantially cylindrical profile, sized and shaped to closely and slidably fit within the receiver cavity **8061** with the exception of outward extensions or wings, generally **8140**, of the spring tabs **8118** that are located adjacent to the upper surfaces **8122**, each wing extending outwardly away from the respective tab body **8118** and having a curved outward surface **8142** that is substantially cylindrical with a curved or frusto-conical lower portion, the surfaces **8142** being sized and shaped to closely cooperate and frictionally engage the cylindrical surface **8089** of the receiver recess **8088**. Each spring tab **8118** further includes first and second inner planar surfaces **8144** and **8145**, the surface **8144** running from the curved top surface **8122** to the surface **8145** and the surface **8145** running from the surface **8144** to the inner spherical surface **8132**. It is foreseen that in other embodiments of the invention, fewer or greater number of planar or other surfaces with other geometries may extend between the top surface **8122** and the spherical surface **8132**.

[0689] The through slit **8134** of the resilient retainer **8012** is defined by first and second end surfaces, **8146** and **8147** disposed in spaced relation to one another (they may also be touching) when the retainer is in a neutral state. Both end surfaces **8146** and **8147** are disposed substantially perpendicular to the bottom surface **8124**. A width X between the surfaces **8146** and **8147** is very narrow, in some embodiments of about or less than 0.004 inches, the narrow slit functioning to provide stability to the retainer **8012** during operation, specifically retention of the shank upper portion **8008** within the receiver **8010** that must withstand extreme pressure both during assembly and subsequent patient movement. The slit **8134** may be made, for example, by an electrical discharge machining (EDM) process with the resulting surfaces **8146** and **8147** almost touching. Because the retainer **8012** is top loadable in a neutral state and the retainer **8012** does not need to be compressed to fit within the receiver cavity **8061**, the width X may be much smaller than what is often required for a bottom loaded compressible retainer ring. The gap X functions only in expansion to allow the retainer **8012** to expand about the shank upper portion **8** both during assembly and during locking of the polyaxial mechanism. The narrow gap X provides for a stronger retainer that has more surface contact with the shank upper portion **8008** upon locking, resulting in a sturdier connection with less likelihood of failure than a retainer ring having a greater gap. Furthermore, because the retainer **8012** body **8116** is only expanded and not compressed, the retainer **8012** does not undergo the mechanical stress that typically is placed on spring ring type retainers that may be both compressed and expanded more than once during assembly and locking. [0690] It is foreseen that in some embodiments of the invention, the retainer **8012** inner surfaces may include a roughening or additional material to increase the friction fit against the shank upper portion **8008** prior to lock down by the rod **8021** or other longitudinal connecting member. Also, the embodiment shown in FIGS. **343-348** illustrates the surfaces **8146** and **8147** as substantially parallel, however, it is foreseen that it may be desirable to orient the surfaces obliquely or at a slight angle.

[0691] With particular reference to FIGS. **340** and **354-361**, the lock and release insert **8014** is illustrated that is sized and shaped to be received by and down-loaded into the receiver **8010** at the upper opening **8066**. The insert **8014** has an operational central axis that is the same as the central axis B of the receiver **8010**. In operation, an insert **8014** that has been pressed downwardly during the locking of the shank **8004** in a desired angular position with respect to the receiver **8010**, by, for example, compression from the rod **8021** and closure top **8018**, is wedged into engagement with the receiver **8010** at outer edge surfaces of the receiver arms, the insert retaining the shank **8006** in a locked position even if the rod **8021** and closure top **8018** are removed as shown in FIG. **383**. Such locked position may also be released by the surgeon if desired. The insert **8014** is thus preferably made from a resilient material, such as a stainless steel or titanium alloy, so that portions of the insert may be pinched and un-wedged from the receiver **8010**.

[0692] The insert **8014** includes a substantially U-shaped body **8150** having opposed ends, generally **8151**, the body **8150** being sized and shaped to extend completely through the U-shaped channel **8064** between the opposed outer surfaces **8094** of the arms **8062** so as to cooperate with the receiver arm outer side surfaces **8069**, the stops **8092**, and the insert wedging edge surfaces **8096** formed by each curved surface **8095** and the channel seat **8068**. A U-shaped channel surface or saddle **8153** formed in the body **8150** also extends between the insert ends **8151** and when the insert **8014** is assembled with the receiver **8010**, the saddle **8153** substantially aligns with the receiver channel **8064**. The saddle **8153** is formed by the insert body **8150** and by two upstanding arms **8157** and is sized and shaped to closely receive the rod **8021** or other longitudinal connecting member. A bore, generally **8160**, is disposed primarily within and through the insert body **8156** that runs along the axis B and communicates with the U-shaped channel formed by the saddle **8153** and upstanding arms **8157**. The bore **8160** is sized and shaped to provide space and clearance for the shank head portion **8040** to extend therethrough so that a rod **8021** or other connecting member seated on the saddle **8153** also directly frictionally engages the spherical surface **8040**. The bore

8160 is also sized such that in any angular position of the shank **8004** with respect to the receiver **8010**, the spherical surface **8040** does not directly engage the insert **8014**, but rather is in contact with the rod **8021** or other longitudinal connecting member. As best shown in FIGS. **381** and **382**, when the shank **8004** is locked in any angular position by the rod **8021**, the insert **8014** contacts the shank **8004** at the spherical surface **8034** and not the spherical surface **8040**. It is foreseen that an alternative insert embodiment may be configured to include planar holding surfaces that closely hold a square or rectangular bar as well as hold a cylindrical rod-shaped, cord, or sleeved cord longitudinal connecting member.

[0693] The arms **8157** disposed on either side of the saddle **8153** and extend upwardly therefrom are sized and configured for ultimate placement above the retainer spring tabs **8118** and beneath the cylindrical run-out surface **8082** located below the receiver guide and advancement structure **8072**. The arms **8157** include outer curved, convex surfaces **8163** that is illustrated as partially cylindrical and curved top surfaces **8164** that are ultimately positioned in spaced relation with the closure top **8018**, so that the closure top **8018** frictionally engages the rod **8021** only, pressing the rod **8021** downwardly against both the shank top surface **8040** and the insert saddle **8153**, the shank **8004** upper portion **8008** then pressing against the retainer **8012** to lock the polyaxial mechanism of the bone screw assembly **8001** at a desired angle. The partially cylindrical surface **8163** extends from the top surface **8164** to a bottom surface **8165** of the insert **8014**. The surface **8163** is sized and shaped to generally fit within the receiver surface **8084**. Formed in each surface **8163** and extending through the saddle **8153** surface is a through bore **8166**, the bore **8166** used for manipulation and removal of the insert **8014** from the receiver through the receiver bore **8074**. A recessed surface portion **8167** located beneath each bore **8166** is sized and shaped to receive the curved upper surface **8122** of a wing **8140** of a retainer **8012** spring tab **8118**. A portion of the recessed portion **8167** extends completely through the insert **8014** and is defined by a lower notched surface **8169**. The recessed portion **8167** is further defined by an upper arched surface **8170** that communicates with the bore **8166** and a flat or slightly convex surface **8171** that extends from the arched surface **8170** to the lower notched surface **8169**.

[0694] The insert **8014** extends from the substantially cylindrical outer arms surfaces **8163** equally outwardly to each end **8151**. Substantially planar outer side surfaces **8172** extend from each arm surface **8163** to a substantially planar surface **8174** disposed perpendicular thereto, the surfaces **8174** substantially defining each of the ends **8151**. Also, adjacent to the side surfaces **8172**, substantially planar upper surfaces **8175** run from the arms **8157** to the end surface **8174**. A recess, generally **8176**, is located directly beneath the side surfaces **8172** and is also formed in each end surface **8174**. Each recess **8176** extends all the way from the end surface **8174** to the arm surface **8163** and is substantially defined by a substantially planar tapering surface **8177** and an upper lip **8178**. Pairs of opposed surfaces **8177** are sized and shaped to wedge against and between opposed surfaces **8068** forming the seat of the receiver channel **8064** to lock the insert **8014** against the receiver **8012** and thus lock the polyaxial mechanism of the assembly **8001** as best shown in FIGS. **378** and **379**, for example. Portions of the surfaces **8177** and respective adjacent end surfaces **8174** terminate at a lower surface **8179** that curves or tapers downwardly to the base rim **8165**. Further cut-outs, tapers or bevels may be made to the surfaces to provide adequate clearance and ease of manipulation of the insert **8014** within the receiver **8010**, such as the angular surfaces **8179'** running from the surface **179** to each of the surfaces **8177**.

[0695] The insert bore, generally **8160**, is substantially defined at the body **8150** by an inner cylindrical surface **8180** that communicates with the saddle **8153** and a lower concave substantially spherical surface **8181** having a radius the same or substantially similar to a radius of the surface **8034** of the shank upper portion **8008**. The surface **8181** terminates at the base **8165** and the lower notched surface **8169**. The through bore **8160** is not completely cylindrical at the saddle surface **8153**, with portions of the bore extending outwardly towards each end **8151** to provide more than adequate clearance for the shank upper portion surface to fully and directly engage the rod **8021** or

other longitudinal connecting member at any and all angular positions of the shank **8004** with respect to the receiver **8010**. The bore **8160** is also sized and shaped to receive the driving tool (not shown) therethrough that engages the shank drive feature when the shank body **8006** is driven into bone with the receiver **8010** attached. Also, the bore **8160** receives a manipulation tool (not shown) used for releasing the insert **8014** from a locked position with the receiver, the tool pressing down on the shank and also gripping the insert **8014** at the opposed through bores **8166** or with other tool engaging features. A manipulation tool for un-wedging the insert **8014** from the receiver **8010** may also access the bores **8166** from the receiver through bores **8074**. The illustrated insert **8014** may further include other features, including grooves and recesses for manipulating and holding the insert **8014** within the receiver **8010** and providing adequate clearance between the retainer **8012** and the insert **8014**.

[0696] As will be discussed in greater detail below, frictional engagement between the insert **8014** and the receiver **8010**, more particularly, the wedging of the tapered surfaces **8177** into the edge **8096** defined by the seat surfaces **8068**, provides independent locking of the polyaxial mechanism of the assembly **8001**, maintaining the upper shank portion **8008** in locked engagement by and between the retainer **8012** and the insert **8014** even if the closure top **8018** and/or rod **8021** are thereafter removed from the receiver **8010**.

[0697] With reference to FIGS. **340** and **377-382**, the illustrated elongate rod or longitudinal connecting member **8021** (of which only a portion has been shown) can be any of a variety of implants utilized in reconstructive spinal surgery, but is typically a cylindrical, elongate structure having the outer substantially smooth, cylindrical surface **8022** of uniform diameter. The rod **8021** may be made from a variety of metals, metal alloys and deformable and less compressible plastics, including, but not limited to rods made of elastomeric, polyetheretherketone (PEEK) and other types of materials, such as polycarbonate urethanes (PCU).

[0698] Longitudinal connecting members for use with the assembly **1** may take a variety of shapes, including but not limited to rods or bars of oval, rectangular or other curved or polygonal cross-section. The shape of the insert **8014** may be modified so as to closely hold the particular longitudinal connecting member used in the assembly **8001**. Some embodiments of the assembly **8001** may also be used with a tensioned cord. Such a cord may be made from a variety of materials, including polyester or other plastic fibers, strands or threads, such as polyethylene-terephthalate. Furthermore, the longitudinal connector may be a component of a longer overall dynamic stabilization connecting member, with cylindrical or bar-shaped portions sized and shaped for being received by the compression insert **8014** of the receiver having a U-shaped, rectangular- or other-shaped channel, for closely receiving the longitudinal connecting member. The longitudinal connecting member may be integral or otherwise fixed to a bendable or damping component that is sized and shaped to be located between adjacent pairs of bone screw assemblies **8001**, for example. A damping component or bumper may be attached to the longitudinal connecting member at one or both sides of the bone screw assembly **8001**. A rod or bar (or rod or bar component) of a longitudinal connecting member may be made of a variety of materials ranging from deformable plastics to hard metals, depending upon the desired application. Thus, bars and rods of the invention may be made of materials including, but not limited to metal and metal alloys including but not limited to stainless steel, titanium, titanium alloys and cobalt chrome; or other suitable materials, including plastic polymers such as polyetheretherketone (PEEK), ultra-high-molecular weight-polyethylene (UHMWP), polyurethanes and composites, including composites containing carbon fiber, natural or synthetic elastomers such as polyisoprene (natural rubber), and synthetic polymers, copolymers, and thermoplastic elastomers, for example, polyurethane elastomers such as polycarbonate-urethane elastomers.

[0699] With reference to FIGS. **340** and **377-382**, the closure structure or closure top **18** shown with the assembly **8001** is rotatably received between the spaced arms **8062** of the receiver **8010**. It is noted that the closure **8018** top could be a twist-in or slide-in closure structure. The illustrated

closure structure **8018** is substantially cylindrical and includes a an outer helically wound guide and advancement structure **8182** in the form of a flange that operably joins with the guide and advancement structure **8072** disposed on the arms **8062** of the receiver **8010**. The flange form utilized in accordance with the present invention may take a variety of forms, including those described in Applicant's U.S. Pat. No. 6,726,689, which is incorporated herein by reference. Although it is foreseen that the closure structure guide and advancement structure could alternatively be a buttress thread, a square thread, a reverse angle thread or other thread like or non-thread like helically wound advancement structure, for operably guiding under rotation and advancing the closure structure **8018** downward between the arms **8062** and having such a nature as to resist splaying of the arms **8062** when the closure structure **8018** is advanced into the channel **8064**, the flange form illustrated herein as described more fully in Applicant's U.S. Pat. No. 6,726,689 is preferred as the added strength provided by such flange form beneficially cooperates with and counters any reduction in strength caused by the any reduced profile of the receiver **8010** that may more advantageously engage longitudinal connecting member components. The illustrated closure structure **8018** also includes a top surface **184** with an internal drive **8186** in the form of an aperture that is illustrated as a star-shaped internal drive such as that sold under the trademark TORX, or may be, for example, a hex drive, or other internal drives such as slotted, tri-wing, spanner, two or more apertures of various shapes, and the like. A driving tool (not shown) sized and shaped for engagement with the internal drive **8186** is used for both rotatable engagement and, if needed, disengagement of the closure **8018** from the receiver arms **8062**. It is also foreseen that the closure structure **8018** may alternatively include a break-off head designed to allow such a head to break from a base of the closure at a preselected torque, for example, 8070 to 8140 inch pounds. Such a closure structure would also include a base having an internal drive to be used for closure removal. A base or bottom surface **8188** of the closure is planar and further includes a point **8189** and a rim **8190** for engagement and penetration into the surface **8022** of the rod **8021** in certain embodiments of the invention. The closure top **8018** may further include a cannulation through bore (not shown) extending along a central axis thereof and through the top and bottom surfaces thereof. Such a through bore provides a passage through the closure **8018** interior for a length of wire (not shown) inserted therein to provide a guide for insertion of the closure top into the receiver arms **8062**.

[0700] An alternative closure top **8018'** for use with a deformable rod **8021'**, such as a PEEK rod, is shown in FIGS. **384** and **385**. The top **8018'** is identical to the top **8018** with the exception that a point **8189'** is located on a domed surface **8190'** in lieu of the planar bottom with point and rim of the closure top **8018**.

[0701] Preferably, the receiver **8010**, the retainer **8012** and the insert **8014** are assembled at a factory setting that includes tooling for holding and alignment of the component pieces and pinching or compressing of the retainer **8012** spring tabs **8118** and manipulating the insert **8014**. In some circumstances, the shank **8004** is also assembled with the receiver **8010**, the retainer **8012** and the insert **8014** at the factory. In other instances, it is desirable to first implant the shank **8004**, followed by addition of the pre-assembled receiver, retainer and insert at the patient's insertion point. In this way, the surgeon may advantageously and more easily implant and manipulate the shanks **8004**, distract or compress the vertebrae with the shanks and work around the shank upper portions or heads without the cooperating receivers being in the way. In other instances, it is desirable for the surgical staff to pre-assemble a shank of a desired size and/or variety (e.g., surface treatment of roughening the upper portion **8008** and/or hydroxyapatite on the shank **8006**), with the receiver, retainer and compression insert. Allowing the surgeon to choose the appropriately sized or treated shank **8004** advantageously reduces inventory requirements, thus reducing overall cost.

[0702] Pre-assembly of the receiver **8010**, the retainer **8012** and the insert **8014** is shown in FIGS. **362-369**. With particular reference to FIG. **362**, first the retainer **8012** is inserted into the upper receiver opening **8066**, leading with one of the spring tabs **8118** with both of the spring tab top

surfaces **8122** facing one arm **8062** and the retainer bottom surface **8124** facing the opposing arm **8062**. With reference to FIG. **362** and also FIGS. **363** and **364**, the retainer **8012** is then lowered in such sideways manner into the channel **8064** and partially into the receiver cavity **8061**, followed by tilting the retainer **8212** such that the top surface **8122** and thereafter the outer tab or wing **8140** of the leading spring tab **8118** is moved into a nearby receiver arm through bore **8078**. With reference to FIG. **365**, the retainer **8012** is then further tilted or turned and manipulated within the receiver to a position within the cavity until the retainer **8012** bottom surface **8124** is directed toward the receiver cavity **8061** and the spring tab upper surfaces **8122** are facing upwardly toward the receiver channel opening **8066**. To accomplish such tilting and turning of the retainer **8012**, the spring tab arm **8118** located within the receiver bore **8078** is manipulated downwardly and then upwardly within the bore **8078** and finally shifted out of the bore **8078** when the opposed spring tab arm **8118** outer tab or wing **8140** moves past and clears the cylindrical surface **8084** of the receiver **8010**. Once the retainer bottom surface **8124** seats on the receiver surface **8104**, both of the spring tab wings **8140** are partially located in opposed receiver bores **8078**.

[0703] With reference to FIGS. **365** and **366**, the compression insert **8014** is then downloaded into the receiver **8010** through the upper opening **8066** with the bottom surface **8179** facing the receiver arm top surfaces **8073** and the insert arms **8157** aligned with the receiver arms **8062**. The insert **8014** is then lowered toward the channel seat **8068** until the insert **8014** arm upper surfaces **8164** are adjacent the run-out area below the guide and advancement structure **8072** defined in part by the cylindrical surface **8082**, with the U-shaped channel or saddle surface **8153** of the insert **8014** aligned with the channel **8064** of the receiver **8010**. With reference to FIG. **366**, at this time, the side surfaces **8172** at the insert ends **8151** are located above the four stops **8092** located on the receiver inner side surfaces **8069** with the lips **8178** resting on each of the stops **8092**. With reference to FIG. **367**, the insert **8014** is then pushed downwardly toward the receiver base **8060**, the resilient u-shaped saddle **8153** being slightly compressed inwardly until the surfaces **8172** pass over the stops **8092**. At this time, the insert **8014** is captured within the receiver **8010** between the stops **8092** and the retainer **8012**.

[0704] With reference to FIGS. **368** and **369**, a tool (not shown) is then used to grip the retainer spring tab arms **8118** at outer surfaces thereof and squeeze or press the tabs **8118** toward one another while moving the retainer **8012** in an upward direction away from the receiver surface **8104**. With reference to FIG. **369**, when the spring tab wing surface projections **8142** face the receiver surface **8089**, the tool (not shown) is released and a portion or portions of each spring tab **8118** spring out to engage the surface **8089**. The retainer **8012** and the insert **8014** are now in a desired position for shipping and also for assembly with the shank **8004**. The insert **8014** recessed areas **8167** are located adjacent to the retainer spring tab top surfaces **8122**. The insert **8014** is fully captured within the receiver **8010** by the stops **8092** and the geometry of the insert **8014** that extends fully within the channel **8064** of the receiver **8010** advantageously provides an insert **8014** with a saddle **8153** fully aligned with the receiver channel **8064** that cannot be rotated out of alignment as may occur with known inserts that are substantially cylindrical in form.

[0705] Typically, the receiver, insert and retainer combination are shipped or otherwise provided to the end user with the spring tab outer wings **8140** wedged against the receiver as shown in FIG. **369**. The receiver **8010**, retainer **8012** and insert **8014** combination is now pre-assembled and ready for assembly with the shank **8004** either at the factory, by surgery staff prior to implantation, or directly upon an implanted shank **8004** as will be described herein.

[0706] As illustrated in FIG. **370**, the bone screw shank **8004** or an entire assembly **8001** made up of the assembled shank **8004**, receiver **8010**, retainer **8012** and compression insert **8014**, is screwed into a bone, such as the vertebra **8017**, by rotation of the shank **8004** using a suitable driving tool (not shown) that operably drives and rotates the shank body **8006** by engagement thereof at the drive **8042**. Specifically, the vertebra **8017** may be pre-drilled to minimize stressing the bone and have a guide wire (not shown) inserted therein to provide a guide for the placement and angle of

the shank **8004** with respect to the vertebra. A further tap hole may be made using a tap with the guide wire as a guide. Then, the bone screw shank **8004** or the entire assembly **8001** is threaded onto the guide wire utilizing the cannulation bore **8050** by first threading the wire into the opening at the bottom **8028** and then out of the top opening at the drive feature **8042**. The shank **8004** is then driven into the vertebra using the wire as a placement guide. It is foreseen that the shank and other bone screw assembly parts, the rod **8021** (also having a central lumen in some embodiments) and the closure top **8018** (also with a central bore) can be inserted in a percutaneous or minimally invasive surgical manner, utilizing guide wires. When the shank **8004** is driven into the vertebra **8017** without the remainder of the assembly **8001**, the shank **8004** may either be driven to a desired final location or may be driven to a location slightly above or proud to provide for ease in assembly with the pre-assembled receiver, compression insert and retainer.

[0707] With further reference to FIG. **370**, the pre-assembled receiver, insert and retainer are placed above the shank upper portion **8008** until the shank upper portion is received within the opening **8110**. With particular reference to FIGS. **371** and **372**, as the shank upper portion **8008** is moved into the interior **8061** of the receiver base, the shank upper portion **8008** presses upwardly against the retainer **8012** in the recess partially defined by the cylindrical surface **8099**. As the portion **8008** continues to move upwardly toward the channel **8064**, the surface **8034** forces outward movement of the retainer **8012** towards the cylindrical surface **8099** defining the receiver expansion recess. The retainer **8012** begins to contract about the spherical surface **8034** as the center of the sphere (shown in dotted lines) passes beyond the center of the retainer expansion recess. At this time also, the spherical surface **8034** moves into engagement with the surfaces **8132** of the retainer flex tabs **8117**, the tabs **8117** expanding slightly outwardly to receive the surface **8034**. With reference to FIG. **372**, the spherical surface **8034** then enters into full frictional engagement with the panel inner surfaces **8132**. At this time, the retainer **8012** panels and the surface **8034** are in a fairly tight friction fit, the surface **8034** being pivotable with respect to the retainer **8012** with some force. Thus, a tight, non-floppy ball and socket joint is now created between the retainer **8012** and the shank upper portion **8**.

[0708] With reference to FIGS. **373** and **374**, the shank **8004** and attached retainer **8012** are then moved partially downwardly and then into a fully locked desired position (FIG. **375**) with the retainer **8012** bottom surface **8124** seated on the receiver surface **8104**. This may be accomplished by either an upward pull on the receiver **8010** or, in some cases, by driving the shank **8004** further into the vertebra **8017**. Also with reference to FIGS. **375** and **376**, the insert **8014** may be pressed downwardly by a tool (not shown) and/or ultimately by a rod and closure top as shown in FIGS. **377-380**. Also, in some embodiments, when the receiver **8010** is pre-assembled with the shank **8004**, the entire assembly **8001** may be implanted at this time by inserting the driving tool (not shown) into the receiver and the shank drive **8042** and rotating and driving the shank **8004** into a desired location of the vertebra **8017**. With further reference to FIGS. **375** and **376**, at this time, the receiver **8010** may be articulated to a desired angular position with respect to the shank **8004**, such as that shown in FIG. **382**, that will be held, but not locked, by the frictional engagement between the retainer **8012** panels **8117** and the shank upper portion **8008**.

[0709] With particular reference to FIGS. **374-376**, prior to assembly with the rod **8021** and the closure top **8018**, the compression insert **8014** upper end surfaces **8175** are located directly below the receiver stops **8092** (see FIG. **374**) and the sloping or tapering surfaces **8177** are resting on or near the edge **8096** that defines the beginning of the receiver channel seat **8068** that is either substantially vertical or may also have an inward slope. With particular reference to FIGS. **378** and **379**, as the closure top and rod press down upon both the shank upper portion **8008** and the insert saddle **8153**, the surfaces **8177** of the insert are wedged against the receiver edges **8096**, pressing the insert into a full frictional engagement with the receiver **8010**. With reference to FIG. **380**, at this time, the insert through bores **8166** are aligned with the upper arched portion **81** of the receiver keyhole like through bores **8078**. Thus, a tool (not shown) may be used to press inwardly on the

insert **8014** at either side thereof at the bores **8166** and pull the insert **8014** upwardly away from the receiver seat **8068** and edge surface **8096**.

[0710] The rod **8021** is eventually positioned in an open or percutaneous manner in cooperation with the at least two bone screw assemblies **8001**. The closure structure **8018** is then inserted into and advanced between the arms **8062** of each of the receivers **8010**. The closure structure **8018** is rotated, using a tool engaged with the inner drive **8186** until a selected pressure is reached at which point the rod **8021** engages the bone screw shank **8004** at the upper surface **8004** as well as the saddle **8153** of the compression insert **8014**, pressing the insert spherical surface **8181** against the shank spherical surface **8034**, the rod **8021** pressing the shank upper portion **8008** into locked frictional engagement with the retainer **8012**. Specifically, as the closure structure **8018** rotates and moves downwardly into the respective receiver **8010**, the point **8189** and rim **8190** engage and penetrate the rod surface **8022**, the closure structure **8018** pressing downwardly against and biasing the rod **8021** into direct compressive engagement with the shank upper portion **8008** toward the retainer **8012** and into locking engagement therewith, the retainer **8012** frictionally abutting the surface **8104** and expanding outwardly against the cylindrical surface **8101**. For example, about 8080 to about 8120 inch pounds of torque on the closure top may be applied for fixing the bone screw shank **8006** with respect to the receiver **8010**. As best shown in FIGS. **381** and **382**, as the retainer **8012** expands outwardly against the receiver cylindrical surface **8101**, the panels **8117** are pulled away from the shank upper portion **8008**, pulling the friction fit surface **8132** away from the spherical surface **8034**. This is not of concern at this time as the friction fit feature, temporarily advantageous for articulation and placement of the shanks **8004** with respect to the receivers **8010** during the surgical process, is no longer required.

[0711] With reference to FIGS. **381** and **382**, two different angular configurations of the shank **8004** and receiver **8010** are shown. With respect to both of the drawing figures, the rod **8021** bears down directly on the shank upper surface **8040** when the assembly **8001** is in a locked position. Also, when in a locked position, the insert surface **8181** directly engages a portion of the shank spherical surface **8034**. Thus, the closure top **8018** can then be loosened without loosening the lock on the polyaxial mechanism provided by the insert **8014** pressing on the shank surface **8034**. With reference to FIGS. **383-385**, the rod **8021** and closure **8018** are shown removed at FIG. **383** and replaced by a deformable rod **8021'** and cooperating closure top **8018'** to result in an alternative assembly **8001'**.

[0712] If a user wishes to unlock the insert **8014** from the receiver **8010**, a tool (not shown) may be used that includes extensions or prongs that are received by and through the opposed through bores **8078** of the receiver **8010** and received into the through bores **8166** of the insert **8014**. Such tool is then pulled upwardly in a direction along the axis B away from the receiver base **8060**, thereby pulling the insert slightly upwardly and away from the receiver base **8060** and releasing the surface **8177** from the receiver surface **8096**. Alternatively, if both the closure top **8018** or **8018'** and the rod **8021** or **8021'** are already removed from the receiver **8010**, another manipulation tool (not shown) may be used that is inserted into the receiver at the opening **8066** and between the insert arms **8157**, with prongs or extensions thereof extending outwardly into the insert through bores **8166**; a piston-like portion of the tool thereafter pushing directly on the shank upper portion **8008**, thereby pulling the insert **8014** surface **8177** away from the receiver surface **8096** and thus releasing the polyaxial mechanism. At such time, the shank **8004** may be articulated with respect to the receiver **8010**, and the desired friction fit returns between the retainer **8012** and the shank surface **8034**, so that an adjustable, but non-floppy relationship still exists between the shank **8004** and the receiver **8010**. If further disassembly of the assembly **8001** is desired, such is accomplished in reverse order to the procedure described previously herein for assembly.

[0713] With reference to FIGS. **386-394**, an alternative polyaxial bone screw **8011''** according to the invention is shown that includes the shank **8004**, receiver **8010** retainer **8012**, rod **8021** and closure top **8018** of the assembly **8001** previously described herein. An insert **8014'** is included in

the assembly **8001''** that is substantially similar to the insert **8014** previously described herein. Thus, the insert **8014'** includes a body **8150'**, opposed ends **8151'**, a saddle **8153'**, upstanding arms **8157'**, a through bore **8160'** the same or similar to the respective body **8150**, opposed ends **8151**, saddle **8153**, upstanding arms **8157** and through bore **8160** previously described herein with respect to the insert **8014**. The insert **8014'** also includes pairs of side surfaces **8172'**, a pair of outer end surfaces **8174'**, a recess with tapered surfaces **8177'** and a lip **8178'** located between the surfaces **8177'** and the side surfaces **8172'** that are substantially similar to the respective side surfaces **8172**, end surfaces **8174**, tapered surfaces **8177** and lip **8178** previously described herein with respect to the insert **8014** with the exception that the surfaces **8177'** are located further inwardly than the similar surfaces **8177** such that the insert **8014'** does not lock up against the receiver edge **8096** when the insert **8014'** is pressed downwardly toward the receiver base **8060**. Thus, with particular reference to FIGS. **392-394**, when the closure top **18** presses the rod **8021** into direct locking engagement with the shank top surface **8040**, the insert surfaces **8177'** move downwardly in spaced relation with the receiver channel seat surfaces **8068** and do not wedge against or otherwise engage the edge surface **8096**. When the closure top **8018** is removed from the assembly **8001''**, the insert **8014'** loosens also and the polyaxial mechanism is unlocked. As with the assembly **8001**, once the shank upper portion **8008** is unlocked from the retainer **8012**, the retainer flex panels **8117** resiliently move back into engagement with the shank surface **8034**, once again providing a friction fit relationship between the shank upper portion **8008** and the retainer **8012**.

[0714] With reference to FIGS. **395-397**, a polyaxial bone screw assembly **8001'''** is shown having the bone screw shank **8004**, retainer **8012**, rod **8021** and closure top **8018** identical or substantially similar to the assembly **8001** and **8001''** previously described herein. The assembly **8001'''** however, does not include an insert **8014** or **8014'**.

[0715] It is to be understood that while certain forms of the present invention have been illustrated and described herein, it is not to be limited to the specific forms or arrangement of parts described and shown.

Claims

1. A pivotal bone anchor assembly intended for securing a longitudinal connecting member to a bone of a patient, the pivotal bone anchor assembly comprising: a receiver comprising a lower portion defining an inner cavity and an upper pair of opposed arms defining an open channel configured to receive the longitudinal connecting member, the inner cavity including an interior seating surface adjacent a bottom opening and an upper region communicating with the open channel to define a central bore, the central bore including a downward-facing abutment stop surface above the interior seating surface; a shank comprising a partial spherical shank head dimensioned for uploading into the inner cavity through the bottom opening of the lower portion of the receiver, and an anchor portion opposite the shank head configured for attachment to the bone; an insert configured for uploading into the central bore of the receiver through the bottom opening and having an upward-facing surface engageable with the downward-facing abutment stop surface and an insert channel configured to receive the longitudinal connecting member upon the longitudinal connecting member being received in the open channel; and an expandable open ring retainer having a non-expanded configuration and an expanded configuration, the retainer configured for uploading into the inner cavity through the bottom opening and for positioning in the inner cavity after the insert is uploaded into the receiver and before the shank head is uploaded through the bottom opening, wherein the retainer is configured to expand to the expanded configuration as the shank head is uploaded therethrough and thereafter return to the non-expanded configuration and engage the interior seating surface of the inner cavity to capture the shank head of the shank within the lower portion of the receiver.

2. The pivotal bone anchor assembly of claim 1, wherein the lower portion is integrally formed

with the receiver.

3. The pivotal bone anchor assembly of claim 1, wherein the insert further comprises a pair of insert arms projecting upwardly from an insert base.
4. The pivotal bone anchor assembly of claim 3, wherein the pair of insert arms define the insert channel configured to receive the longitudinal connecting member.
5. The pivotal bone anchor assembly of claim 3, wherein a top surface of the pair of insert arms further comprises the upward-facing surface engageable with downward-facing abutment stop surface.
6. The pivotal bone anchor assembly of claim 3, further comprising opposite alignment notches formed into the pair of insert arms.
7. The pivotal bone anchor assembly of claim 6, wherein the opposite alignment notches are formed into outer side surfaces of the pair of insert arms.
8. The pivotal bone anchor assembly of claim 7, wherein the opposite alignment notches are configured to receive opposing spring tabs integrally formed with the receiver and extending inward into the central bore, so as to maintain an alignment of the insert with respect to the receiver.
9. The pivotal bone anchor assembly of claim 1 and further comprising the longitudinal connecting member and a closure top, wherein the closure top is configured for positioning entirely within the open channel of the receiver above the longitudinal connecting member and in engagement with a guide and advancement structure formed into interior surfaces of the upright arms to apply a downward pressure to a top of the longitudinal connecting member, so as to secure the longitudinal connecting member to the bone of the patient.
10. A method of assembling a pivotal bone anchor assembly intended for securing a longitudinal connecting member to a bone of a patient, the method comprising: uploading an insert into a central bore of a receiver through a bottom opening of the receiver, the receiver comprising a lower portion defining an inner cavity and an upper pair of opposed arms defining an open channel configured to receive the longitudinal connecting member, the inner cavity including an interior seating surface adjacent the bottom opening and an upper region communicating with the open channel to define the central bore, the central bore including an abutment stop surface above the interior seating surface, the insert having an upward-facing surface engageable with the downward-facing abutment stop surface and an insert channel configured to receive the longitudinal connecting member upon the longitudinal connecting member being received in the open channel; uploading an expandable open ring retainer into the inner cavity through the bottom opening after uploading the insert, the retainer having a non-expanded configuration and an expanded configuration and uploading a partial spherical shank head of a shank into the inner cavity of the receiver through the bottom opening after uploading the insert and the retainer, the shank including an anchor portion opposite the shank head configured for fixation to the bone; capturing the shank head within the inner cavity of the lower portion of the receiver, wherein the retainer is configured to expand into the expanded configuration as the shank head is uploaded therethrough and thereafter return to the non-expanded configuration and engage the interior seating surface of the inner cavity to capture the shank head within the lower portion of the receiver.
11. The method of claim 10, wherein the lower portion is integrally formed with the receiver.
12. The method of claim 10, wherein the insert further comprises a pair of insert arms projecting upwardly from an insert base.
13. The method of claim 12, wherein the pair of insert arms define the insert channel configured to receive the longitudinal connecting member.
14. The method of claim 12, wherein a top surface of the pair of insert arms further comprises the upward-facing surface engageable with the downward-facing abutment stop surface.
15. The method of claim 12, further comprising opposite alignment notches formed into the pair of insert arms.

16. The method of claim 15, wherein the opposite alignment notches are formed into outer side surfaces of the pair of insert arms, and wherein the opposite alignment notches are configured to receive opposing spring tabs integrally formed with the receiver and extending inward into the central bore, so as to maintain an alignment of the insert with respect to the receiver.

17. The method of claim 10, further comprising: positioning the longitudinal connecting member into the open channel of the receiver; and positioning a closure top entirely within the open channel of the receiver above the longitudinal connecting member and in engagement with a guide and advancement structure formed into interior surfaces of the upright arms to apply a downward pressure to a top of the longitudinal connecting member, so as to lock the pivotal bone anchor assembly and secure the longitudinal connecting member to the bone of the patient.
